



Technical Documentation Version 9.6

Data Management Interface (DMI)



**Center for Advanced Decision Support
for Water and Environmental Systems**
UNIVERSITY OF COLORADO BOULDER

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Chapter 1

General DMI Information

The RiverWare Data Management Interface (DMI) provides a mechanism to transfer data into and out of RiverWare in a highly configurable and automated manner. For example, you can configure a DMI that, with the click of a button, brings all of the observed inflows and initial pool elevations into the model.

Topics

- [Types of DMIs, page 1](#)
- [DMI User Interface, page 2](#)

Types of DMIs

There are three types of DMIs.

Control File-Executable

In the Control File-Executable approach, you develop a “control file” that specifies the data to move into or out of RiverWare. Also, you develop an external “executable” which interacts with the control file, RiverWare, and the external data repository. See [“Control File-Executable DMI”](#) on page 37.

Trace Directory DMI

The Trace Directory DMI is similar to the Control File-Executable approach where you develop a “control file” that specifies the data to move into or out of RiverWare. However, instead of an executable, you specify a top-level directory, under which the data files for traces are held in subdirectories named with the trace number. See [“Trace Directory DMI”](#) on page 59.

Database DMI

RiverWare supports a direct connection between the following “databases”:

- the Hydrologic Engineering Center’s Data Storage System (HEC-DSS),
- the Hydrologic Data Base (HDB),
- Microsoft Office Excel,
- the Demand Input Tool (DIT), and
- Web services
 - CDSS Surface Water Time Series - Day: Colorado’s Decision Support Systems,
 - CWMS Data API web service,

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General DMI Information

- HDB web service,
- Reclamation Hydromet (Missouri Basin / Great Plains) and
- USGS web service.

In the Database DMI configuration, all of the slot specification is done within RiverWare using Datasets, Name Maps, and Database DMIs. See [“Database DMI”](#) on page 65.

In this section, we describe the DMI user interface components that are general to all of the approaches. Then we describe the Control File-Executable approach and its specific user interface components and the Trace Directory DMI approach and its specific user interface components. Finally, we describe the Database DMI approach and its specific user interface components.

DMI User Interface

This section describes the user interface used to execute and create DMIs. This section applies to all of the DMI approaches. See [“Database DMI”](#) on page 65 for details on the user interface for the Database DMI.

Accessing DMIs and the DMI Manager

The Workspace provides a DMI menu and a DMI toolbar button.  Selecting either the **DMI** menu, then **DMI Manager** or the toolbar button opens the DMI Manager. Once one or more DMIs are defined, the DMI menu also contains these DMIs, which can be invoked directly from the workspace. See [“Invoking DMIs and DMI Groups”](#) on page 29 for details.

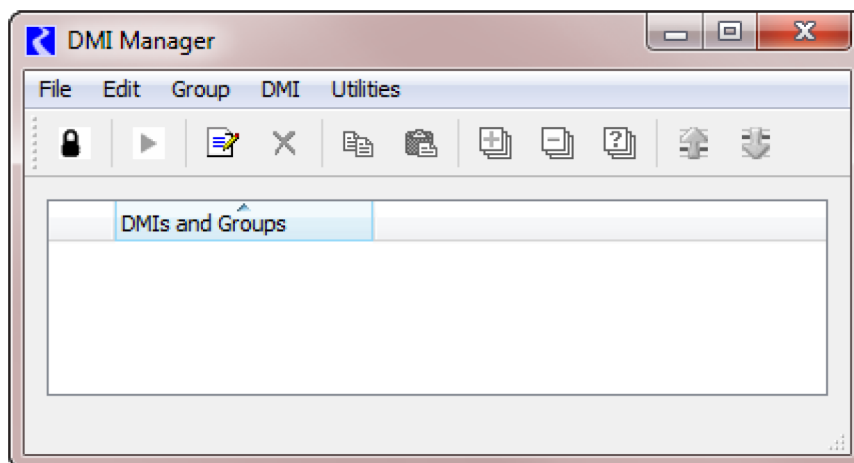


DMI Manager

The DMI Manager enables is used for the following:

- Create and delete DMIs
- Open the DMI Editor
- Create and delete DMI groups
- Edit DMI groups
- Sort DMIs and DMI groups
- Invoke DMIs and DMI groups
- Copy and paste DMIs and DMI groups

- Open the DMI Parameters
- Lock the DMI Manager
- View records of DMI invocations



In the DMI Manager and related windows, actions can be performed using one or more of the following, each of which acts on the current selection:

- A main menu item
- A toolbar button
- A context (right mouse button) menu item
- A keyboard accelerator

Main menu items, toolbar buttons and context menu items performing the same action share the same icon. For example, [Figure 1.1](#) and [Figure 1.2](#) illustrate that from the DMI Manager a user can invoke the selected DMI identically using the **DMI**, then **Invoke** main menu item, the **Invoke** toolbar button, the DMI **Invoke** context menu item and the **Control-I** keyboard accelerator.

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Figure 1.1

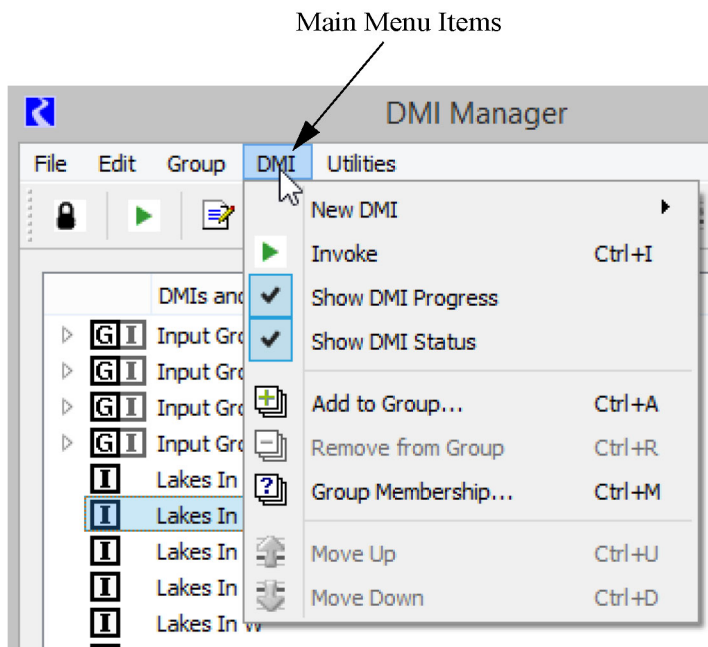
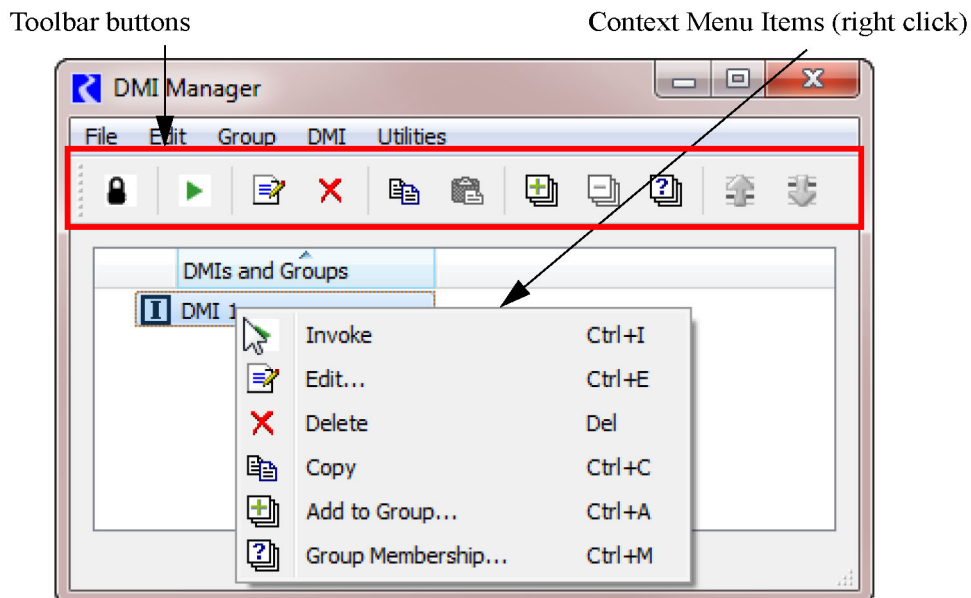


Figure 1.2



Actions which aren't valid for the current selection are disabled. (That is, their main menu items, toolbar buttons and context menu items are disabled.)

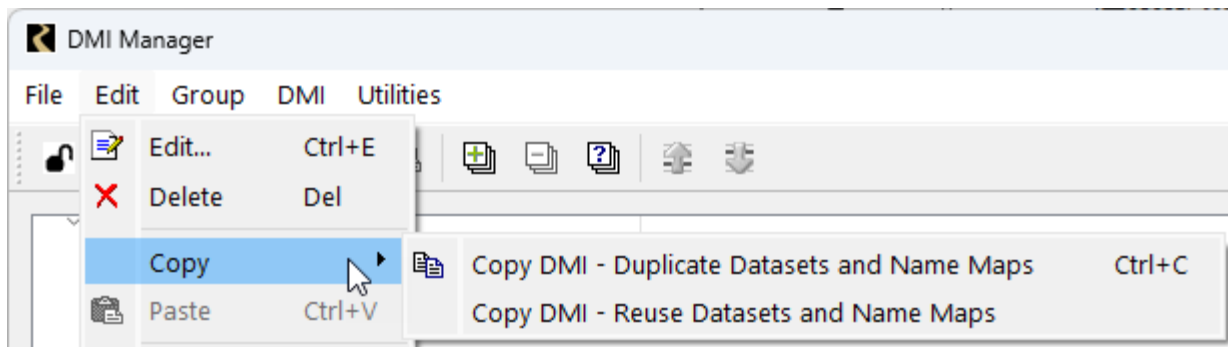
The document refers to main menu items when describing actions; be aware that in most cases toolbar buttons, context menu items and keyboard accelerators are also available for the actions.

Copy and Paste DMIs

The **Edit** menu and right-click shortcut menu have different **Copy** operations based on the type of DMIs. The options are:

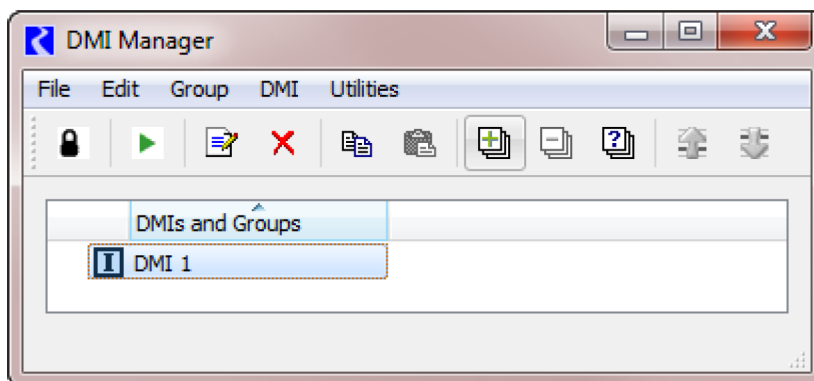
- **Copy**. For Control File - Executable DMIs and Trace Directory DMIs, this is the standard copy operation. Only the DMI is copied.
- **Copy DMI - Duplicate Datasets and Name Maps**. For Database DMIs, this operation provides a “deep copy” of the Database DMI and all referenced Datasets and Name Maps accessed. When pasted, the new Database DMI is connected to the newly duplicated Datasets and similarly, Name Maps.
- **Copy DMI - Reuse Datasets and Name Maps**. For Database DMIs, this is a “shallow” copy of the Database DMI only. None of the accessed Datasets or Name Maps are duplicated, only the Database DMI itself. When pasted, the new Database DMI refers to the original Datasets and Name Maps, that is, they are reused.
- **Copy Group**. For DMI groups, this operation creates an identical group with the same member DMIs. The DMIs are not duplicated.

A **Paste** operation will add the copied items with the name “Copy of ...” terminology.



Creating, Deleting, and Sorting DMIs

Create a new DMI under the **DMI**, then **New DMI** main menu item and choose the type of DMI to create; the DMI has a default name.



I indicates an input DMI; similarly **O** indicates an output DMI.

Delete a DMI with the **Edit**, then **Delete** main menu item.

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Open the DMI Editor with the **Edit**, then **Edit** main menu item or double-click the row. The DMI Editor is different for the different types of DMIs. See “[DMI Editor Dialog](#)” on page 53 for details on the DMI Editor for the Control File-Executable approach. See “[Trace Directory DMI Editor Dialog](#)” on page 61 for details on the Trace Directory DMI. See “[Creating a Database DMI](#)” on page 68 for details on the DMI Editor for the Database DMI.

Sorting of DMIs and DMI groups is limited to sorting by name or type by selecting column header.

DMI Groups

A DMI group is an ordered collection of one or more DMIs. When a DMI group is invoked, its DMIs are invoked in order, until all have completed or one has failed. There are two kinds of groups:

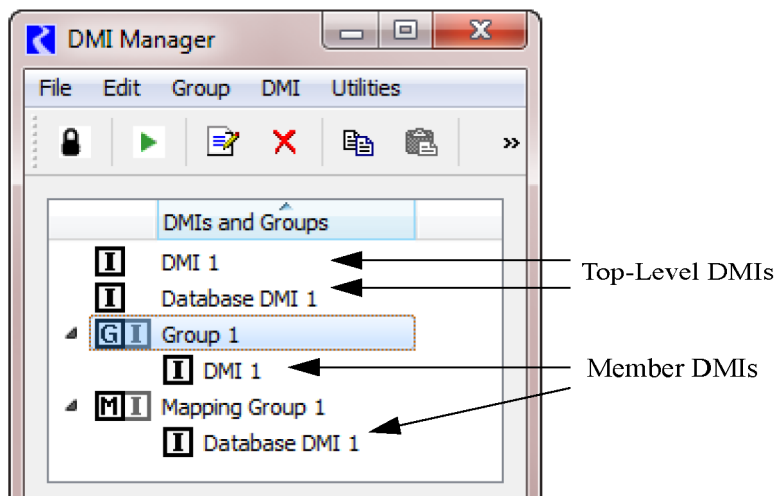
- **GI DMI Group**. This is a collection of DMIs to be executed in order.
- **MI Year Mapping Group**. This special type of group maps run years to years in the database. See “[Year Mapping Groups](#)” on page 7 for details.

Additionally, an icon indicates whether the group contains all Input DMIs **I**, all Output DMIs **O** or a mix of Input and Output DMIs **IO**

Create a new DMI group with the **Group**, then **New Group** or **New Year Mapping Group** main menu item.

DMIs can appear one or more times in the DMI Manager; once as a top-level DMI and zero or more times as a member of a group. Actions which pertain to a DMI globally (for example, editing it) are only valid when a top-level DMI is selected; actions which pertain to a DMI within a group (for example, moving it up or down) are only valid when a member DMI is selected. See [Figure 1.3](#).

Figure 1.3



When the distinction is important, a DMI is referred to as either a top-level DMI or a member DMI.

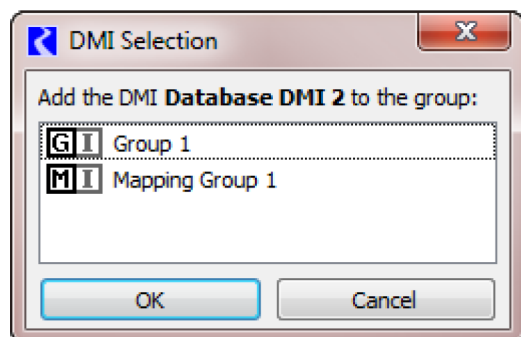
Editing DMI groups includes:

- Editing a DMI group’s name
- Adding a top-level DMI to a group

- Removing a member DMI from a group
- Moving a member DMI within a group (e.g. ordering the DMIs within the group)
- Querying a DMI's group membership

To edit a DMI group's name double-select the name cell, to edit the name in place.

To add a top-level DMI to a group, select the **DMI**, then **Add to Group** main menu item. You are then presented with a dialog listing the available groups; select the desired group and select **OK**.

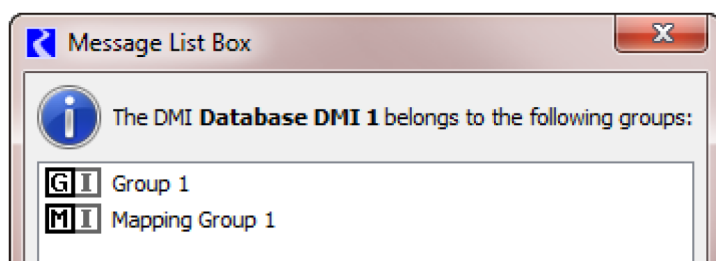


If no groups are available (e.g. there are no groups or the DMI is a member of all groups) you will be notified.

To remove a member DMI from a group, select the **DMI**, then **Remove from Group** main menu item.

To move a member DMI within a group, select either the **DMI**, then **Move Up** main menu item or the **DMI**, then **Move Down** main menu item.

To query a group membership for a DMI, select the **DMI**, then **Group Membership** main menu item; the group membership is displayed.



Year Mapping Groups

A **Year Mapping Group** **MI** allows you to import data from the specified years in your database to years in your run. For example, for a 5-year run with run dates from 1/1/2000 to 12/31/2005 you might want to use data from 1992, 1936, 1952, 1962, 1932. This tool allows you to create the map between the data in the database and the years of your run. It has a specific algorithm for dealing with leap days; see [“Year Mapping Algorithm”](#) on page 12 for details.

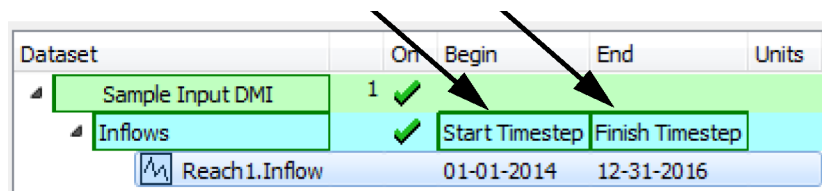
Limitations on Year Mapping Groups

- Year Mapping Groups can only contain Input DMIs. They are not relevant for Output DMIs.

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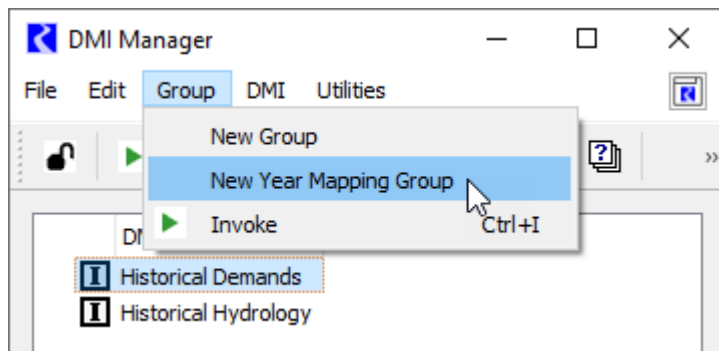
- Year Mapping Groups support Control File-Executable DMIs and Database DMIs. They do not support Trace Directory DMIs.
- Year Mapping Groups can be used for run timesteps of 1 Day and smaller. They do not support 1 Month or 1 Year timesteps.
- Year Mapping Groups will always import data from the run Start Timestep to Finish Timestep. For a Control File-Executable DMI, if the control file contains the start_timestep or end_timestep keyword (see [“Time Range Specification”](#) on page 42), a warning will be issued, and the keyword will be ignored. For a Database DMI, if the Begin Timestep and End Timestep are something other than Start Timestep and Finish Timestep, they will be ignored.



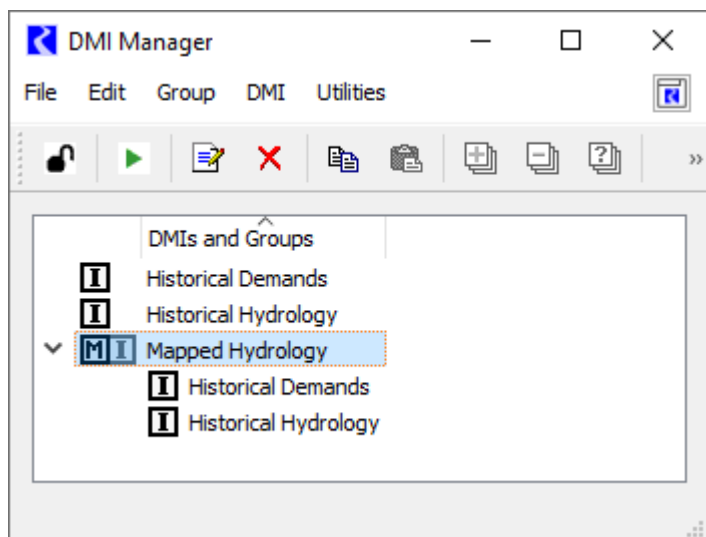
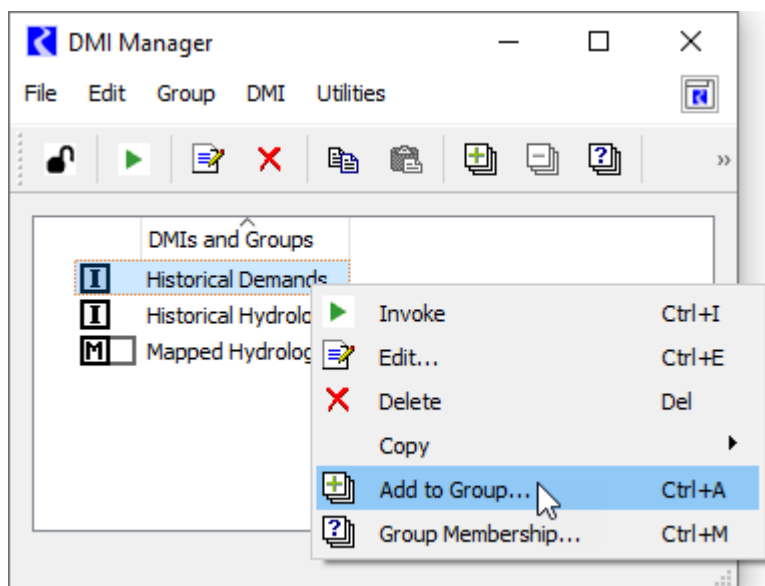
Dataset	On	Begin	End	Units
Sample Input DMI	1			
Inflows		Start Timestep	Finish Timestep	
Reach1.Inflow		01-01-2014	12-31-2016	

Creating and Using a Year Mapping Group

1. **Add a Year Mapping Group:** In the DMI Manager, select **Group**, then **New Year Mapping Group**.



2. **Add DMIs to the Group:** Select an input DMI. Then select **DMI**, then **Add to Group**; select the desired group, and select **OK**. Alternatively, right-click on the input DMI. Then select **Add to Group**.



3. **Define the map:** Edit the group by double-clicking the item. This brings up a simple dialog where you may edit the mapping group's name, the **Year in Database**, and **Comment** for each run year.
 - **Run Year:** The year in the RiverWare run. This is not editable and is taken from the Run Control.
 - **Year in Database:** The year of data you wish to import for the given Run Year.

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- **Comment:** A text field to optionally annotate the entry.

DMI Year Mapping Group

Name:

Select: ☐ Consecutive Years ☒ Individual Years

First Year:

Run Year	Year in Database	Comment
2014	1933	Year Type II
2015	1935	Year Type I
2016	1939	Year Type I
2017	1954	Year Type II
2018	1978	Year Type I
2019	1945	Year Type I
2020	1938	Year Type I
2021	1964	Year Type I
2022	1971	Year Type II
2023	1948	Year Type II
2024	1956	Year Type I

The ordered sequence represents dry years of history. Dry is type I and II out of seven.

Show: ☒ **Group Comment**

You can copy and paste the information from and to external applications using the right-click context menus shown.

Run Year	Year in Database	Comment
2014	2007	Year Type I
2015	2000	Year Type II
2016	2005	
2017	<6	
2018	<6	
2019	1948	Year Type III

You can optionally select the **Consecutive Years** option. Then enter the **First Year**, and all run years will be filled in automatically with consecutive years.

DMI Year Mapping Group

Name:

Select: ☒ Consecutive Years ☐ Individual Years

First Year:

Run Year	Year in Database	Comment
2014	1976	
2015	1977	
2016	1978	
2017	1979	
2018	1980	
2019	1981	
2020	1982	
2021	1983	
2022	1984	
2023	1985	
2024	1986	

The ordered sequence represents dry years of history. Dry is type I and II of seven.

Show: ☒ Group Comment

Note: If the workspace run period changes, editing a Year Mapping Group adjusts the run years to match the workspace run period. For years which overlap, the Year in Database and Comment are preserved. You are notified that the mapping group's run period has been adjusted. If you cancel out of the confirmation dialog the adjustment is not canceled (only the changes made after the dialog opened are canceled).

4. **Invoke the Year Mapping Group** like any other group. Select the group. Then use one of the following options.
 - Select **DMI**, then the **Invoke** menu item,
 - Select the **Invoke** toolbar button ,
 - Right-click on the group. Then select the **Invoke** context menu item,
 - The **Control-I** keyboard accelerator.

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Year Mapping Groups can also be invoked from the **DMI** menu on the main workspace or SCT or from an Execute DMI script action in the same manner as a DMI (see [“Execute DMI” in Automation Tools](#)).

Year Mapping Algorithm

So what does this Year Mapping Group actually do? Following is the algorithm and description of errors.

For each DMI in the Year Mapping Group,

For each Run Year, X

Obtain the Year in the Database from the table, N

Query the database for N to get the series of data

Set the slot's values for year X with returned data from year N, dealing with leap days as shown in the following table.

End For

End For

[Table 1.1](#) describes how the mapping deals with Leap Years.

Table 1.1

Year in Database	Run Year	Result
Non Leap Year	Non Leap Year	Bring in data exactly from database
Leap Year	Leap Year	Bring in data exactly from database
Leap Year	Non Leap Year	Remove the February 29th values
Non Leap Year	Leap Year	Repeat the February 28th values on February 29th (unless the run Start Timestep is February 29th)

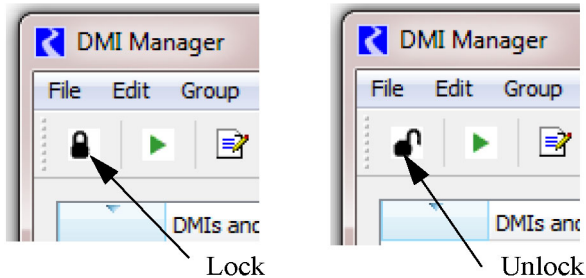
One exception to the cases in [Table 1.1](#) is when the run Start Timestep is February 29 of a leap year, and the year in the database is a non leap year. In that case, the values from March 1 are used for February 29 and March 1 because February 28 is not in the run range.

Following are some possible error conditions:

- If the workspace run period has changed and you have not opened the Year Mapping Group to update the run years, invoking a Year Mapping Group results in an error. Open the Year Mapping Group and specify a map for the new run range.
- Executing an Output DMI in a Year Mapping Group leads to an error.
- Unspecified Years in the Database lead to an error.
- Specifying a year in the Mapping Group that is not fully included in the database leads to an error.

Locking the DMI Manager

To lock the DMI Manager, select **Edit**, then **Lock** main menu item. Before locking, you must close the DMI Parameter and DMI Editors (and will be warned if they are open).



When locked, you cannot edit the DMI or DMI group configuration.

To unlock the DMI Manager, select the **Edit**, then **Unlock** main menu item.

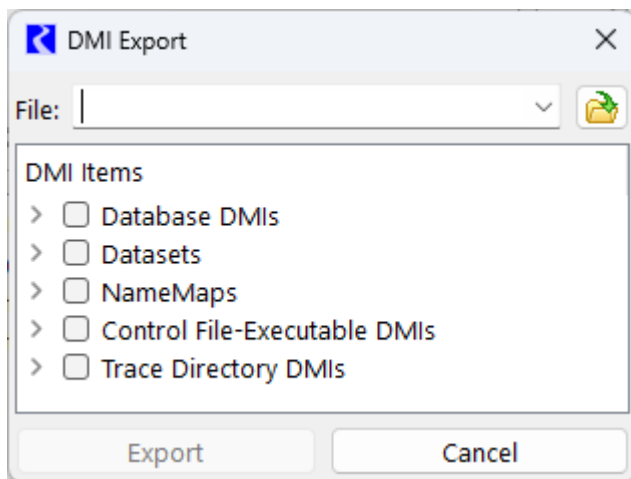
Sharing DMIs

The DMI Manager includes a feature that allows for the Export to and Import from a file. You are able to export and import DMIs, Datasets, and Name Maps. Following are instructions and descriptions of both exporting and importing DMI configurations.

Export

To export the DMI configuration, select **File**, then **Export DMIs** from the DMI Manager. The DMI Export window appears.

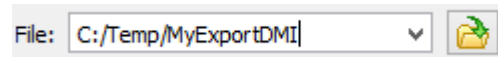
Figure 1.4 DMI Export with no selections



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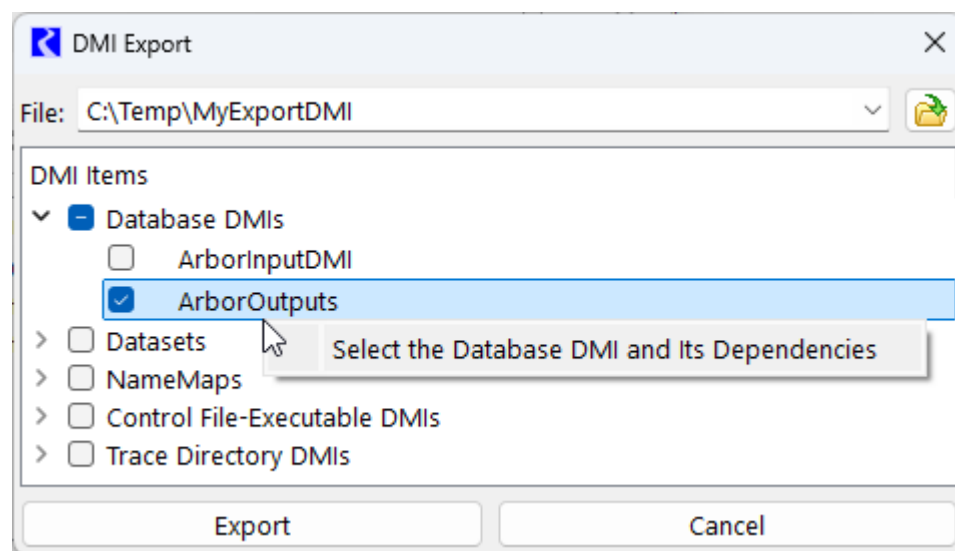
Type a file path into the **File** field or select an existing file using the browser button.



Using the tree-view options, select the check boxes to select the DMIs, Datasets or NameMaps to export. Selecting the top tree level multiple times cycles through selecting all of the items, selecting none of the item, and the last selection made.

Tip: Right-click on a Database DMI or a Dataset name to get a context menu that allows you to **Select the Database DMI and Its Dependencies** as shown in [Figure 1.5](#). This allows you to select not only the Database DMI, but also the Dataset and/or NameMap associated with the selection.

Figure 1.5 DMI Export with menu shown to select Database DMI Dependencies

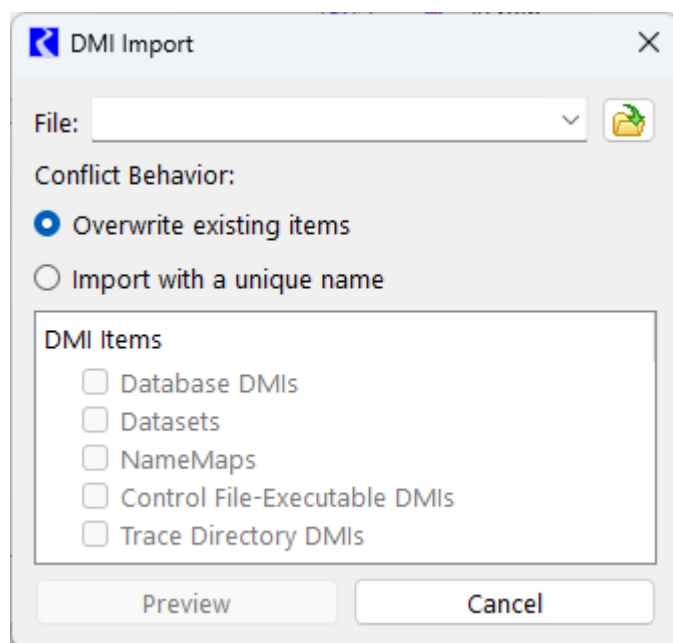


Once a file is entered and a selection is made, the **Export** button becomes active. Selecting **Export** will export the configuration to the file.

Import

Importing a set of configurations is similar to exporting. Select the **File**, then **Import DMIs** menu on the DMI Manager to open the DMI Import window.

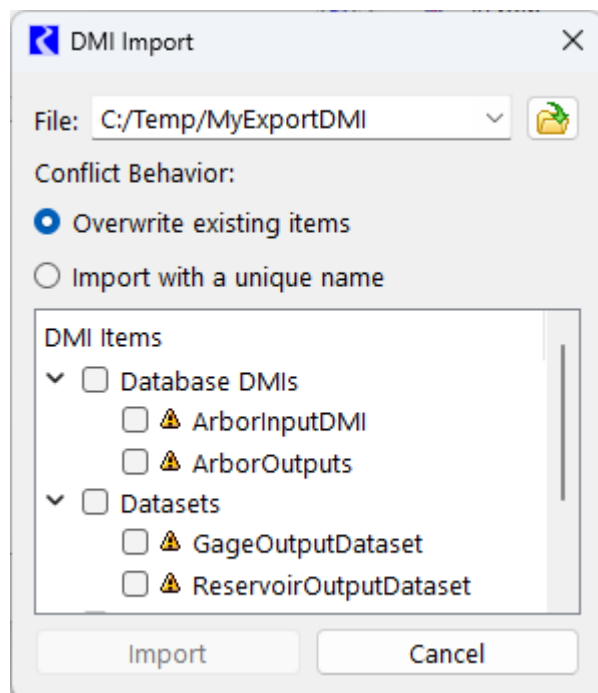
Figure 1.6 DMI Import with nothing selected



Type in the file name or use the browser to select the file to import. Select the **Preview** button to read the file and displays the items in the file.

Select the items to import from the tree-view.

Figure 1.7 DMI Import with conflicts shown

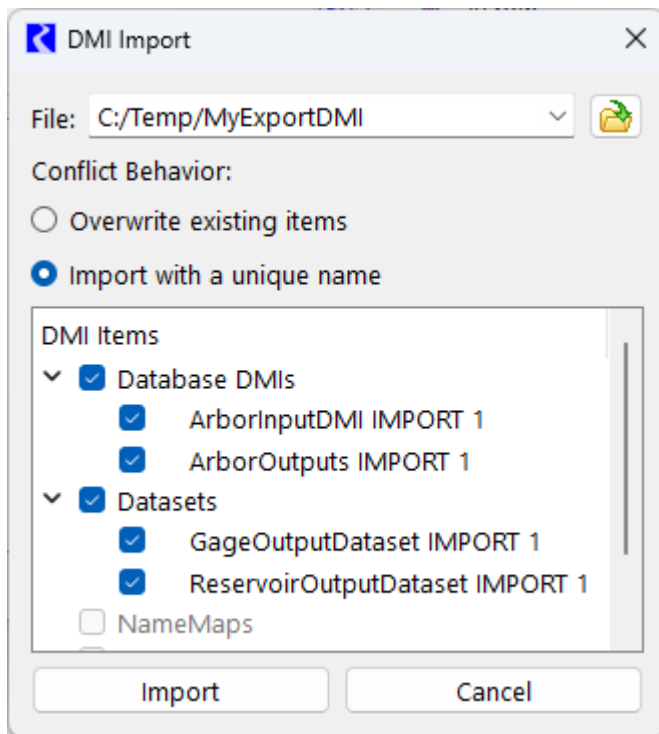


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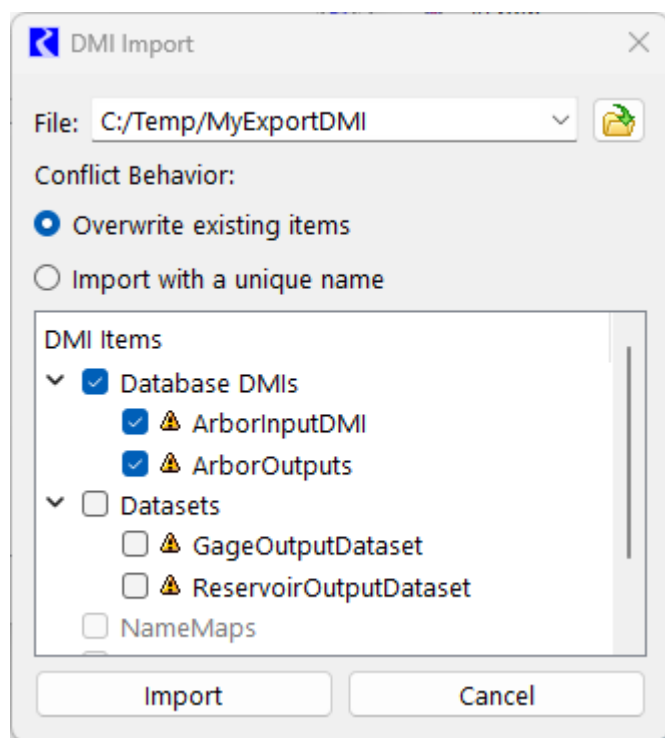
If the item already exists in the model, a small warning sign is displayed. Toggle on the **Import with a unique name** option to import the item and give it a new name, with IMPORT # appended. In addition, when this option is selected, the imported DMIs will be modified to access any newly imported Datasets and Name Maps that are also imported.

Figure 1.8 DMI Import with unique names



If the **Overwrite existing items** option is selected and there is a conflict, the import will overwrite any existing items of the same name in the model.

Figure 1.9 DMI Import that will overwrite items



When satisfied with the selection, select **Import** to bring the items into the model.

Tip: If your DMI has parameters, (“[DMI Parameters](#)”, below) the parameters must be exported to a separate file and then imported into the destination model. Then import the DMIs to the destination model and the parameters will be recognized.

DMI Parameters

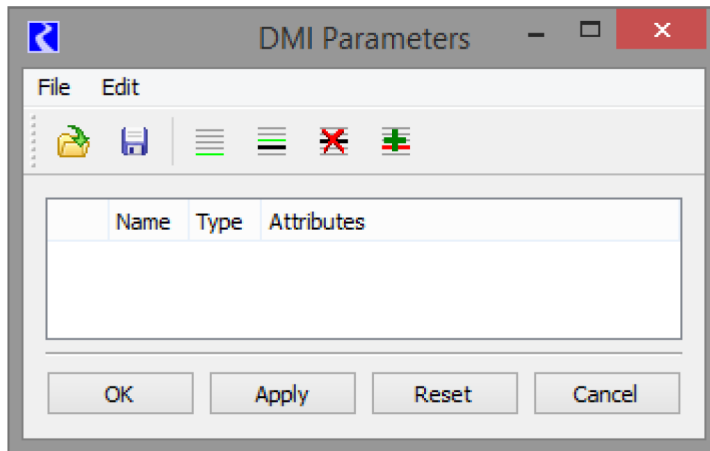
The DMI Parameters is used for the following:

- Create and delete parameters
- Edit a parameter’s name
- Edit a parameter’s type and attributes
- Order parameters

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- Export and import parameters



Creating and Deleting Parameters

To create a parameter, either:

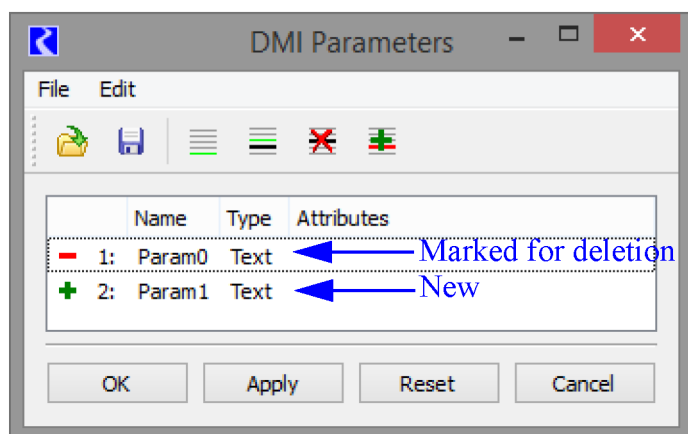
- Append the new parameter to the end of the list (selects the **Edit**, then **Append Row** main menu item)
- Insert the new parameter into the list, before the currently selected parameter (selects the **Edit**, then **Insert Row Before** main menu item)

The parameter has a default name, and **+** marks it as a new parameter. When the edits are applied the **+** mark is removed from the parameter.

To delete a parameter, select the **Edit**, then **Delete Row** main menu item. If the parameter is new (marked with **+**) it's removed from the list; otherwise **-** marks it for deletion. When the edits are applied parameters marked with **-** are removed from the list. See [Figure 1.10](#).

To restore a row, select the **Edit**, then **Undelete Row** main menu item; the **-** mark is removed from the parameter.

Figure 1.10



Editing a Parameter

When you edit a parameter as described in the following sections ▲ marks the parameter as modified. When the edits are applied the ▲ mark is removed from the parameter.

Editing a Parameter Name

To edit a parameter name, click twice in the **Name** cell, to edit the name in place. When the edits are applied the new parameter name is validated; the possible errors are:

- A blank name
- A name containing invalid characters (other than letters, digit and underscore)
- A duplicate name (the same name as another parameter)

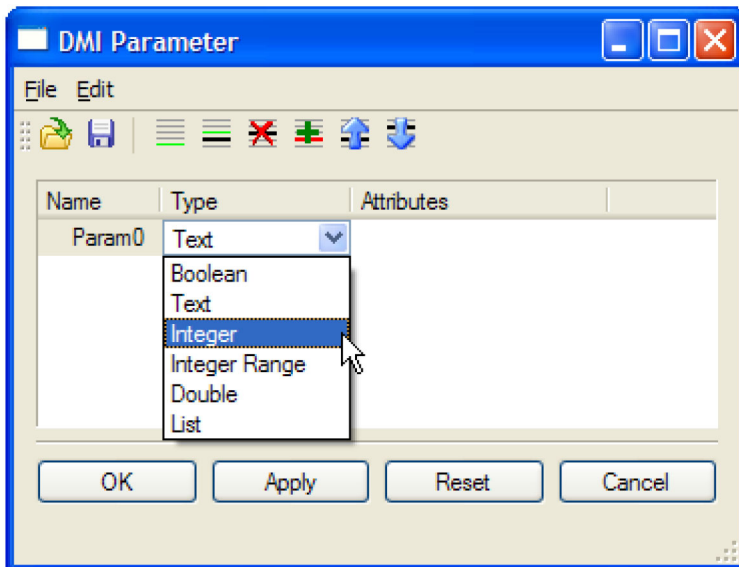
Editing a Parameter Type and Attributes

Type	Attributes	DMI Editor Control
Boolean	None	Menu containing “False” and “True”
Text	None	One line text editor, constrained to non-white space characters
Integer	None	Integer spinner, unbounded
Integer List	Minimum and maximum integer values (Default is 0, 1)	Integer spinner, bounded by minimum and maximum integer values
Double	None	One line text editor, constrained to syntactically valid double precision floating point values
List	List of text items, constrained to non-white space characters (Default is empty list)	Menu containing list of text items

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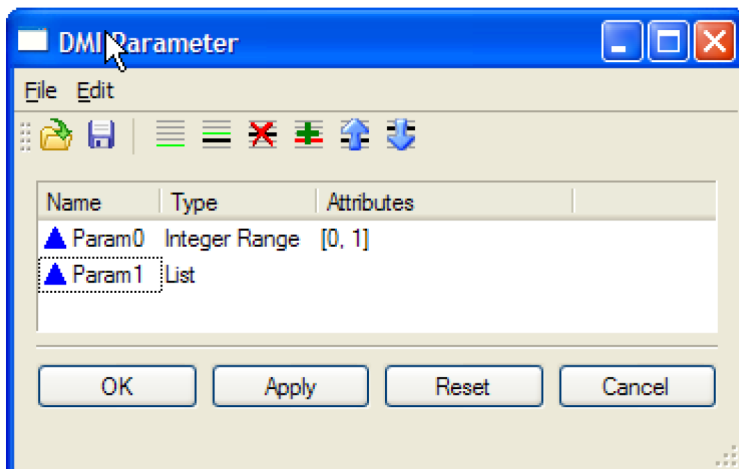
General DMI Information

To edit a parameter type, **Click-Pause-Clicks** in the **Type** cell, to activate the **Type** menu and then selects the parameter type.



If the selected type has attributes the default attributes are displayed in the **Attributes** column.

Note: A List parameter's default attribute is an empty list, so nothing is displayed.



To edit a parameter's attributes, double-click the **Attributes** cell to activate the  button, and then selects the  button to open an attribute-specific window.

For an Integer Range the attribute-specific window enables you to enter the minimum and maximum values, ensuring the maximum value is greater-than the minimum value:

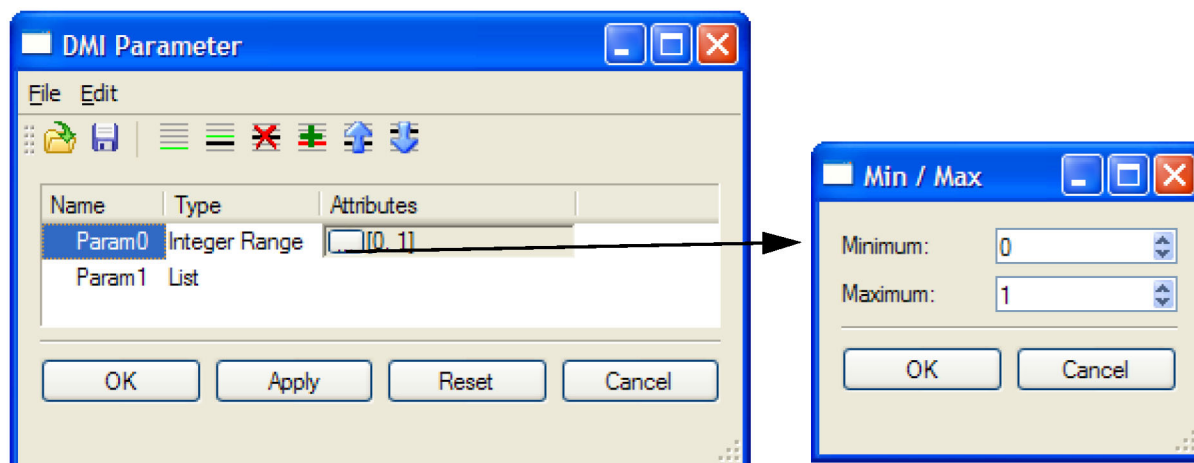


Figure 1.11 shows that for a List, the attribute-specific window enables you to append, insert, delete, edit and order text items, ensuring the text items contain only non-white space characters (first image).

Figure 1.11

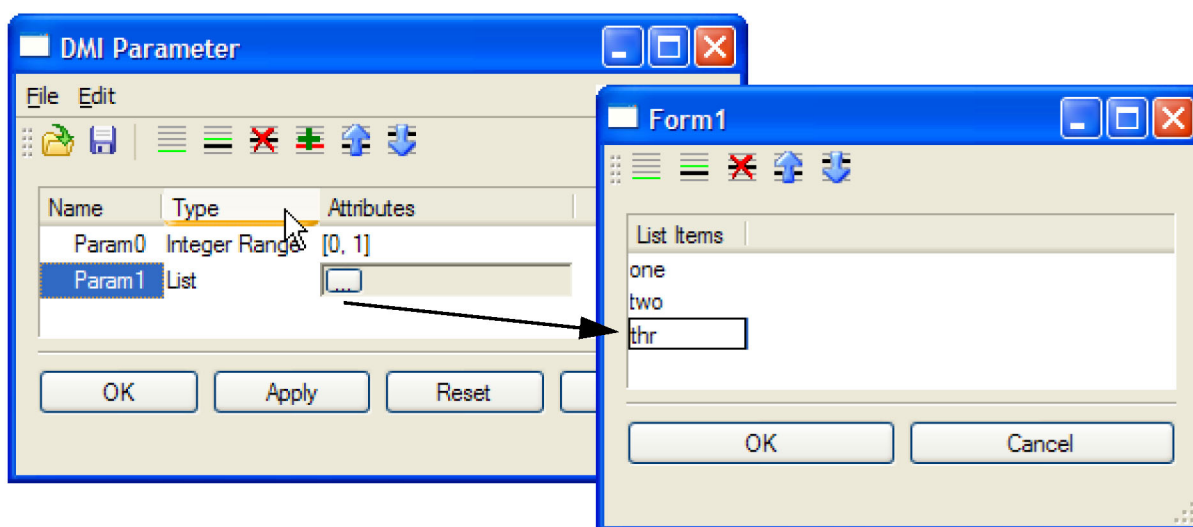
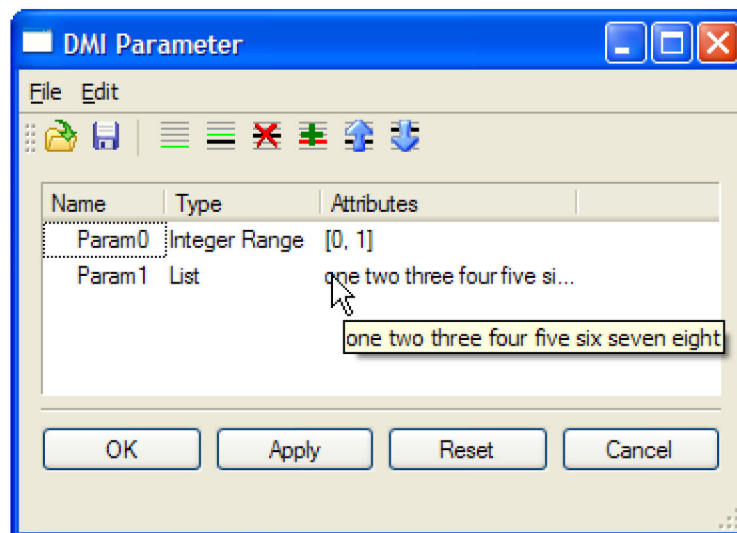


Figure 1.12 shows that if the resulting list contains more text than will fit in the **Attributes** cell, an ellipsis (“...”) is displayed and the entire list is displayed in a tooltip when the mouse pointer is paused on the **Attributes** cell (second image).

Figure 1.12



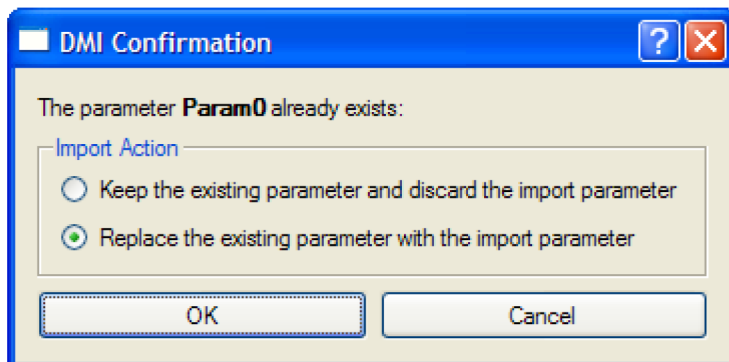
Exporting and Importing Parameters

Before exporting or importing parameters, all edits (+ , - or ▲) must be resolved (applied or canceled).

To export parameters, select the **File**, then **Export Parameters** main menu item, which opens a file chooser. Select the file to export the parameters to, then select **Save**. If the file has a “.gz” suffix the parameters are saved in a compressed format.

To import parameters, select the **File**, then **Import Parameters** main menu item, which opens a file chooser. Select the file to import the parameters from, then select **Open**.

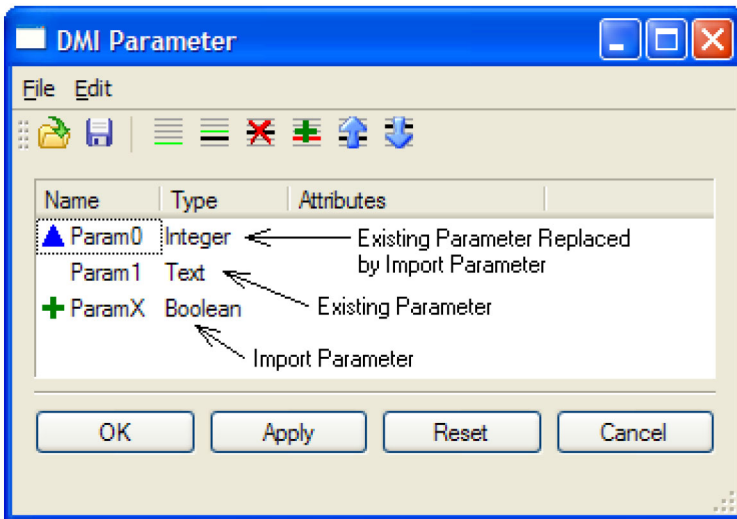
Parameters are imported by name; if the file contains a parameter which already exists, you are asked whether to keep the existing parameter or replace it with the import parameter:



Also, parameters are imported as though they were edited. Import parameters which are new are marked with + ; import parameters which replace an existing parameter are marked with ▲ :

Then, you can:

- Apply the import
- Cancel the import
- Further edit the parameters before applying the import and edits



Tip: If you are also exporting and importing DMIs, the parameters must be exported to a separate file and then imported into the destination model. Then import the DMIs to the destination model and the parameters will be recognized.

Reconciling Parameter Edits

When the DMI Parameter window changes a parameters it can invalidate DMIs using the parameters. Consider Param0 (a Boolean parameter), Param1 (a Text parameter) and Param2 (an Integer Range parameter):

- Deleting Param0 would invalidate a DMI using the parameter.
- Changing Param1's type from Text to Integer would invalidate a DMI using the parameter with a value of "xyz".
- Changing Param2's maximum value from 99 to 89 would invalidate a DMI using the parameter with a value of 93.

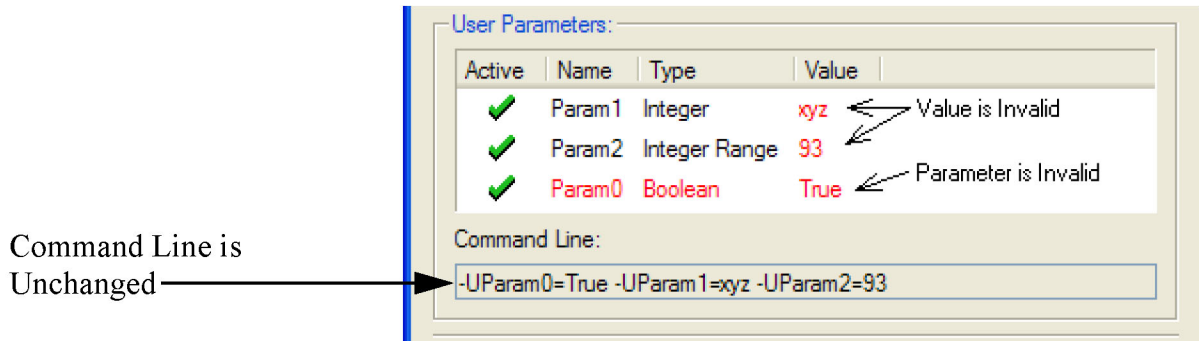
Although the DMI is invalid, it isn't changed; that is, when the DMI is invoked its executable's command line will still include:

```
-UParam0=True -UParam1=xyz -UParam2=93
```

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General DMI Information

However, when you edit the DMI there will be a graphical indication that the DMI is invalid, and you will be constrained in how the DMI can be edited.



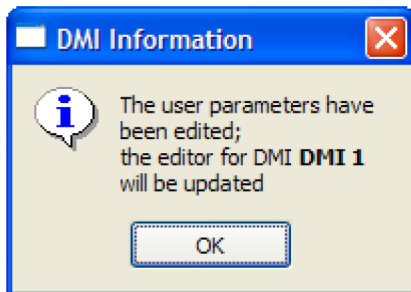
Param0 is shown entirely in *red* to indicate it is deleted. In this case, you are limited in that you:

- Can deactivate the parameter
- Cannot edit the parameter's value

Param1's value and Param2's value are shown in *red* to indicate they are invalid. Here, you are limited in that you:

- Can deactivate the parameter
- Can edit the parameter, constrained to the parameter's new type or attributes. (In other words, Param1 can be edited as an integer, Param2 can be edited as an integer with a maximum value of 89.)

Finally, if the DMI Editor is open when you edit the parameters, a confirmation dialog appears, notifying that the DMI Editor will be updated.



DMI Invocation Manager

The DMI Invocation Manager displays all DMI invocations that have been recorded. From this tool, you can clear values that were set by a specific DMI.

Note: DMI invocations are only recorded for DMIs that have been explicitly configured to do so. See ["Record Invocations"](#) on page 54 for details on Control File - Executable and Trace Directory DMIs. See ["Creating a Database DMI"](#) on page 68 for details on Database DMIs.

The Invocation Manager is opened from the DMI Manager using the **Utilities**, then **Invocation Manager** menu.

Overview

For each invocation the manager displays the following:

- The DMI name.
- The date and time the DMI was invoked.
- The invocation status, one of:
 - “OK” - The invocation succeeded, without warnings.
 - “Warnings” - The invocation succeeded, with warnings.
 - “Errors” - The invocation failed, with errors.
- The user who invoked the DMI.
- The slots on which the DMI invocation set values and the number of values.

The DMI Invocation Manager is a singleton; that is, no other edits or views to any RiverWare window can be made while the DMI Invocation Manager is open. It must be closed first. Use the **File**, then **Close Window** menu, **Ctrl-W** shortcut or **Close** button.

Values set by a DMI with Invocations Recorded are given the “Z” flag. Tooltips on the slot window displays indicate which DMI set each values on the slot. For example, in [Figure 1.13](#), on Jan 2, 2020 the value 1685.71 was set by the “Arbor Basin Operations” DMI.

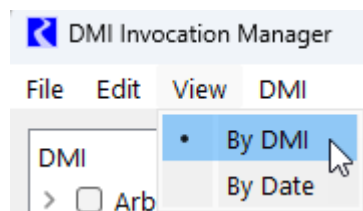
Figure 1.13

01-01-2020 Wed	1,684.35	Z
01-02-2020 Thu	1,685.71	Z
01-03-2020 Fri	1,687.01	

Set by DMI: Arbor Basin Operations

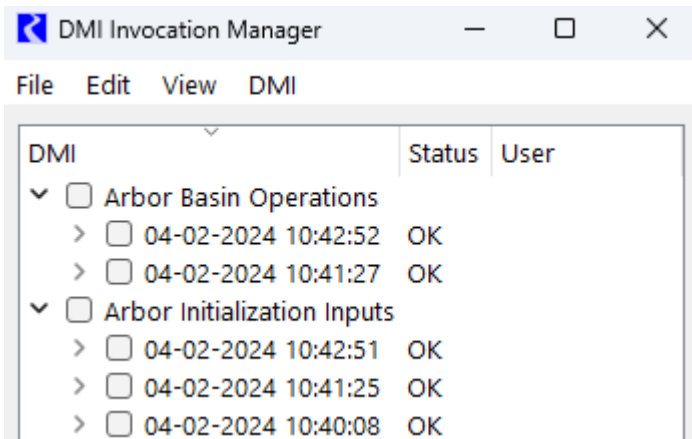
Views

The manager has two views, **By DMI** and **By Date** accessible from the View menu.



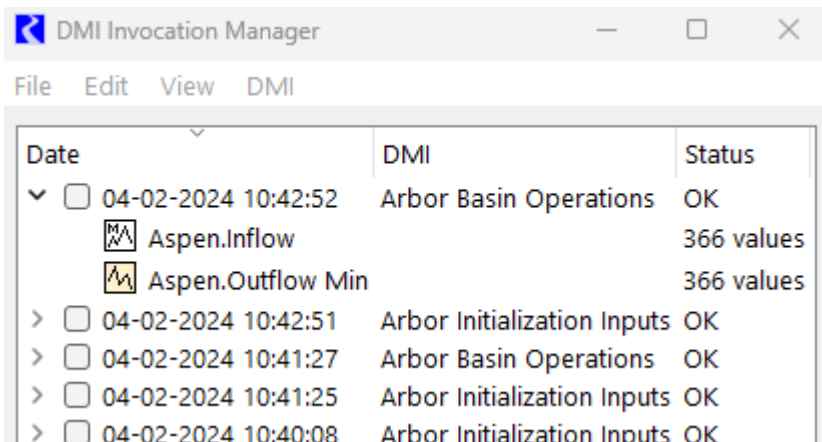
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In the **By DMI** view, the invocations are grouped by DMI, with the invocations in reverse-chronological order (most recent is on the top).



In the **By Date** view, the invocations are displayed in a list, in reverse chronological order.

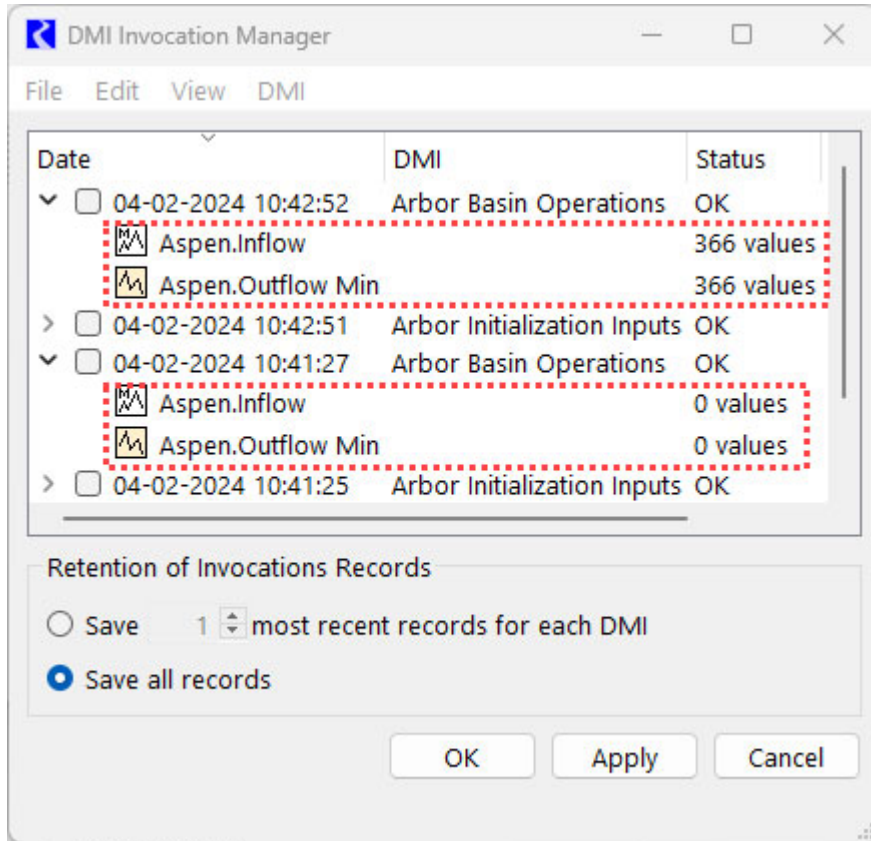
Note: The columns are sortable by selecting the column heading.



In both views, the slots set by an invocation are displayed in a tree view in alphabetical order.

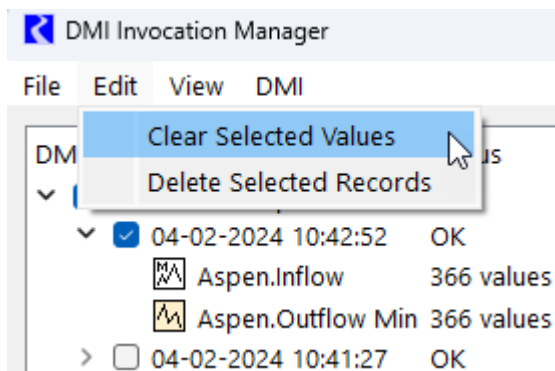
It is possible an invocation of a DMI will be superseded by other events, perhaps another invocation of the DMI or user inputs from the user interface. In , the first invocation of “Arbor Basin Operations” (bottom) has been

superseded by the second invocation (top), and consequently the first invocation (bottom) no longer is responsible for setting the values that are on the slot. This is indicated with the “0 values” in the Status column.



Clearing Values

To clear numerical values that were set by a DMI invocation, select the invocations, and then from the **Edit** menu **Clear Selected Values**.



The **Edit**, then **Clear Selected Values** feature clears values set by DMIs on a per-invocation basis; you select the DMI invocations to clear, and only the values set during those invocations will be cleared. Suppose a DMI is invoked twice, setting values on a SeriesSlot at timesteps t_1 and t_2 during the first invocation and at timesteps t_3

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General DMI Information

and t_4 during the second invocation. If a user selects the first invocation, only the values set at timesteps t_1 and t_2 will be cleared.

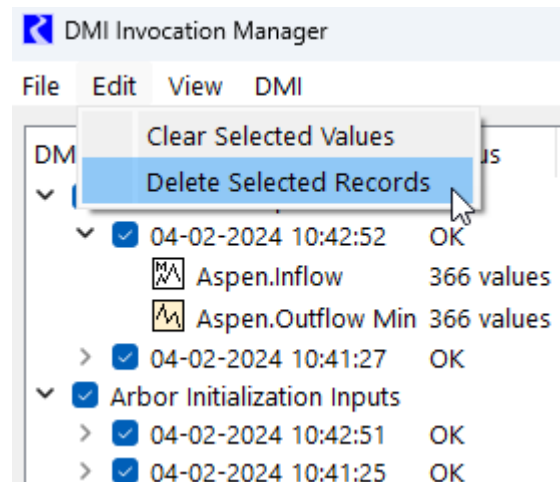
Following are some limitations on the clearing of DMI values:

- Control File-Executable type DMIs which set values AND flags are not clear-able.
- You are able to clear values set by DMIs invoked from the user interface (the DMI Manager and the workspace menu) and from batch mode - not from RPL blocks or multiple run management.
- The feature clears values set by a DMI (as opposed to restoring values overwritten by a DMI). Specifically, if a DMI overwrites a non-NaN value, the clear feature will replace the value the DMI wrote with NaN - it will not restore the previous non-NaN value.
- The feature does not clear any “side effects” of a DMI. For example, if a DMI sets a Series Slot’s start date, the feature will not restore the previous start date.
- The feature clears values set by DMIs on series slots on a per-timestep basis. If a subsequent action (e.g., user input) overwrites a value set by a DMI, it will not be cleared.
- The feature clears table slots and scalar slots in their entirety. If a subsequent action (e.g., user input) overwrites a table value set by a DMI, it will be cleared.

DMIs can then be executed directly from the DMI Invocation Manager using the **DMI** menu.

Deleting Invocation Records

To delete Invocation Records, select the desired invocation records, and then delete by selecting **Edit** and then **Deleted Selected Records**. This operation deletes the record from the list of invocations but does not delete any of the values.



Retention of Invocation Records

When recording invocations, the DMI information is written to the model file during model save. There are two options to specify how much information to retain over time:



Retention of Invocations Records

☒ Save 1 most recent records for each DMI

☐ Save all records

- **Save N most recent records for each DMI.** This is the default option with the default value set to 1 record. You can specify how many invocation records to save.
- **Save all records.** This was the original default but led to saving unnecessary information in the model file. Use this option to save all invocation records.

When switching from **All** to **N** records, a confirmation is presented if records will be deleted.

Invoking DMIs and DMI Groups

There are five ways to invoke a DMI or DMI Group as described below:

DMI Manager

To invoke a top-level DMI, select **DMI**, then **Invoke**; to invoke a DMI group, select **Group**, then **Invoke**.

Workspace

The Workspace **DMI** menu. You can alternate between a conventional menu ([Figure 1.14](#)) and a cascading menu ([Figure 1.15](#)). In the conventional menu the DMIs and DMI groups are sorted by type (first input DMIs then output DMIs then DMI groups). Within the type, they are sorted alphabetically.

Figure 1.14 DMI Menu using Conventional Menus

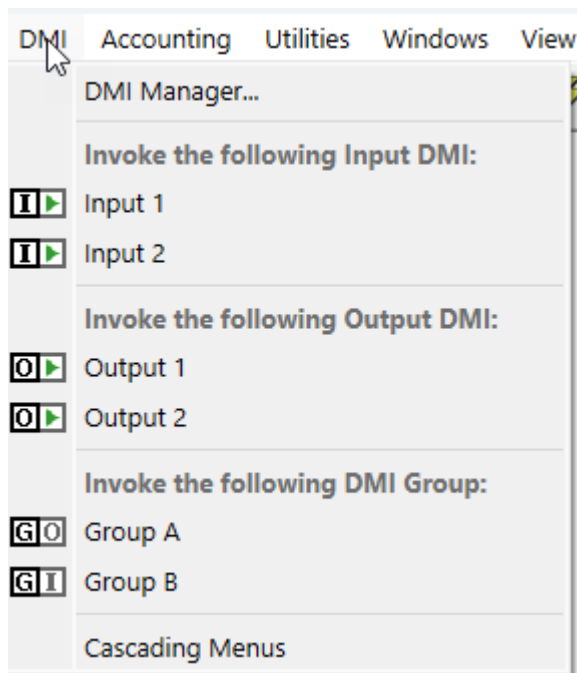
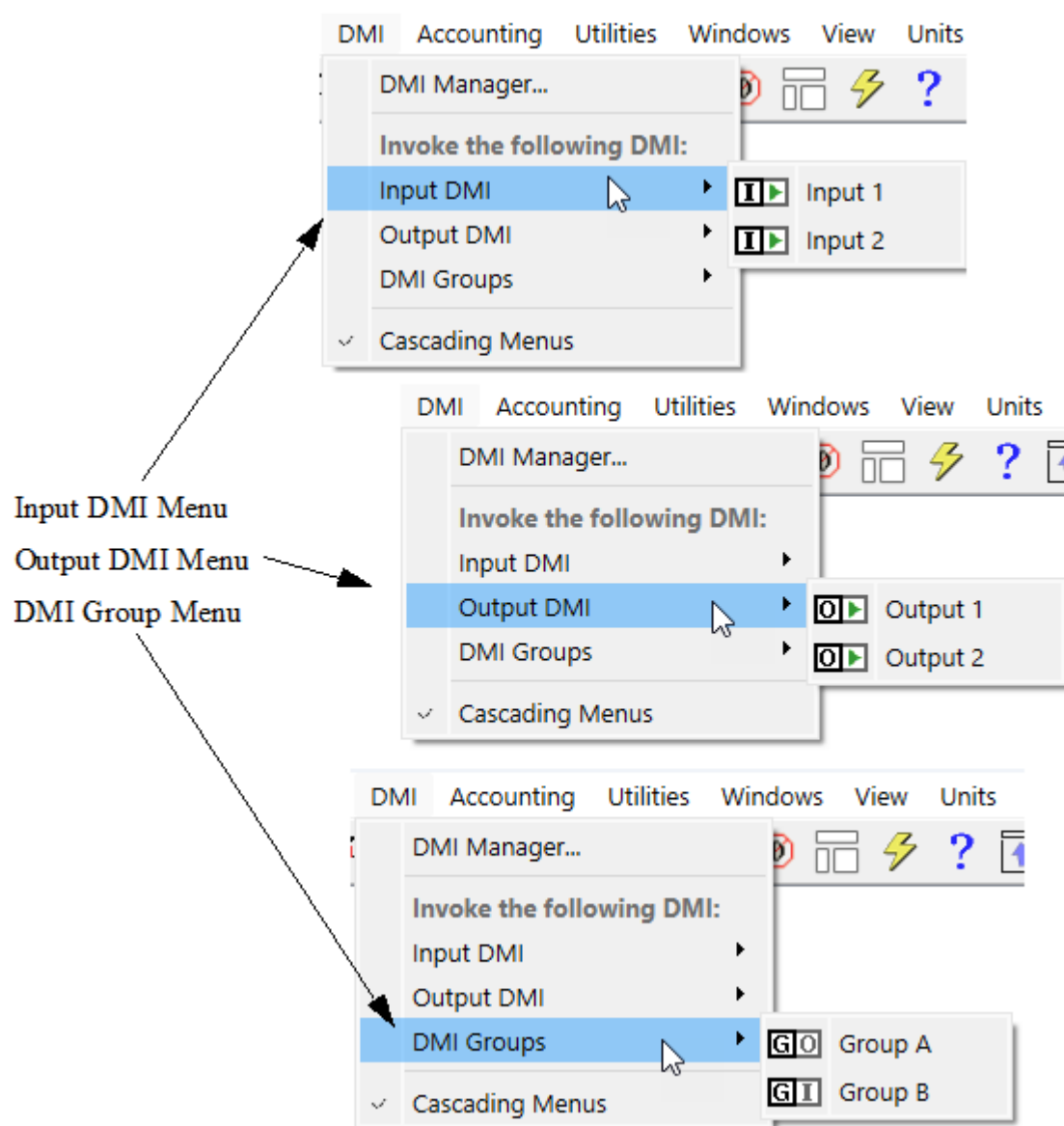


Figure 1.15 DMI Menu using Cascading Menus



SCT

The SCT has the same **DMI** menu as the Workspace (described above).

DMI Invocation Manager

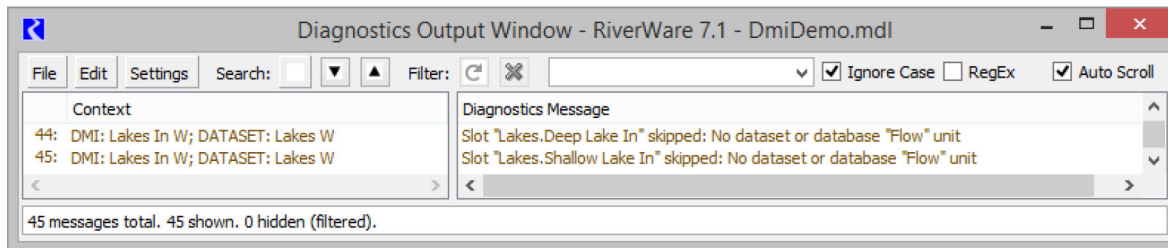
The DMI Invocation Manager has a similar menu to the Workspace **DMI** menu.

RPL Blocks

A DMI or a DMI group can be executed from a RPL block (typically a rule) as either a pre-execution or post-execution. See [“Executing DMIs From Blocks”](#) in *RiverWare Policy Language (RPL)* for details.

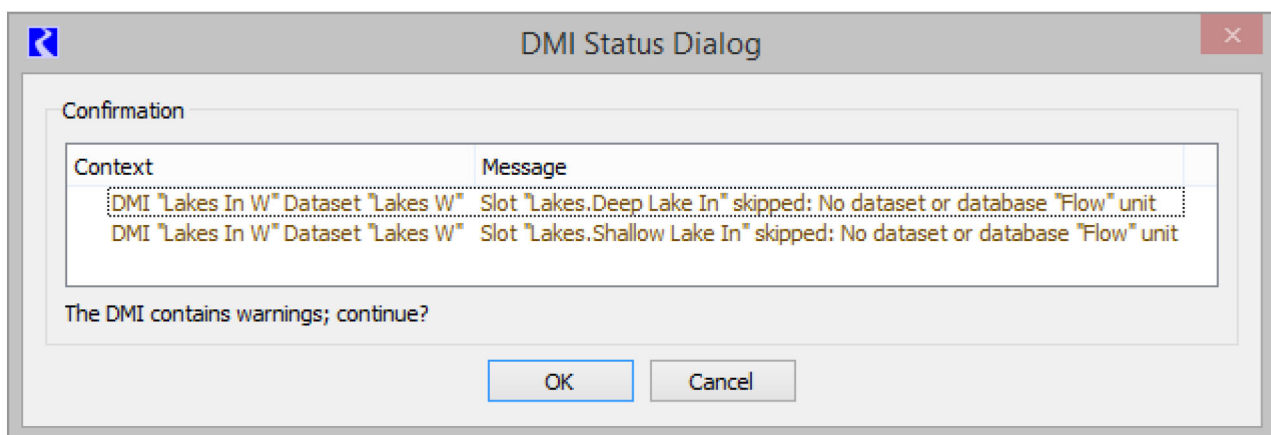
Warning and Error Messages

When a DMI is invoked, each warning is displayed in the Diagnostic Output.



If the DMI is configured to Confirm Warnings (see “[Name and Type](#)” on page 71 and “[Confirm Warnings Checkbox](#)” on page 54) or an error occurs, a separate DMI Status Dialog will open. [Figure 1.16](#) shows the warning or error messages.

Figure 1.16



If the messages are just warnings and you wish to continue execution of the DMI (**Confirm Warnings** is selected), select **OK**. Otherwise, select cancel to stop execution.

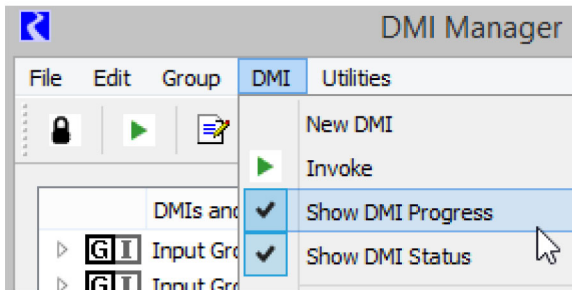
Any errors generated by the DMI are displayed in the DMI Status Dialog and in the Diagnostic Output in red text. After an error, the DMI stops executing.

Note: There are diagnostics available for DMIs. See “[Workspace Diagnostics](#)” in *Debugging and Analysis* for details.

DMI Status and Progress

The status of an executing DMI is shown in the bottom of the DMI manager. You can also configure to show a separate DMI Status Dialog that has optional panels for the Progress, Status, and Confirmation. This shows information on the DMI as it executes including the slots that it is processing and the location of the data being processed. To show this status dialog, from the DMI Manager, select **DMI**, then **Show DMI Status** or **DMI**, then **Show DMI Progress**. The selections are saved with the model file. In [Figure 1.17](#), both **Show DMI Progress** and **Show DMI Status** are selected. In addition, there are Confirmation messages that must be confirmed before this DMI can continue.

Figure 1.17



In Figure 1.18, the Progress panel shows that the DMI is executing including the Group (if there is one), the DMI, and the slots being processed.

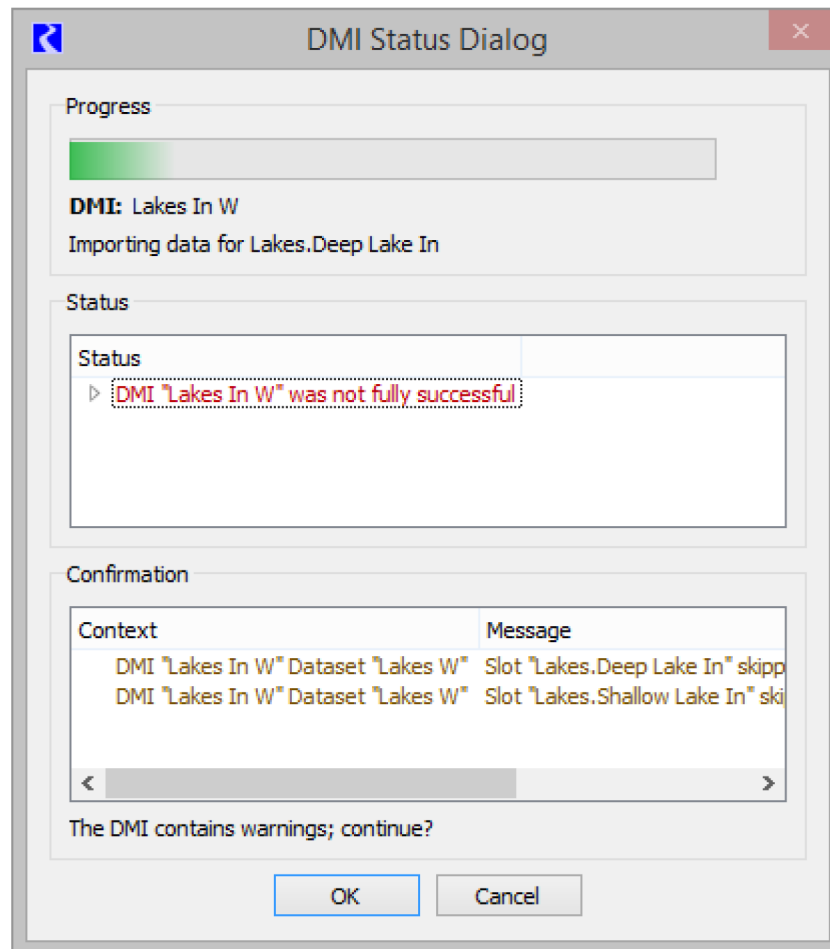
Figure 1.18

Progress
Panel

Status
Panel

Confirmation
Messages

Continue?



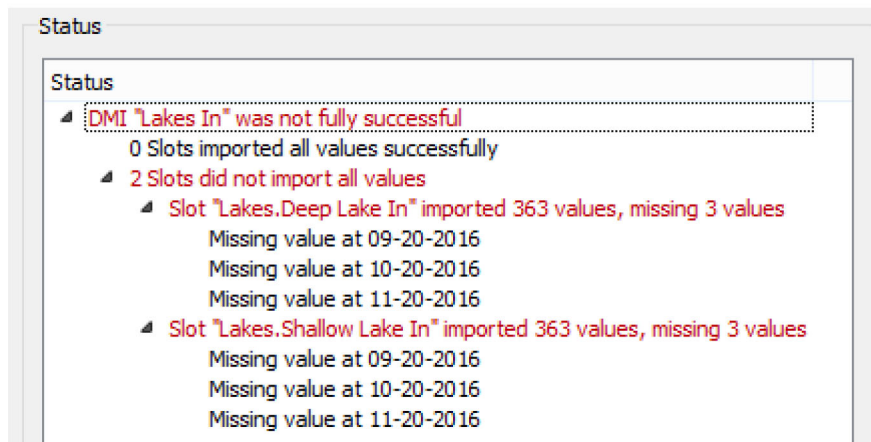
The Status Panel shows the results (for an Input DMI only). For example, in Figure 1.19, the treeview has been expanded to show all of the information. This indicates that 363 values were imported for each slot, but three

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values in each slot were missing. The red messages are also copied to the Diagnostics Output Window so you can see the messages after closing the DMI Status dialog.

Figure 1.19



Icon Quick Reference

Table 1.2 DMI Manager

Icon	Meaning
	Invoke the selected top-level DMI or DMI group
	Edit the selected top-level DMI
	Delete the selected top-level DMI or DMI group
	Copy the selected top-level DMI or DMI group
	Paste the previously selected top-level DMI or DMI group, creating a new top-level DMI or DMI group
	Add the selected top-level DMI to a DMI group
	Remove the selected member DMI from its DMI group
	Display the selected DMI's group membership
	Move the selected member DMI up within its DMI group
	Move the selected member DMI down within its DMI group

Table 1.2 DMI Manager

Icon	Meaning
	Select to unlock the DMI Manager
	Select to lock the DMI Manager

Table 1.3 DMI Parameter

Icon	Meaning
	Append a row to the list
	Insert a row into the list, before the selected row
	Delete the selected row
	Undelete the selected (and previously deleted) row
	Move the selected row up one position
	Move the selected row down one position
	Import parameters from a file
	Export parameters to a file
	Imported parameters which replace an existing parameter are marked with this icon
	Imported parameters which are new are marked with this icon

Table 1.4 Database DMI

Icon	Meaning
	Add a new Name Map
	Edit the selected Name Map or Dataset
	Copy the selected Name Map or Dataset
	Paste a copied Name Map or Dataset
	Delete or remove the selected Name Map or Dataset
	Choose a file using a file selector

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Table 1.4 Database DMI

Icon	Meaning
	The name map, dataset or DMI has changes that have not been applied
	Indicates the name map, dataset, or DMI is locked by another window and cannot be edited.
	Indicates the name map, dataset, or DMI is scheduled for deletion and locked by another window and cannot be edited.

Chapter 2

Control File-Executable DMI

The Data Management Interface (DMI) provides an efficient means of transferring large amounts of data between RiverWare and an external data source or sink. There are two directions in which a DMI can send data:

- An Input DMI, which transfers data from an external data source to RiverWare.
- An Output DMI, which transfers data from RiverWare to an external data sink.

Topics

- [About the Control File-Executable DMI, page 37](#)
- [Control File, page 38](#)
- [Data Files, page 47](#)
- [DMI Executable, page 51](#)
- [DMI Editor Dialog, page 53](#)

About the Control File-Executable DMI

The Control File-Executable DMI consists of several components. Refer to [Figure 2.1](#) for the relation between these components.

- The user-specified Control File is created by the user and contains a list of slots which the user wants imported or exported from the RiverWare model. The control file is an ASCII file which maps data between object.slots in the Workspace and data files
- The external DMI Executable is a separate software component which is called by the RiverWare DMI facility. For import DMIs, this external executable provides the data files which RiverWare loads into the model. For output DMIs, the external executable processes data files exported by RiverWare and performs such functions as saving the data in the database, reprocessing the data into reports, and preparing the data for use by statistical analysis programs.
- The RiverWare DMI Data Files are a set of files used to transfer the data to and from the RiverWare model. Each of these files contains the data for a single slot in the model. The files are created by either the External DMI Executable or RiverWare, depending on whether it is an input or an output DMI.

As shown in [Figure 2.1](#), the user must minimally create a control file and an executable. Following is an example of how the DMI works. In this example we assume it is an Input DMI, but it is the same, just reversed for an Output DMI.

1. The user Invokes the Input DMI. RiverWare verifies the DMI to make sure it is valid.

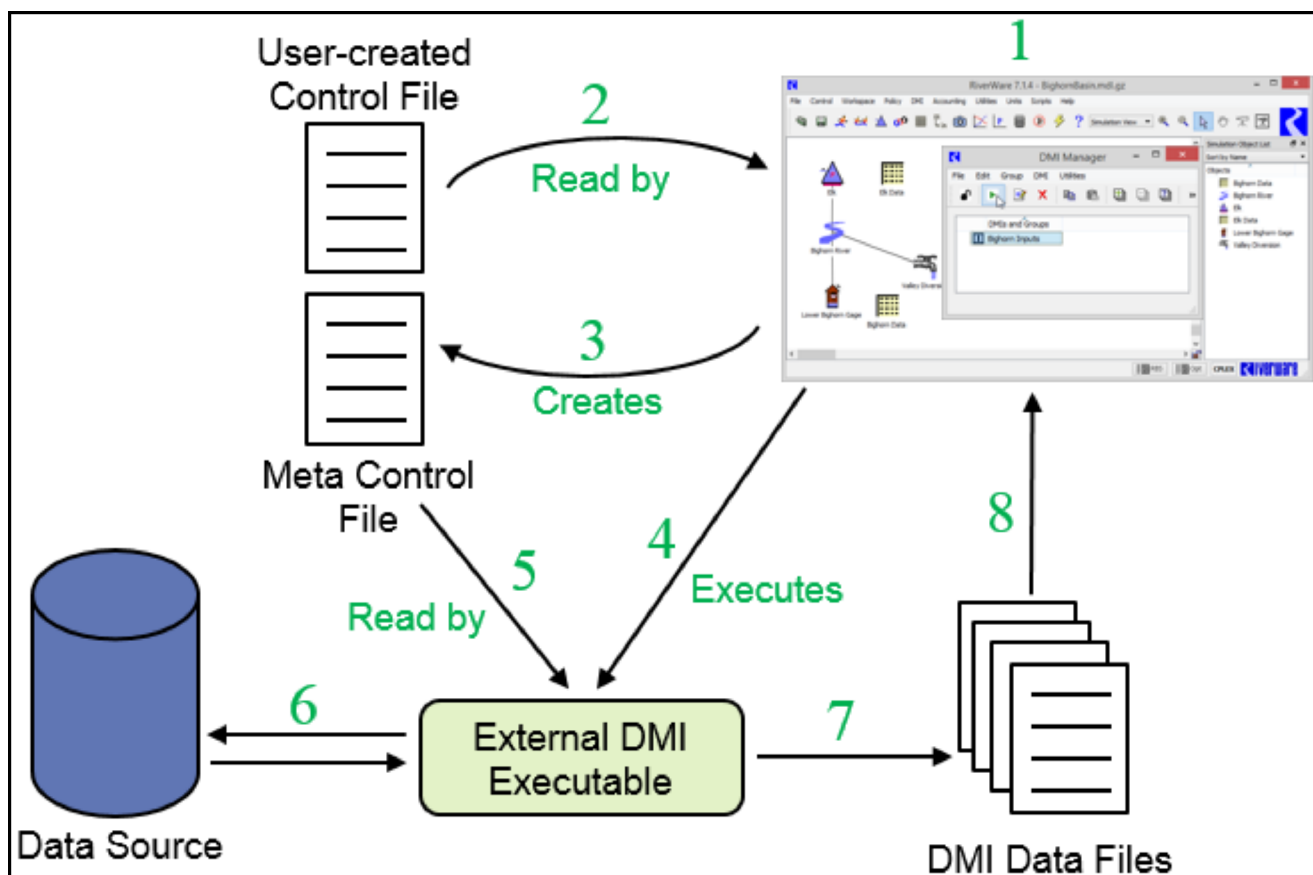
Chapter 2

Control File-Executable DMI

2. RiverWare reads the user Control File and resolves all wild carding.
3. RiverWare writes the Meta Control File which has a fully specified line for each slot in the Control File.
4. RiverWare runs the External DMI Executable.
5. The executable reads the Meta Control File.
6. The executable interacts with the Data Source (i.e. the database) to retrieve the data.
7. The executable creates the DMI Data Files. .
8. Once the executable is complete, RiverWare reads the DMI Data Files and places data in the slots.

Note: Alternatively, the user can create the DMI Data Files separately without using an executable. In this case steps 4-7 will be skipped

Figure 2.1 Control File - Executable Input DMI schematic



Control File

The control file is an ASCII file which the user provides. The purpose of the control file is as follows:

- Map between object.slots in the Workspace and data files. In the case of an import DMI, data is imported from the data files to the object.slots; in the case of an export DMI, data is exported from the object.slots to the data files.
- Provide information about the data files.

Each control file entry contains three pieces of information:

- An object.slot specification.
- One or more file names listed as file = name
- Optional keyword = value pairs.

The control file format is:

```
object.slot: file=name keyword=value keyword=value ...
```

If a file name contains spaces then it must be enclosed in double quotations marks (“”), for example:

```
object.slot: file="file name" keyword=value keyword=value ...
```

The DMI must also be configured to allow spaces in file paths; see [“Allow Spaces in File Paths”](#) on page 55 for details.

Object.Slot Specifications

The object.slot specifications indicate which slots will import from or export to data files. There are four ways of specifying the object:

- <object name> - A specific object name, e.g., “FtLoudoun,” which matches the object with that name. Aggregate object elements are specified as <aggregate name>:<element name>.
- <object type> - An object type, e.g., “LevelPowerReservoir,” which matches all objects of that type.
- <sub-basin name> - A sub-basin name, which matches all objects in the sub-basin.
- * - A wild card which matches all objects of all types.

There are three ways of specifying the slot:

- <slot name> - A specific slot name, e.g., “Inflow,” which matches the slot with that name.
- <slot type> - A slot type, e.g., “SeriesSlot,” which matches all slots of that type.
- dispatch - Any dispatching slots.

There are two ways of specifying a column within multi-column slots:

- <column name> - A specific column name, which matches one of the columns in the selected slot.
- no name - If no name is given, the DMI defaults to the first column.

An account or supply can be specified like so:

- <object name>^<account name>
- <supply name>.Supply - the “.Supply” portion of the string is necessary to tell RiverWare that this is a supply in the accounting system and not a slot in the physical system. If a DMI is used to bring in values, it should be

Chapter 2

Control File-Executable DMI

setting supplies in the accounting system rather than accounting slots. See [“Setting Slots Versus Setting Supplies” in *Accounting*](#) for details.

Also, the slots in a Slot Set can be specified using the SlotSet keyword, as follows. See [“Slot Sets” in *User Interface*](#) for details.

SlotSet.<slot set name>

For example, SlotSet.Reservoir Inflows would expand to all of the slots in the Reservoir Inflows set.

The above slot specifications lead to many ways of specifying the object.slot pairs, as follows:

```
<object name>.<slot name>.<column name>
<object name>.<slot name>
<object name>.<slot type>.<column name>
<object name>.<slot type>
<object name>.dispatch.<column name>
<object name>.dispatch
<object type>.<slot name>.<column name>
<object type>.<slot name>
<object type>.<slot type>.<column name>
<object type>.<slot type>
<object type>.dispatch.<column name>
<object type>.dispatch
<sub-basin name>.<slot name>.<column name>
<sub-basin name>.<slot name>
<sub-basin name>.<slot type>.<column name>
<sub-basin name>.<slot type>
<sub-basin name>.dispatch.<column name>
<sub-basin name>.dispatch
*.<slot name>.<column name>
*.<slot name>
*.<slot type>.<column name>
*.<slot type>
*.dispatch.<column name>
*.dispatch
<object name>^<account name>
<supply name>.Supply
SlotSet.<Slot Set Name>
```

These pairs are listed from most exact to least exact. It is possible that more than one control file entry will select a specific object.slot pair in the Workspace. In this case, the more exact entry will always take precedence. For example, in the control file:

```
# All Confluence objects, Inflow1 slot.
Confluence.Inflow1: file=~/%o.inflow1.dat
# Confluence01 object, Inflow1 slot.
```

Confluence01.Inflow1: file=~/.confluence01.inflow1.dat

Both entries will select Confluence01's Inflow1 slot (the first through an object type, the second through an object name). The more exact entry, the second entry, will take precedence.

There are two other concepts this example introduces: the control file may contain comments (lines beginning with #) and the file name may contain special characters (~ and %), described below.

Finally, note that object and slot names and types must be specified exactly as they appear in the user interface, including case and whitespace.

File Name and Directory Specification

The file name indicates from which data files the object.slots will be imported or to which data files the objects.slots will be exported. The file name may be either relative (beginning with ~/) or absolute (beginning with C:/). If the file name is relative, then it is relative to the DMI's working directory. The DMI's working directory is where it creates its temporary files. If the RIVERWARE_DMI_DIR environment variable is set, the DMI's working directory is \$RIVERWARE_DMI_DIR/dmi_name; otherwise, the DMI's working directory is /tmp/dmi_name. In both cases, dmi_name is the name that appears in the DMI Selector list.

The file name may contain the following directives (currently, it may contain at most one of each):

- %o - The %o is replaced with the object name. These are useful when using symbolic references to object in the file specifications.
- %s - The %s is replaced with the slot name. These are useful when using symbolic references to slots in the file specifications.
- %t - The %t is replaced with the current MRM trace number. If there is no trace number because the DMI is not executed as part of MRM, a trace of 1 is assumed.
- %tempdir - The %tempdir is replaced by a guaranteed unique temporary directory as described below.

Note: Because dmi names, object names and slot names are mapped to directory and file names, characters which are not valid in directory and file names (currently all characters other than A-Za-z0-9/_.) are replaced with _ when the directory or file name is created.

Consider the following example:

RIVERWARE_DMI_DIR is C:\temp\local\riverware

The DMI name is "test dmi"

The control file contains the following entry:

Canal.Flow 1: file=~\%o%t.%s.dat

The file which the "Tellico Canal.Flow 1" object.slot for trace 23 or an MRM would import from or export to would be:

C:\temp\local\riverware\test_dmi\Tellico_Canal23.Flow_1.dat

The ~ has been replaced with the value of the working directory, RIVERWARE_DMI_DIR<dmi name>, %o has been replaced with the object name, %t has been replaced with the trace number, and %s has been replaced with the slot name (with the spaces having been replaced with _).

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Multiple files can be specified on a single control file line entry for input dmi's. This enables the user to import data into a slot from two or more files. Use of the `data_date` file keyword allows the user to specify where in the series slot each file should be placed. The syntax for multiple files is

```
object.slot: file=name file=name
```

The data files are imported from left to right. If the `import = resize` is specified, the slot will grow to accommodate the data. Overlapping data is allowed but values will be overwritten by the later data files. If the incoming data does not overlap and there are gaps, any untouched values that existed on the slot will remain.

Another control file directive available is `%tempdir`, which is replaced by a guaranteed unique temporary directory. When the DMI is invoked RiverWare generates the directory name, makes the directory and passes the directory to the DMI executable via the `RW_DMI_TEMPDIR` environment variable. The executable should be able to recognize `RW_DMI_TEMPDIR`. After the DMI finishes RiverWare removes the directory. The directory name will be similar to the following:

```
b05741ad-2b57-4f82-8044-bda67606e4c9
```

Because RiverWare removes the directory you most likely won't see these directories, but if you kill RiverWare or RiverWare crashes a directory might be left behind, in which case you can safely remove it.

Time Range Specification

The time range of data that is imported or exported by a Control File-Executable DMI is controlled in the control files and possibly in the data file. If not specified, the default is to export/import the range of the slot. But if you want to limit the data that is imported or exported use one or more of the following options:

- For an output DMI, you can limit the output to only the run range using the “Limit Output to Run Range” check box described in [“Limit Output to Run Range”](#) on page 55. See also the note below.
- For an input DMI, specify the timestep of the data that should be imported in the data file using the `data_date` keyword. This specification is in the data file itself so it is on a per-slot basis. See the description of `data_date` in [“Data Files”](#) on page 47.
- For either an input or output DMI, specify the `start_timestep` and `end_timestep` in the control file for each slot specification. Both of these keywords allow you to also specify the timestep symbolically using RPL syntax. For example, the following are valid entries:
 - `start_timestep=“24:00 November Max DayOfMonth, 1996”`
 - `end_timestep=“Start Month Start DayOfMonth, Start Year - 2 days”`
 - `start_timestep=“Current Timestep”` (Current timestep is only supported for rule executed DMIs as they know at which timestep the DMI is executing, see [“Executing DMIs From Blocks”](#) in [RiverWare Policy Language \(RPL\)](#)).
 - `end_timestep=“24:00 3/1/2021”`

See [“DATETIME”](#) in [RiverWare Policy Language \(RPL\)](#) for details on datetimes. Basically, any fully specified datetime can be used. No `@` symbol is necessary but quotes are necessary when specifying datetimes that contain spaces.

When using these keywords on an input DMI, the data file does not have to have the same `start_date` and `end_date` as the control file. The data file should encompass the desired data. Otherwise a warning message

will be issued, and no data will be imported. If the data file does not have a `start_date` keyword, the data will be imported starting with the first value in the data file imported to the `start_timestep`. Also, using the `data_date` keyword in the data file is possible. It is recommended to create data files that match the control file's `start_timestep` and `end_timestep` for improved performance.

For an input DMI, the keyword `end_timestep` will not grow a slot. Use the `import=resize` to grow a slot to fit the imported data. (See “`import=<fixed>|<resize>`” on page 44.)

Note: The time range of the slot after import will be based on the `start_date` and `end_date` keywords in the *data file* consistent with the description for “`import=<fixed>|<resize>`” on page 44. Data will be imported only for the range specified by the `start_timestep` and `end_timestep` in the control file, and any additional timesteps will contain NaN.

On an output DMI, if you specify `start_timestep` and `end_timestep` (a symbolic range) in the control file and **Limit Output to Run Range** in the DMI interface, the following applies: If the run range and symbolic range are disjoint, constrain the data transfer to the symbolic range. Otherwise, constrain the data transfer to the *intersection* of the run range and symbolic range.

Optional keyword=value Pairs

The optional keyword=value pairs provide additional information about the data files the slots will import. (The one exception to this is the file keyword, which identifies the data file itself.) Although the keywords are not case sensitive, the values can be.

The currently supported keywords are:

- **file=<data file>**

The data file the slot will import from or export to, as described above in “[File Name and Directory Specification](#)” on page 41.

- **start_timestep=<datetime>**

An optional datetime that indicates the timestep at which to start importing/exporting, as described above in “[Time Range Specification](#)” on page 42. If not specified for an input DMI and the data file has a `start_date` keyword, the data are imported beginning with the data file's `start_date`. If not specified for an input DMI and the data file does not have a `start_date`, the slot's start timestep is used. If not specified for an output DMI, the slot's start timestep is used. The datetime can be specified symbolically using RPL syntax.

- **end_timestep=<datetime>**

An optional datetime that indicates the timestep at which to end the importing/exporting, as described above in “[Time Range Specification](#)” on page 42. If not specified or an input DMI, the end of the import range is based on the combination of the `end_date` keyword in the data file and the `import=<fixed>|<resize>` keyword in the control file as described in “`import=<fixed>|<resize>`” on page 44. If not specified for an output DMI, the end timestep of the slot is used. The datetime can be specified symbolically using RPL syntax.

- **units=<units>**

The units of the data the slot will import or export. <units> must match exactly one of the units defined in the Units List dialog (see “[Display Units](#)” in *User Interface* for details on the Units List).

Note: If **units** are not specified in the control file, then it defaults to the current setting in the user interface as specified by the active Unit Scheme; (see “[Unit Schemes](#)” in *User Interface* for details). Since the unit scheme can change from invocation to invocation, it is *strongly recommended* that the control file always contain units.

- **scale=<scale>**

The scale of the data the slot will import or export.

Note: If **scale** is not specified in the control file, then it defaults to the current setting in the user interface as specified by the active Unit Scheme; (see “[Unit Schemes](#)” in *User Interface* for details). Since the unit scheme can change from invocation to invocation, it is *strongly recommended* that the control file always contain the scale (even if the scale is 1.0).

- **precision=<display>|<model>**

The precision of the data the slot will import or export. The <display> precision is that specified in the Unit Scheme for the slot. The <model> precision is 17, the same precision as all internal calculations. The default is <display>.

- **import=<fixed>|<resize>**

This keyword controls how the DMI sets the time series range of the series slot. The default is fixed.

For either import option, the Begin Timestep of the slot is set in the same manner. If the data file contains a **start_date** header, the slot Begin Timestep is set to the **start_date** from the data file. If the data file does not contain a **start_date** header, the slot maintains its existing Begin Timestep. The data import always starts on the new Begin Timestep of the slot with the first value in the data file.

For **import=resize**, if the data file contains an **end_date** header, the End Timestep of the time series range of the series slot will be set to the *later* of the **end_date** header in the data file or the number of timesteps necessary hold the number of data values in the data file given the updated Begin Timestep of the slot from above. If the data file does not contain an **end_date** header, the End Timestep of the time series range of the slot will be extended, if necessary, to hold the number of data values in the data file given the updated Begin Timestep of the slot from above. The End Timestep of the slot will never be moved earlier if there is no **end_date** header in the data file.

For **import=fixed**, if the data file contains an **end_date** header, the End Timestep of the series slot is set to the **end_date** from the data file. Data will not be imported past the new End Timestep of the slot. If the data file does not contain an **end_date** header and the existing End Timestep on the slot is *later* than the updated Begin Timestep, the slot maintains its existing End Timestep. Data will not be imported past the End Timestep of the slot. If the existing End Timestep on the slot is equal to or earlier than the updated Begin Timestep, then the slot will be expanded to hold all of the data values in the data file starting with the updated Begin Timestep.

Note: If the control file uses the `start_timestep` and `end_timestep` keywords, the time range for the slot after import will still be as described here, but data will only be imported for the time range corresponding to the `start_timestep` and `end_timestep` in the control file. Any additional timesteps will contain NaN. (See “[Time Range Specification](#)” on page 42.)

- **flags=<true>|<false>**

The flags keyword specifies whether or not Series Slot flag values (Input, Target, Best Efficiency, Max Capacity, Drift, and Unit Values) will be exported or imported with the data. The syntax is `flags=true`; the default is `<false>`. If an input DMI, the keyword pairs in the data file should be specified as “value flag” with the flag being an integer value, e.g. Input = 1, Output = 2, etc.

- **aggregate=<true>|<false>**

The aggregate keyword is used to specify whether or not the DMI should import/export all time series slots on an Aggregate Series slot or just the first time series column. If present, with the value “true”, and if the slot indicated is an Aggregate Series slot, the data file is expected to (on import) or will (on export) contain data for all subslots (columns) in the Aggregate Series slot. If the keyword=value pair is absent, or if its value is “false”, only the data for the first subslot (column) of the Aggregate Series slot are present. [Table 2.1](#) shows the resulting metacontrol file entry for the range of possible control file entries and , the aggregate keyword values

Table 2.1 Possible entries when using the aggregate keyword.

Control File entry	aggregate keyword	Metacontrol result	Comment
Obj.Slot	false	Obj.Slot.ColumnLabel	Only the first column of the agg is import/exported
Obj.Slot	true	Obj.Slot	All columns are imported/exported
Obj.Slot.ColumnLabel	false	Obj.Slot.ColumnLabel	Only the specified column is imported/exported
Obj.Slot.ColumnLabel	true	Obj.Slot	A warning is posted that the keyword is inconsistent and will be ignored.

When Aggregate Series slot data are imported or exported in total, the data are in row-major format, one time-step per line. The subslots of the aggregation must be homogeneous in start-date, end-date and step size, and all subslots (columns) must be represented on each line.

If used with the ‘flags=true’ keyword-value pair, the data will appear in the form:

value flag value flag value flag...

The flag is an integer number representing the flag. If the named slot is in fact the first subslot (column) of an Aggregate Series slot, the behavior is the same as if the subslot (column) name were absent.

- **slot_anno = <true>|<false>**

The slot_anno keyword indicates whether to import or export series slot notes; see “[Exporting and Importing Notes Using DMIs](#)” in *User Interface* for details. The default value for slot_anno is false, meaning no notes will be written.

When slot_anno is set to true for an output DMI, each timestep that has a note will have the following content in the data file to the right of the slot value:

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```
anno {<Note Group>} {<Note Text>} {<Timestamp>} {<Username>}
```

For an input DMI, a note in the data file should have the same format. The timestamp corresponds to the time at which the note instance was applied and has the format MM-DD-YYYY HH:MM:SS. For an input DMI, if no timestamp is included in the data file, the note instance will be given a timestamp corresponding to the time of the data import, otherwise it will apply the timestamp from the data file. The username corresponds to the user who applied the note instance.

Note: The `slot_anno` keyword cannot be used in combination with the `slot_set_info = <true>|<false>` or `slot_opt_sol_info = <true>|<false>` keywords.

• `slot_set_info = <true>|<false>`

The `slot_set_info` keyword indicates whether to export series slot value information. The default value for `slot_set_info` is false, meaning no information will be written. When true, the information corresponds to the tool tips seen on slot values on slot and SCT dialogs () will be written to the file; see “[Series Slots](#)” in *User Interface* for details. The string appears after the value. Examples of the information are “Set by DMI: Input DMI”, “Set by Rule: (7) Flood Control”.

Note: The `slot_set_info` keyword cannot be used in combination with the `slot_anno = <true>|<false>` or `slot_opt_sol_info = <true>|<false>` keywords.

• `slot_opt_sol_info = <true>|<false>`

The `slot_opt_sol_info` keyword indicates whether to export series slot optimization solution information. The default value for `slot_opt_sol_info` is false, meaning no information will be written. When true, the available per slot/value optimization solution information will be written to the file. The string appears after each value. Examples of the information are “Frozen at Lower Bound”, or “Frozen by (3) Minimum Load 47.6% between limits set by 3.1.1.1 and 2.1.1.1.”. See “[Tooltips on Variables](#)” in *Optimization* for details.

Note: The `slot_opt_sol_info` keyword cannot be used in combination with the `slot_anno = <true>|<false>` or `slot_set_info = <true>|<false>` keywords.

User keyword=value Pairs

The user keyword=value pairs provide an additional means by which the user can pass parameters to the DMI executable. The user keyword=value pairs are specified in the control file as:

```
!keyword=value
```

The keyword=value pairs are written to the metaControl file, with the ‘!’ removed, on all lines they apply to. The DMI executable can then use the keyword=value pairs to influence its behavior.

Example 2.1

```
#
# All LevelPowerReservoir Outflows
#
LevelPowerReservoir.Outflow: file=~/%o.Outflow units=cfs scale=1.0
#
```



```
# All LevelPowerReservoir Pool Elevations,  
# but for Hiwassee use a different scale.  
#  
LevelPowerReservoir.Pool Elevation: file=~/%o.%s units=ft scale=1.0  
Hiwassee.Pool Elevation: file=~/%o.%s units=ft scale=10.0  
#  
# All LevelPowerReservoir Storages.  
#  
LevelPowerReservoir.Storage: file=~/%o.%s units=cfs-day scale=1.0  
#  
# All LevelPowerReservoir Hydrologic Inflows.  
#  
LevelPowerReservoir.Hydrologic Inflow: file=~/%o.%s units=cfs scale=1.0  
#  
# Finally, all SeriesSlots which are not selected by one of  
# the preceding lines. (Remember that *.<slot type> is the  
# least exact specification.) The file ~/start_end is a data  
# file which contains only the start_date: and end_date:  
# attributes. This will serve to change the start date and  
# end date for all series slots which import this file.  
#  
*.SeriesSlot: file=~/start_end
```

Data Files

The data files are the files from which the slots will be imported or to which the slots will be exported. They contain both the data attributes and data values. The attributes are specified as keyword:value pairs, and all attributes must come before the first data value. Although the keywords are not case sensitive, the value in the keyword:value pair can be.

The currently supported keywords are:

- The start date of a series slot:

start_date: yyyy-mm-dd hh:mm

start_date: 1996-04-15 06:00

On import, the slot will be expanded to include this timestep. On export, the start_date indicates the first time-step at which data was first.

- The end date of a series slot:

end_date: yyyy-mm-dd hh:mm

end_date: 1996-04-22 06:00

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On import, this keyword is not used. Resizing a slot is configured using the import keyword in the control file. On export, the end_date indicates the last timestep at which data was exported.

- The start date of where the data should be imported.

data_date: yyyy-mm-dd hh:mm

data_date: 1996-10-22 06:00

data_date specifies where in the series slot the data is to be imported. Implicitly, data_date is only useful with input DMIs. It was designed for use with the multiple data file feature in control files: Object.Slot: file=... file=... in which data files are imported from left to right. If all data files were imported at the beginning of the series slot then values from later data files would overwrite values from earlier data files. Therefore, all data files (except possibly the first) should contain the data_date keyword, to import their values at the correct location in the series slot and to prevent their values from overwriting values from earlier data files.

- The timestep of the data being exported from a series slot:

timestep: <count> <timestep>

<count> is an integral value.

<timestep> is one of HOUR, DAY WEEK, MONTH, YEAR.

timestep: 6 HOUR

- The units of the data in the data file.

units: units

units: acre-feet/month

For multi-column table slots:

units: <column 0 units> <column 1 units> ...

units: ft cfs

- The scale of the data in the data file.

scale: scale

scale: 100

For multi-column table slots:

scale: <column 0 scale> <column 1 scale> ...

scale: 1 1000

- The new display units for the slot importing the data. Also, if the units keyword is not specified, then the units of the data being imported is the new_units:

set_units: new_units

set_units: cms

- The new scale or the slot importing the data. Also, if the scale keyword is not specified, the scale of the data being imported is the new_scale.

set_scale: new_scale

set_scale: 10

- The new Lower Bound for the slot. In the units and scale specified on the slot.

set_min: new_min

set_min: 10

- The new Upper Bound for the slot. In the units and scale specified on the slot.

set_max: new_max

set_max: 1000

- The new Optimization Min Value for the table slot. In the units and scale specified on the slot.

opt_min: <column 0 min> <column 1 min> ...

opt_min: 10 10

- The new Optimization Max Value for the table slot. In the units and scale specified on the slot.

opt_max: <column 0 max> <column 1 max> ...

opt_max: 10000 100000

- The new display format of the slot importing the data.

display_format: <new_format>

<new_format> is one of float, integer, scientific.

display_format: scientific

- The new display precision of the slot importing the data.

display_precision: new_precision

display_precision: 4

- The new period of the periodic slot importing the data.

periodic_period: <count> <timestep>

<count> <timestep> is one of 6 HOUR, 12 HOUR, 1 DAY, 1 MONTH, N YEAR where N is 1 to 99.

periodic_period: 2 YEAR

- The new base year for the multi-year periodic slot importing the data.

periodic_base: new_year

periodic_base: 1998

- The new interval type for the periodic slot importing the data.

periodic_interval: <new_interval_type>

<new_interval_type> is one of regular, irregular.

periodic_interval: irregular

- The new interval size for the regular periodic slot importing the data.

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periodic_regular_size: <count> <timestep>

<count> <timestep> is one of 1 HOUR, 6 HOUR, 12 HOUR, 1 DAY, 1 MONTH.

periodic_regular_size: 1 MONTH

- The new data interpolation method for the periodic slot importing the data.

periodic_data_interp: <new_method>

<new_method> is one of lookup, interpolate.

periodic_data_interp: lookup

- The new header units for a periodic slot with numeric column headers that is importing data.

column_units: new_units

column_units: cfs

- The new header scale for a periodic slot with numeric column headers that is importing data.

column_scale: new_scale

column_scale: 10

- New symbolic row dates for the irregular periodic slot that is importing the data.

periodic_rows: <new_row_dates>

<new_row_dates> is a space separated list of <symbolic_date> entries.

<symbolic_date> takes the form of minute/hour/day/month/multi-year index, where each piece is represented by an index number (minute (0), hour (0-23), day (1-31), month (1-12), year (1-99)). Symbolic dates need only be complete to the longest appropriate interval for the period of the particular slot. For example, 0/3 is 3:00, 0/3/21 is 3:00 Day 21, 0/3/21/4 is 3:00 April 21, and 0/3/21/4/2 is 3:00 April 21, Year 2.

periodic_rows: 0/0/16/2 0/0/2/3 0/0/16/6 0/0/2/10 0/0/16/12

- New column header numbers for a periodic slot with numeric column headers that is importing data.

periodic_columns: <new_column_entries>

<new_column_entries> is a space separated list of numeric values.

periodic_columns: 1000 4000 10000 40000 100000

The data values are specified either one per line (for series, list, and scalar slots) or one row per line (for table and periodic slots). For table slots, the row may include a new row label before the data values. If the label contains spaces, it must be enclosed in double quotes. As with the control file, the data files may contain comments (lines beginning with #).

Series slot notes (see [“Notes on Series Slots” in User Interface](#)) can also be represented in data files for import or export. They are specified as:

<value> anno {NoteGroup} {Note}

where <value> will be a numeric value, “anno” is a keyword, NoteGroup is the name of the NoteGroup and Note is the actual note. The “<” and “>” will not exist in the file, while the “{” and “}” do exist, delimiting the Note Group name from the note text itself. For Example, a line in the file might be:

102.031 anno {Storm Events} {4.3 Inches}

where Storm Events defines the Note Group to which the note belongs and 4.3 Inches is the note itself.

DMI Executable

The DMI executable is invoked by RiverWare, which then waits for it to exit. For an import DMI, the executable creates data files from an external data source. For an export DMI, the executable populates an external data sink from data files.

Nearly any programming language can be used to write the DMI executable. Often the choice of tools will be based on user experience and existing tools that can communicate with the database.

Note: A Windows batch (.bat) script could work as a DMI executable. It does need to include an exit statement: `exit /b 0` The convention for a DMI executable is a zero exit status indicates success, a non-zero exit status indicates an error. Without the exit statement the bat script would return the "DMI executable failed: Exited with 1." error.

The DMI executable is passed the following command line arguments:

Meta-control File

The path to the meta-control file. The meta-control file is a copy of the control file described above, with wildcards resolved. Each line is of the form:

`object.slot: file=<file name> [optional keyword=value pairs]`

The DMI executable should use the object.slot specification to determine for which objects and slots it must provide data (import DMI), or for which objects and slots data is being provided (export DMI). The executable should use the file specification to determine to where to write the data to be read by RiverWare (import DMI), or from where to read the data exported by RiverWare (export DMI).

If the start_timestep and/or end_timestep are supplied in the control file, the meta-control file will contain start_date, start_time, end_date and end_time information.

Working Directory

The working directory is the DMI's working directory, as described above. It is an artifact of an earlier implementation and can safely be ignored by the DMI executable.

Start Data (YYYY-MM-DD)

The start date of the data the executable must provide (import DMI), or of the data which is being provided (export DMI).

Start Time (HH:MM)

The start time of the data the executable must provide (import DMI), or of the data which is being provided (export DMI).

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End Data (YYYY-MM-DD)

The end date of the data the executable must provide (import DMI), or of the data which is being provided (export DMI).

End Time (HH:MM)

The end time of the data the executable must provide (import DMI), or of the data which is being provided (export DMI).

Timestep (6HOUR)

The timestep of the data the executable must provide (import DMI), or of the data which is being provided (export DMI).

System Parameters

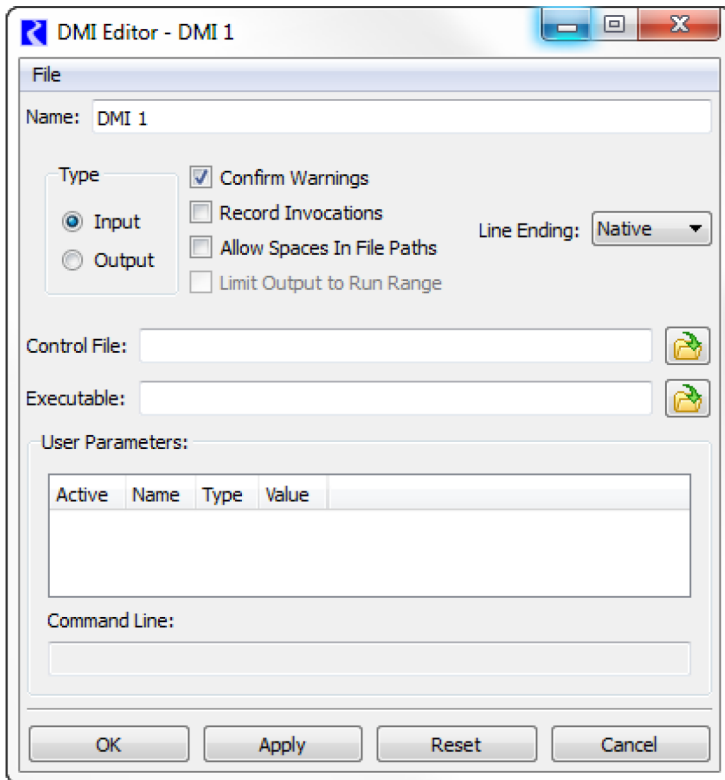
Not implemented.

User Parameters

The user parameters, in the form: -Ukeyword=value The intention of the user parameters is described above.

DMI Editor Dialog

The DMI Editor dialog enables the user to edit the DMI name, type, control file path, executable file path and other configuration options.



Name

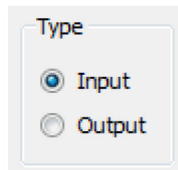
To edit the DMI name, the user selects the **Name** field to edit the name in place:

When the edits are applied the new DMI name is validated; the possible errors are:

- A blank name
- A duplicate name (the same name as another DMI)

Type

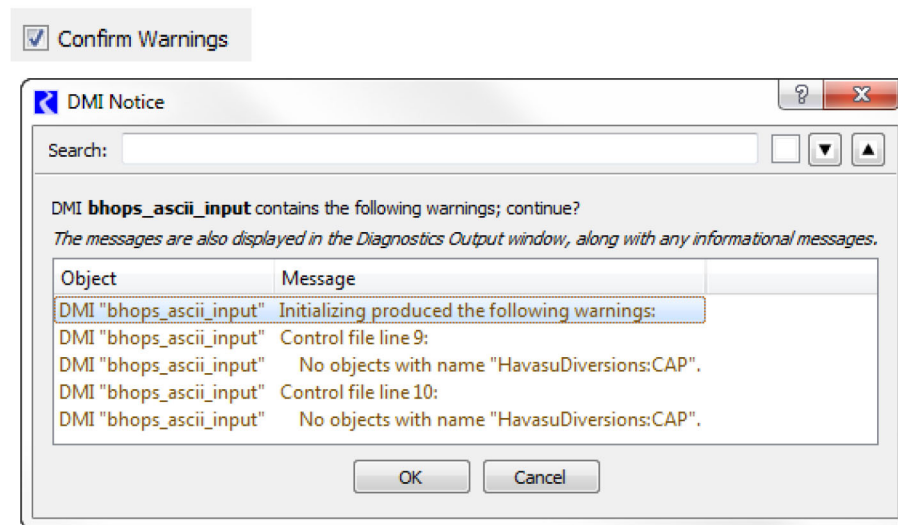
To edit the DMI type, the user selects the appropriate Type.



Confirm Warnings Checkbox

The **Confirm Warnings** check box controls whether you wish for the DMI to continue when warnings are generated. If **Confirm Warnings** is selected, the user is prompted to confirm any warning messages that appear; if not checked, the DMI continues after any warnings. [Figure 2.2](#) is a sample dialog that appears during execution when the **Confirm Warnings** check box is selected.

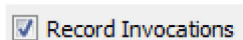
Figure 2.2



Record Invocations

The **Record Invocations** check box allows RiverWare to maintain information which chronicles input DMI invocations. With this information, values set by DMIs can be cleared on a per-invocation basis.

If a user wishes to clear all values that are input by a DMI, this box should be checked. If a user knows values set by a particular DMI won't be cleared then it is not necessary to maintain information about the DMIs invocations and this box should remain unchecked. (The information occupies memory and could, over time, degrade performance.)



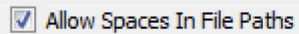
The check box is only enabled for input DMIs and the default is unchecked. See [“DMI Invocation Manager”](#) on page 24 for details on clearing values set by an input DMI.

Note: When the **Record Invocations** box is checked, series values imported by the input DMI are given the “Z” flag indicating they were set by a DMI.

Allow Spaces in File Paths

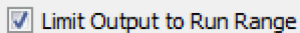
The **Allow Spaces in File Paths** allows the user to specify paths that have spaces. Without this box checked, the DMI would replace spaces in paths with ‘_’.

Note: If users have written their DMI executable to recognize white space as the end of a path, and they begin using spaces in path names, then the executable should be modified.



Limit Output to Run Range

For an output DMI, the **Limit Output to Run Range** toggle specifies that output data should only be written for the run range (Start Timestep to Finish Timestep). Without this box checked, the DMI will write the entire series of data. This configuration is only available for output DMIs. See [“Time Range Specification”](#) on page 42 for additional options.



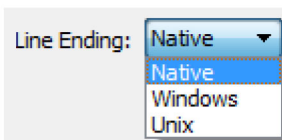
Line Ending

The **Line Ending** menu allows the user to specify the type of line ending sequence to write for output DMIs. This option is useful when you are running RiverWare on one platform, e.g. Windows, but your output data processing utilities and scripts are on another platform, e.g. Solaris.

The choices are:

- Native (default, uses the format for the current platform)
- Windows (CR NL)
- Unix (NL)

CR = carriage return, NL = newline



Control File

To edit the DMI control file, the user either:

- Selects  to open a file chooser dialog and selects the control file
- Selects the **Control File** field to edit the control file in place

Control File: 

The control file can contain environment variable references of the form:

- \$VARIABLE
- \$(VARIABLE)
- \${VARIABLE}

VARIABLE is a letter followed by zero or more letters, digits or underscores. If the variable is followed by a letter, digit or underscore it must be “quoted” with () or {}. For example:

- \$DMI_DIR/Control/Res.ct1
- \$(DMI_DIR)_\$TARGET/Control/Res.ct1

When the edits are applied the control file is validated; the possible errors are:

- A blank control file

All other validation occurs when the DMI is invoked, including:

- Replacing environment variable references with their values
- Verifying the resulting path refers to a readable file

Executable

To edit the DMI executable file, the user either:

- Selects  to open a file chooser dialog and select the executable file
- Selects the **Executable** field to edit the executable file in place

Executable: 

Note: Alternatively, the user can create the DMI Data Files separately without using an executable. In this case the executable field can be left blank.

The executable file can contain environment variable references, the same as the control file.

All validation occurs when the DMI is invoked, including:

- Replacing environment variable references with their values
- Verifying the resulting path refers to an executable file

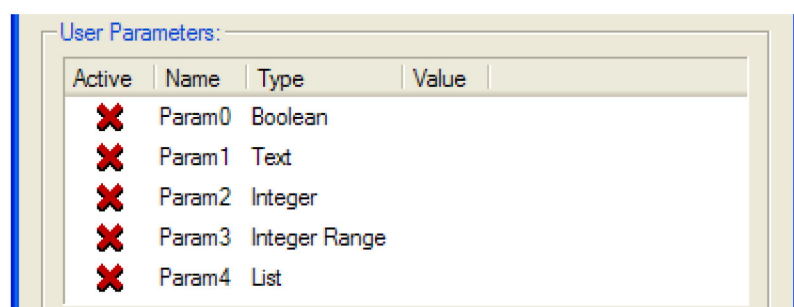
User Parameters

To edit the DMI user parameters the user specifies:

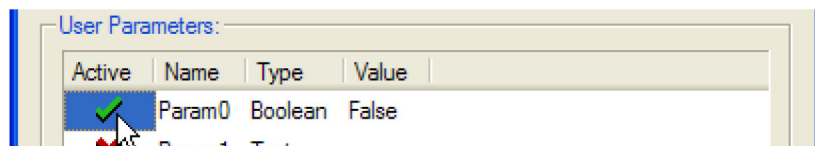
- If a parameter is active (e.g. if it should be included on the DMI executable's command line)
- If a parameter is active, its value

✗ indicates an inactive parameter; ✓ indicates an active parameter; by default parameters are inactive:

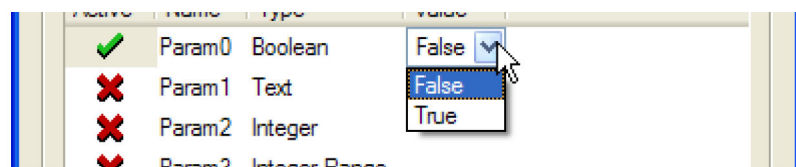
The user toggles a parameter's active state by selecting the **Active** cell; when a parameter is active its value is displayed in the **Value** cell.



The user edits a parameter value by clicking twice in the **Value** cell, activating a control appropriate for the parameter type.

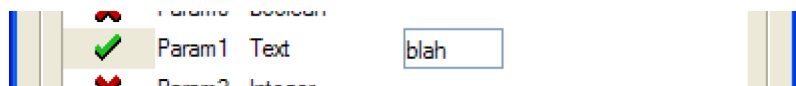


For a Boolean parameter the control is a menu containing False and True.



For a Text parameter the control is a one line text editor. When the edits are applied the parameter's value is validated; the possible errors are as follows:

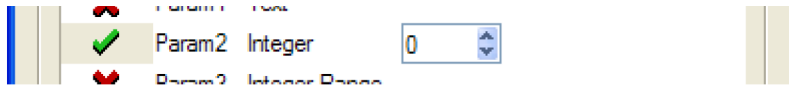
- The value is blank
- The value contains white space



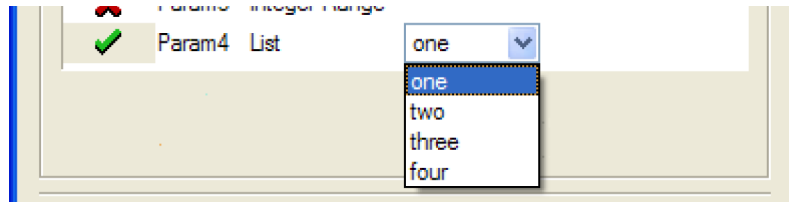
Chapter 2

Control File-Executable DMI

For an Integer parameter the control is an unbounded integer spinner. For an Integer Range parameter the control is an integer spinner, bounded by the parameter's minimum and maximum values.

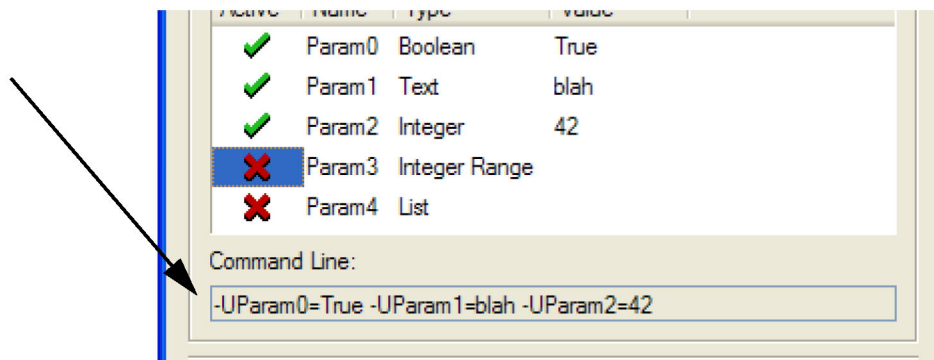


For a List parameter the control is a menu containing the list items defined in the DMI Parameter dialog.



Command Line

As the user edits the user parameters they are displayed as they will appear on the DMI executable's command line:



The command line is not editable, but it can be horizontally scrolled, copied-and-pasted and dragged-and-dropped (to a Microsoft Word document, for example).

Chapter 3

Trace Directory DMI

Topics

- [About the Trace Directory DMI, page 59](#)
- [Trace Directory DMI Editor Dialog, page 61](#)

About the Trace Directory DMI

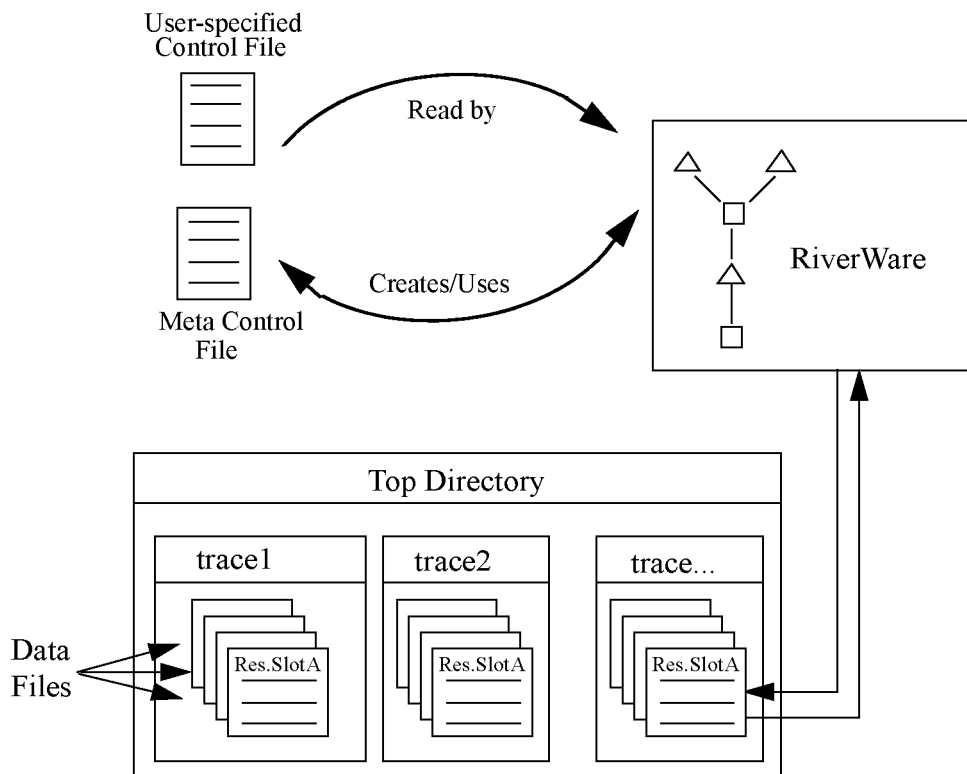
The Trace Directory Data Management Interface (DMI) is similar to the Control File-Executable approach where the user develops a *control file* that specifies the slot data to move into or out of RiverWare. However, instead of an executable, the user specifies a top directory, under which the data files for traces are accessed in subdirectories named with the trace number.

The Trace Directory DMI provides a means to transfer data between RiverWare and data files in a known directory structure. There are two directions in which the DMI can transfer data:

- An Input DMI transfers data from data files into RiverWare.
- An Output DMI transfers data from RiverWare into data files.

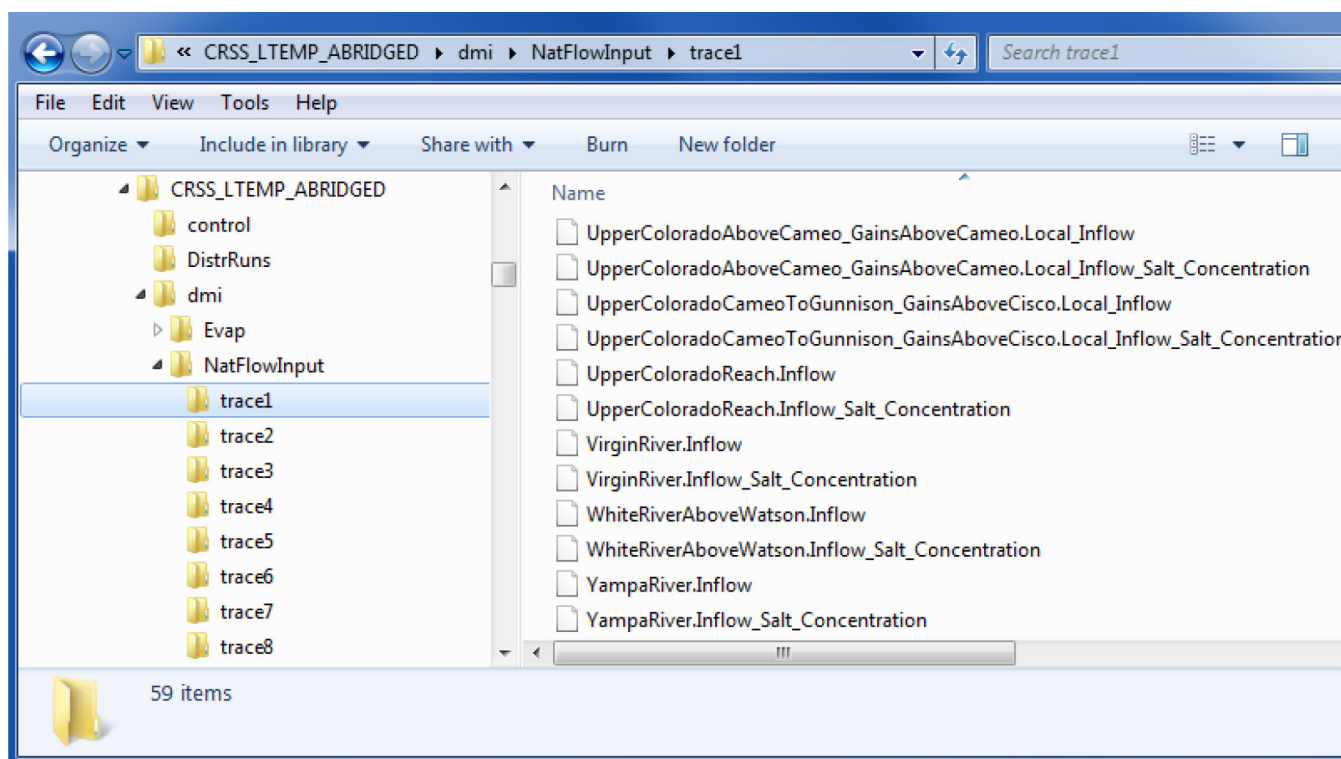
The Trace Directory DMI consists of several components. [Figure 3.1](#) illustrates the relationships among these components.

Figure 3.1 Trace Directory DMI components



- The user-specified Control File is created by the user and contains a list of slots which the user wants imported or exported from the RiverWare model. The control file is an ASCII file which can map data between object.slot names in the Workspace and data file names. See “[Control File](#)” on page 38 for details. But no file= needs to be specified as the Trace Directory DMI can resolve the file name from the trace directory path and the object.slot name. The control file could be as simple as the following entry:
`BigRes.Outflow: units=cfs scale=1`
- The user-specified top directory is a directory under which data files will be located within subdirectories named with the trace number corresponding to the data (i.e. trace1, trace2, etc.) Multiple traces could be associated with input or output DMIs from multiple runs, but running a Trace Directory DMI outside of MRM would use a single trace folder. [Figure 3.2](#) is an example of a top directory (NatFlowInput) and trace subdirectories, each of which contains slot data files.

Figure 3.2



- The RiverWare DMI data files are a set of files used to transfer the data to and from the RiverWare model. Each of these files contains the data for a single slot in the model. In an output DMI RiverWare will create the files, or will read existing files for an input DMI. See “Data Files” on page 47 for details.

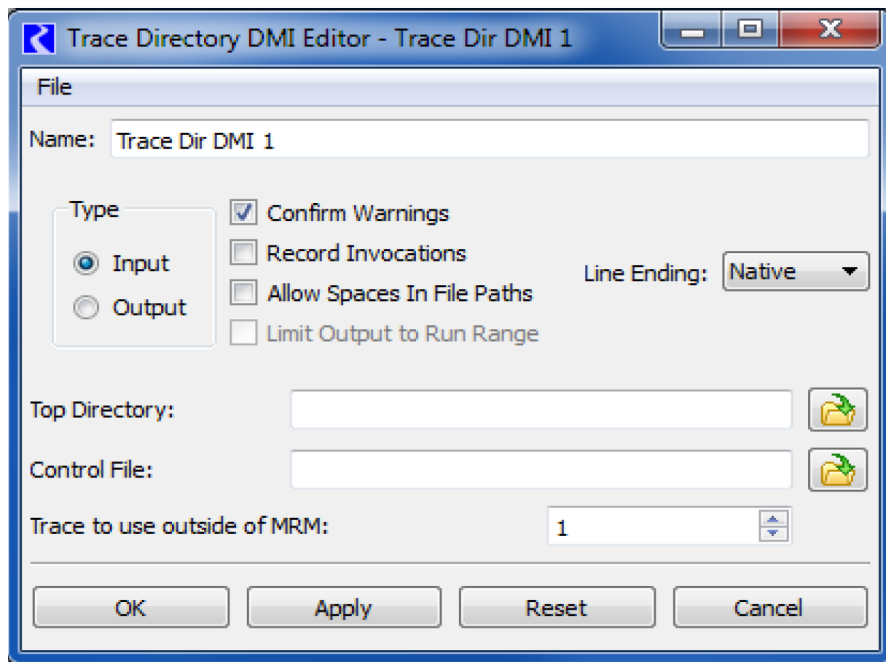
Following is an example of how the DMI works. In this example we assume it is an Input DMI, but it is the same, just reversed for an Output DMI.

1. The user Invokes the Input DMI.
2. RiverWare verifies the DMI to make sure it is valid.
3. RiverWare reads the user Control File and resolves all wild carding. It then substitutes the directory path for the current trace into the file keyword path for each entry in the control file. If a file keyword is not provided, it generates one from the trace directory path and the object.slot name. It then writes the Meta Control File, which has a fully specified line for each slot in the Control File.
4. RiverWare then uses the Meta Control File to read the DMI Data Files from their appropriate directory and imports the data into the slots in RiverWare.

Trace Directory DMI Editor Dialog

A new Trace Directory DMI is created from the main DMI Manager dialog by selecting **DMI**, then **New DMI**, then **Trace Directory DMI**. Edit the DMI by double-clicking in the DMI Manager or by selecting **Edit** from the **Edit** menu; the Trace Directory DMI Editor dialog appears.

Figure 3.3



Many of the configuration options are the same as those for the Control File-Executable DMI. The following list references the discussions in that section for the equivalent controls:

- **Name** - [Name](#), page 53.
- **Type** - [Type](#), page 54.
- **Confirm Warnings Toggle** - [Confirm Warnings Checkbox](#), page 54.
- **Record Invocations** - [Record Invocations](#), page 54.
- **Allow Spaces in File Paths** - [Allow Spaces in File Paths](#), page 55.
- **Limit Output to Run Range** - [Limit Output to Run Range](#), page 55.
- **Line Ending** - [Line Ending](#), page 55.
- **Control File** - [Control File](#), page 56.

To specify a Top Directory and the Control File the user either:

- The  button to open a chooser dialog and select the directory/file.
- Select the field to edit the path in place.

The top directory path can contain environment variable references of the form:

- \$VARIABLE
- \$(VARIABLE)
- \${VARIABLE}

VARIABLE is a letter followed by zero or more letters, digits or underscores. If the variable is followed by a letter, digit or underscore it must be “quoted” with () or {}. For example:

- \$DMI_DIR/EvapData/
- \$(DMI_DIR)_\$TARGET/EvapData/

The **Trace to use outside of MRM** control is an integer spinner where a trace number can be selected or typed. If this DMI is executed outside of a multiple run, this trace number is used to determine the subdirectory under the top directory to use for reading or writing data. For example, if the trace number is 3, the subdirectory “trace3” is used as the location for the DMI input or output files. Within MRM, the trace number is automatically substituted.

Chapter 3

Trace Directory DMI

Chapter 4

Database DMI

Topics

- [About the Database DMI, page 65](#)
- [Database DMI Overview, page 66](#)
- [Creating a Database DMI, page 68](#)
- [Datasets Overview, page 94](#)
- [DSS Datasets, page 95](#)
- [HDB Datasets, page 99](#)
- [DIT Datasets, page 115](#)
- [Excel Datasets, page 119](#)
- [Web Service Datasets, page 138](#)
- [Name Mapping, page 154](#)

About the Database DMI

The Database DMI provides a direct connection between RiverWare and an external database. Databases currently supported are the U.S. Army Corp of Engineers Hydrologic Engineering Center Data Storage System (DSS), the Hydrologic Database (HDB), the Demand Input Tool (DIT), Microsoft Excel, and various web services.

Following are the slot types that can be read or written by each type of DMI:

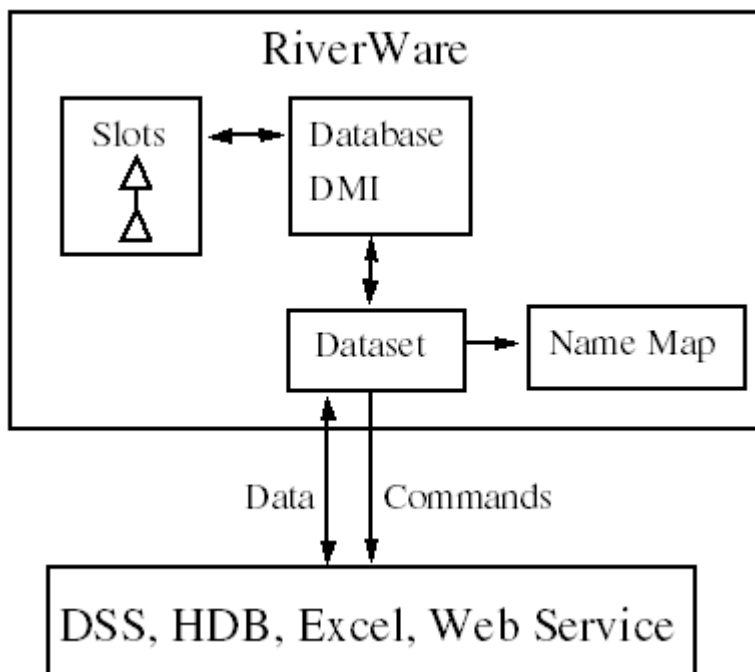
- The DSS interface supports series data and paired data (Table Slots with two columns).
- The HDB interface supports series slot data.
- The Excel interface support series, periodic, table, and scalar data.
- The DIT interface supports series slot data.
- The Web Service interface supports import of series slot data from the following web services:
 - CDSS Surface Water Time Series - Day web service
 - CWMS Data API (CDA) web service
 - HDB web service
 - Reclamation Hydromet (Missouri Basin / Great Plains) web service, and
 - USGS Daily Values web service

This section presents an overview of the Database DMI, then presents details on creating database DMIs, datasets, name maps, and sharing database DMIs.

Database DMI Overview

The Database DMI allows RiverWare to import and export data directly from and to the database. In this document, a *database* refers to a specific database implementation, for example a DSS file or Excel workbook. [Figure 4.1](#) shows the interaction between RiverWare and the database. The user sets up the Name Maps, configures the Datasets, and identifies slots and time intervals at which to import/export data in the Database DMI.

Figure 4.1 Databased DMI Interaction between RiverWare and Databases

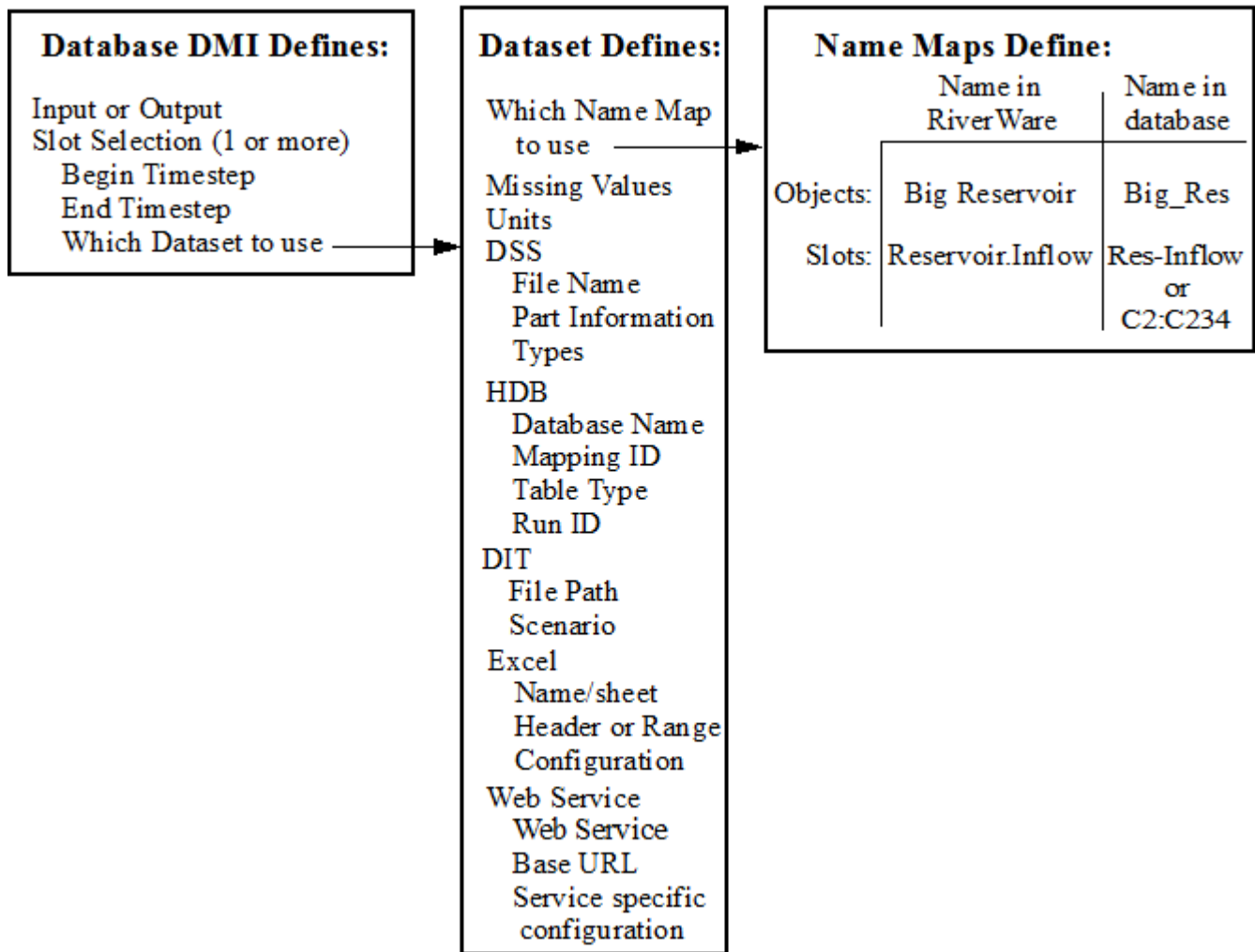


When the Database DMI is invoked, the Dataset and Name Maps are read specifying the database to use, the units and what slots/objects are named in the database. Any wildcarded slot selections are then fully specified and then any Name Maps are applied. Then commands are passed to the database: for an input DMI, the commands locate the data in the database and imports data into RiverWare slots; for an output DMI, the database DMI takes the RiverWare slot data and inserts it into the appropriate place in the database.

[Figure 4.2](#) shows in more detail the three pieces of a database DMI and how they work together to define a DMI.

Note: With the DMI Database Editor you can configure an entire Database DMI in one dialog but these three components still exist and interact with one another. See [“Creating a Database DMI”](#) on page 68 for details.

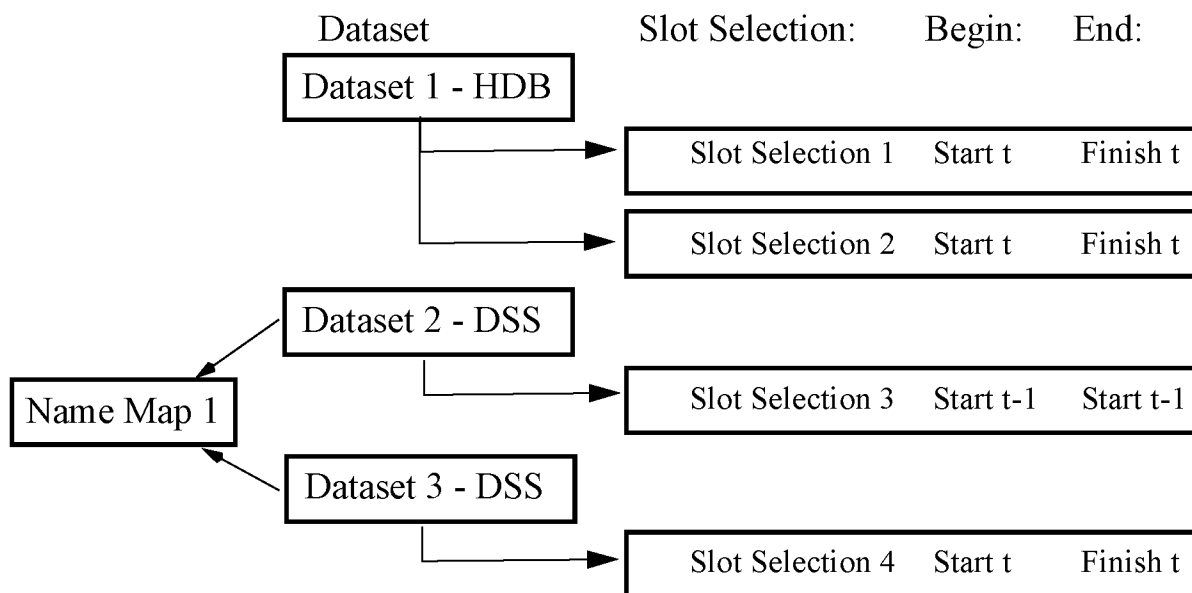
Figure 4.2 Diagram showing the pieces of a Database DMI



The three pieces (slot selections, datasets, and name maps) were designed to be very flexible. In the simple case of importing or exporting 1 slot, the user needs 1 slot selection, 1 dataset, and optionally 1 name map. It is possible to configure the system with more functionality which leads to more complexity.

For example, in [Figure 4.3](#), slot selections 1 and 2 share dataset 1 which is an HDB dataset; no Name Map is required. In this DMI, Slot selection1 is an input from HDB for the full run and then slot selection 2 is an input from HDB for the full run. Slot selection 3 is used to input initial conditions and uses Dataset 2 which points to a separate DSS file in Dataset 2. Slot selection 4 inputs from a DSS file referenced in Dataset 3 for the full run. Both Dataset 2 and 3 are DSS datasets and share Name Map 1.

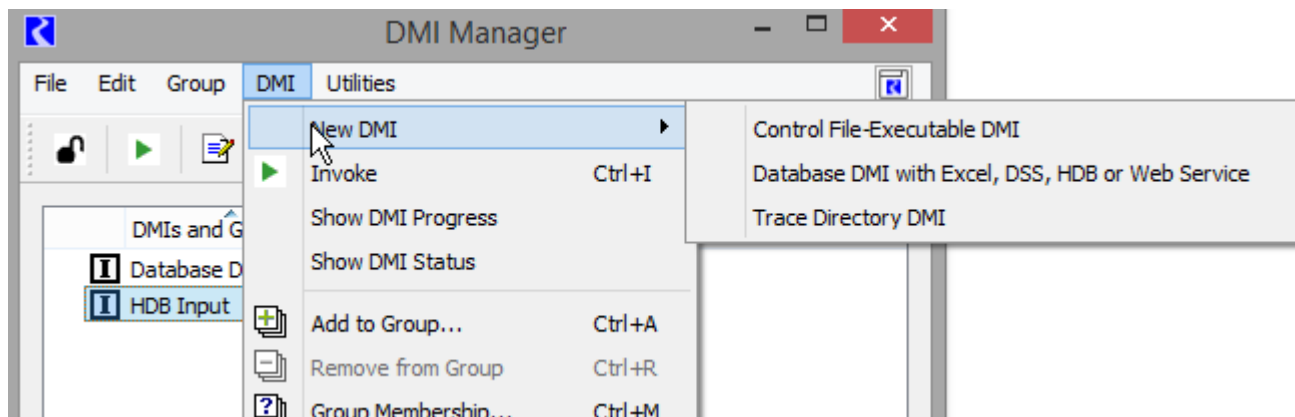
Figure 4.3



We will now look at each component starting with creating a Database DMI, then look at Datasets followed by Name Maps.

Creating a Database DMI

A Database DMI is created from the main DMI Manager dialog using the **DMI**, then **New DMI**, then **Database DMI with Excel, DSS, HDB or Web Service** menu.



There are two ways to define and edit a Database DMI:

- Define a Name Map, Dataset and then use the Database DMI editor. You create a Name Map, then a Dataset, then a Database DMI that pulls it all together.
- Use the Database DMI Editor. The Database DMI Editor allows you to create a fully functional DSS Database DMI from a single dialog and provides access to other dialogs for Excel, HDB, and web service DMIs. It fully

expands the slot selection to show the exact configuration that will be used. For example, it displays that BigReservoir.Inflow will be imported using BigRes Res-Inflow and displays the part information and units that will be used.

Once a new Database DMI is added to the DMI manager, the user can edit it using the Database DMI Editor. The Database DMI Editor dialog provides a convenient place to create, edit, and view database DMIs. It also provides a single dialog for configuration of a fully functional DSS Database DMIs. To that end, it allows the user to specify all required configuration information and as much optional configuration information as is feasible. The real power of this dialog is that the user can see the fully specified information that will be sent to the database, which is especially useful for DSS.

To open the New Database DMI Editor, the user selects **Edit**, then **Edit with New Database DMI Editor**. The Database DMI dialog appears.

This dialog is like an editable view into the Dataset Manager. Thus, the user can see what has already been configured in those dialogs but also make edits that will be applied to those pieces. It is important to note that if the user edits a dataset or Database DMI in the editor window, the user cannot also edit the individual components at the same time.

Note: Because there are many approaches to create and edit Database DMIs, there are multiple ways to edit the same information, e.g. you could edit a dataset from both the dataset manager and the Database DMI Editor. To avoid conflicts, the dialog that first starts editing a name map or dataset creates a “lock” on the information and no other dialog can edit that feature. When a lock has been

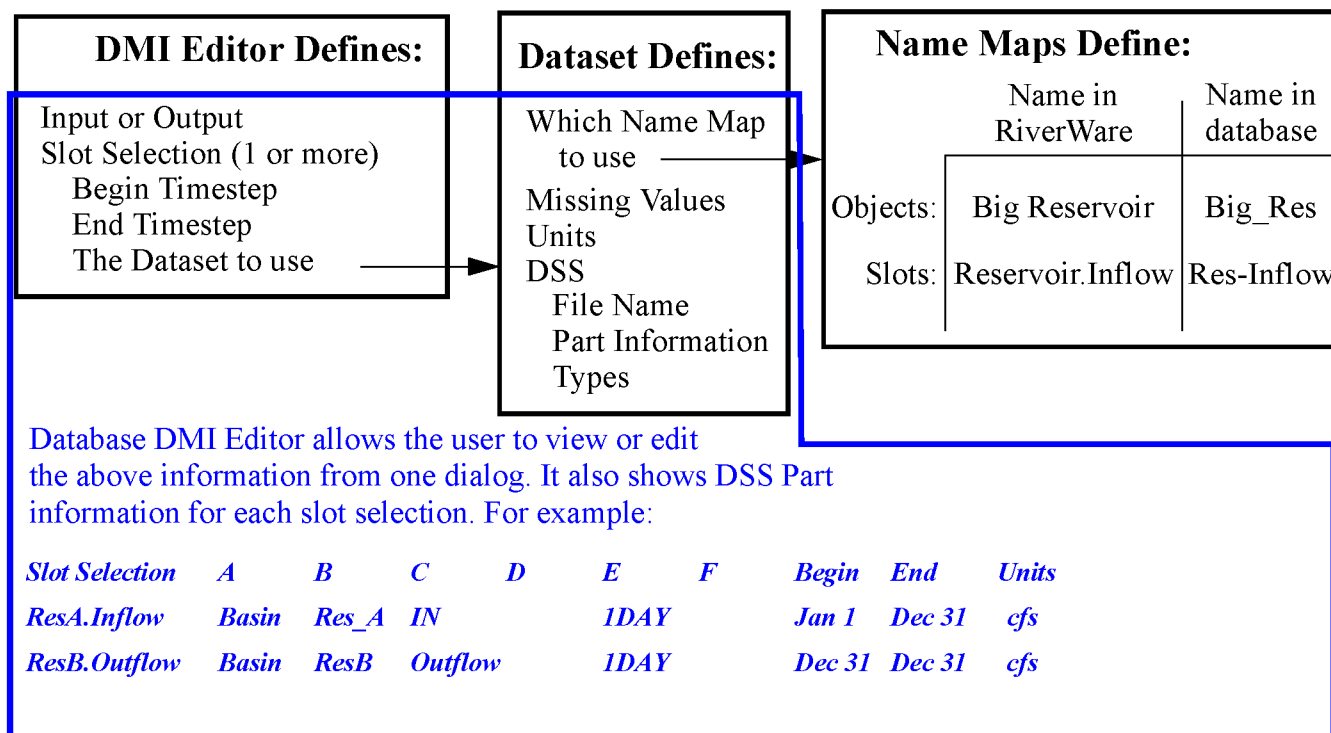
created, the lock icon  is added to any other dialogs that are trying to edit that information. If you see this icon, the information is not editable from that location. Similarly, if a name map or dataset has

been scheduled for removal from another dialog, the “delete lock” icon is shown. .

Figure 4.4 shows a similar figure to Figure 4.2, but with the Database DMI Editor also shown with a DSS dataset example.

Note: The information in each of the components is represented in the dialog, but in a slightly different organization. The parts of the dialog are described in detail in the following section.

Figure 4.4 DMI Pieces of a Database DMI with the parts highlighted that represent the DMI Editor



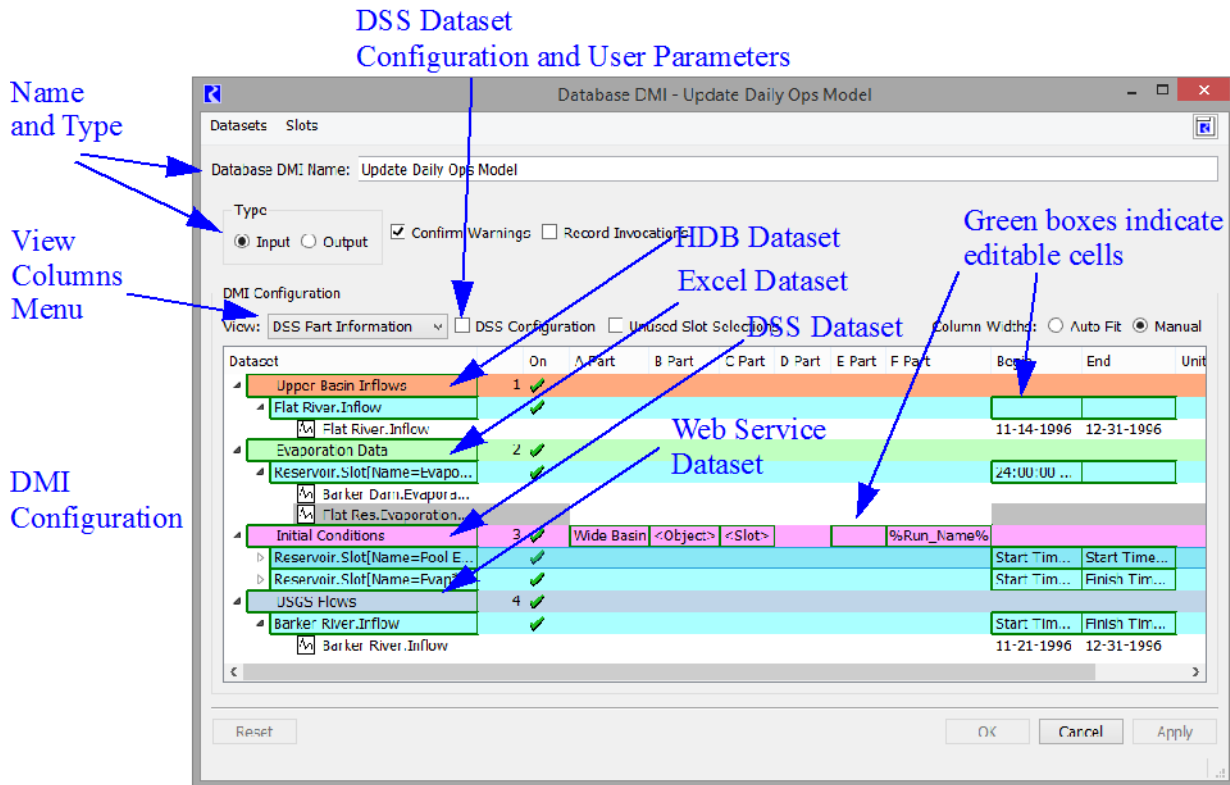
Conceptually the dialog has four sections from the top to the bottom, as shown in [Figure 4.5](#) and described in the following sections:

- Database DMI Name and Type
- DMI Configuration
- DSS Configuration including DMI User Parameters (Only shown when a DSS dataset is present)

In this dialog, each type of dataset has its own color:

- HDB—orange
- Excel—green
- DSS—pink
- DIT—buttermilk (not shown below)
- Web service —blue grey

Figure 4.5 Annotated screenshot of a Database DMI



The cyan rows are slot selections. The green boxes are cells in which data entry is possible.

Name and Type

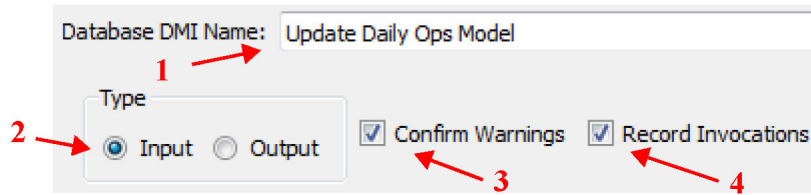
The Database DMI general configuration section (upper section) of the dialog allows you to do the following:

- Specify the **Database DMI Name** in the text field.
- Specify whether the Database DMI is an **Input** or **Output** DMI.
- Specify if you wish to confirm warnings before the Database DMI continues using the toggle.
- Specify whether the DMI manager should record invocations of the DMI. The **Record Invocations** check box allows RiverWare to maintain information which chronicles input DMI invocations. With this information, values set by DMIs can be cleared on a per-invocation basis. If a user wishes to clear out all values that are input by a DMI, this box should be checked. If a user knows values set by a particular DMI won't be cleared then it is not necessary to maintain information about the DMI invocations and this box should remain unchecked. The

Chapter 4

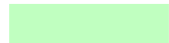
Database DMI

checkbox is only enabled for input DMIs and the default is unchecked. See [“DMI Invocation Manager”](#) on page 24 for details on clearing values set by an input DMI.



DMI Configuration

The DMI configuration section (middle section) displays the configuration of the DMI. In this dialog, each type of dataset has its own color:

- HDB—orange 
- Excel—green 
- DSS—pink 
- DIT —buttermilk 
- Web service—blue grey 

The cyan (light blue) rows  are slot selections. The green boxes are cells in which data entry is possible. This portion of the dialog allows you to:

- Create Datasets using the Dataset menu.
- Remove Datasets using right-click context menus.
- Create and define slots selections and time intervals.
- For DMIs with DSS datasets, see and edit the part information for each slot.
- Select the columns to view. See [“View Columns”](#) on page 79.
- Show or hide Unused Slot Selections; see [“Unused Slot Selections”](#) on page 79.
- Select either **Auto Fit** or **Manual Column Widths**.

Note: The first column in Auto Fit mode, only grows to a fixed maximum width.

- In this dialog, tool tips and the status bar (the lower left) are used to show the contents of cells, indicate directions, or additional information.

The information is presented in a tree-view, with each level representing a more detailed specification, as detailed in [Table 4.1](#).

Table 4.1

Name	What it is	Tree Level	Items shown
Dataset	Entered by user	1: Highest	Name, type and generic Part information. The Name and Part information is editable. All datasets including HDB, Excel, and web service datasets, are listed. HDB, Excel, and web service datasets can be shown in this dialog but cannot be edited.
Slot Selection	Slot Selection Name as shown (or modified) in the Selector dialog. It can include wildcards, e.g. StorageReservoir.Outflow	2: Middle	The begin and end timestep for this selection are shown and can be edited.
Slots in Selection	Slots that were selected by the Slot Selection shown, including resolution of wildcards	3: Lowest	Fully specified information as it will be passed to the database. For DSS, this includes Part information, Begin and End Timestep, and units. No editing is allowed

The following sections describe the editable parts of each of the levels of the treeview.

Dataset

The following actions can be performed for a Slot Selection row:

- Name DSS Datasets. Double-click to edit the name.
- Open HDB, Excel, DIT, and web service datasets. Double-click these datasets to edit.
- Disable a Dataset for debugging; see [“Enabling and Disabling for Debugging”](#) on page 78.
- Specify the part information. Two notes:
 - Part D is not editable
 - The default for Part E is the timestep size such as 1Day or 1Hour. The full options are Second, Minute, Hour, Day, Week, Month, Year. Note, pseudo-regular time series can be identified using "~1Day". See [“Time Zone”](#) on page 97

There are many options for specifying the part information.

The part information can be edited by entering text or by selecting text from a right-click context menu. In addition to typing text, you can use the two following items to generalize DSS DMIs:

- Workspace Entities. You can use generic Workspace Entities for wildcarding any part (except Part D which is not editable). Select the desired entity from the right-click context menu. [Table 4.2](#) lists the options.

Table 4.2

Workspace Entity	Description
Slot	The slot name.
Parent	The item that contains the slot. For a simulation slot, this is the object (e.g. a reservoir). For an accounting slot, this is the account (e.g. a storage account).
SimObj	The object that contains the item. For both simulation and accounting slots, this is the containing object (e.g. a reservoir)

Table 4.2

Workspace Entity	Description
Account	The name of the account containing the object. (Valid when accounting is enabled)
Supply	The name of the supply. For example a supply named, ResASStorageToReachBFish.Supply would result in ResASStorageToReachBFish (Valid when accounting is enabled)
Exchange	The name of the exchange. (Valid when accounting is enabled)
Payback	The name of the payback. (Valid when accounting is enabled)
UpSimObj	Not implemented yet. For a supply, this evaluates to the object that contains the upstream account.
UpAccount	Not implemented yet. For a supply, this evaluates to the account name of the upstream account.
DownSimObj	Not implemented yet. For a supply, this evaluates to the object that contains the downstream account.
DownAccount	Not implemented yet. For a supply, this evaluates to the account name of the downstream account.

Note: In versions prior to 6.1, Part B was by default the object name and Part C was by default the slot name. Now these need to be explicitly defined for new datasets, existing datasets will be given the <parent> and <slot> entities for parts b and c, respectively.

- User Parameters. You can use parameters in the DSS parts (see “[DSS DMI User Parameters](#)” on page 82, for details). To use the parameters in the DSS part information, specify the value with the percent sign:
%Parameter%

Slot Selection

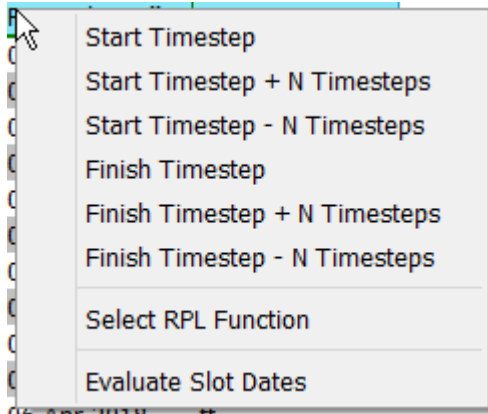
The following actions can be done for a Slot Selection row:

- Select slots by selecting the name of the slot selection. Typically, this is NONE before the slots have been selected. Use the Selector to choose the desired slots. Either create a new selection or use an existing Slot Set (see “[Slot Sets](#)” in *User Interface*). When creating a selection, wildcards are very useful to symbolically specify the slots. For example, all Reservoir Inflows could be specified using a wildcard on reservoirs and on the slot name, Reservoir.Slot[Name=Inflow]. See “[Creating a Name Map of All Reservoir.Inflow Slots](#)” in *User Interface* for a step by step example of this selection.

Note: The Slot Selections are maintained as references to the specified slots. So if you delete the objects/slots to which the selection refers, the references could be lost, particularly if the slots are manually specified, that is, not using wildcards. If you want to export and re-import the same named custom slots, maybe from a different model, without breaking the DMI references, use the Export and Import of Multiple Slots as that utility maintains the slot references. See “[Exporting and Importing Multiple Slots](#)” in *User Interface*.

- Disable a Slot Selection for debugging; see “[Enabling and Disabling for Debugging](#)” on page 78.
- Specify the time interval for each slot selection. The begin and end dates are edited in place; they can be edited by entering text or by selected text from a right-click context menu. In both cases the text is a symbolic date/time. If not entered, the default time range is the slot’s range. In the Begin and End columns, specify the timesteps at which to begin/end the Input or Output data. Either select and manually type in the date, or right-

click and a select from a list of commonly used reference datetimes as shown in the following figure. If selecting one of the reference datetimes, enter the number of Timesteps in place of the letter N as necessary.



The datetime uses the same syntax as the RiverWare Policy Language (RPL) datetimes. For example, as shown in Figure 4.6, the user could enter either of the following as a valid datetime:

- 24:00:00 November Max DayOfMonth, 1996
- 24:00:00 Start Month Start DayOfMonth, Start Year - 24 Hours

Figure 4.6

Dataset	On	Begin	End
Evaporation Data	2 ✓		
Reservoir.Slot[Name=Evaporation Rate]	✓	24:00:00 Start Month Start DayOfMonth, Start Year - 24 Hours	Finish Timestep
Barker Dam.Evaporation Rate		11-20-1996	12-31-1996

See “**DATETIME**” in *RiverWare Policy Language (RPL)* for details on datetimes. Basically, any fully specified datetime can be used

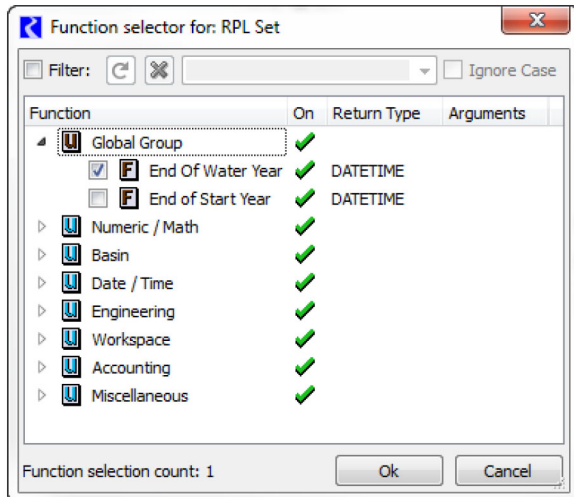
Note: No @ or quotes “ ” are necessary when specifying the datetimes in the DMI slots dialog.

To allow additional flexibility specifying datetimes, the user can choose the **Select Function From Global Function Set** option and then choose a RPL function from an opened Global Function Set; see “**Global RPL Functions**” in *RiverWare Policy Language (RPL)* for details. A RPL Object selector opens and displays a tree

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view of all the global utility groups and predefined function groups; see “[Example: Creating a New RBS Rule-set](#)” in *RiverWare Policy Language (RPL)* for details.



Select a single function by selecting the box to add a check mark. This function **MUST**:

- Not have any arguments
- Return a fully specified DATETIME variable

If either is not true, an error will be posted when the selection is applied or the DMI is run. After you select **Ok**, the function name is displayed in the Begin / End column with “()” appended. Alternatively, you can manually enter the function name, but you must include the “()”.

Tip: The symbolic datetimes are evaluated for each row in the slot selection when the DMI tree is expanded, the datetime specification is edited, and the DMI is executed. If you change the start/finish timestep, the body of the specified RPL function, or data referenced by the RPL function, the dates won’t automatically update. Use one of the above operations, or to manually evaluate, use the Slots (or right-click context) and then Evaluate Slot Dates menu.

Slots in Selection

The following actions can be performed for Slot Selection rows:

- See the slots that were selected using wildcard.
- See the resolved time interval for each slot, as it will be passed to DSS. See the tip above.
- See the units and scale as they will be passed to DSS.

- For DSS parts, mouse over the slot's part columns to see the information that will be passed to DSS. The Status at the bottom shows the entire DSS path.

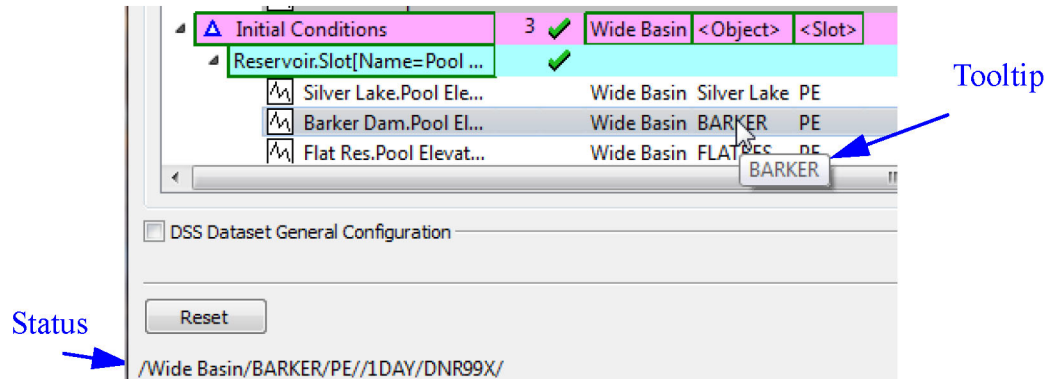


Figure 4.7 Screenshot of Database DMI with DSS dataset (left portion)

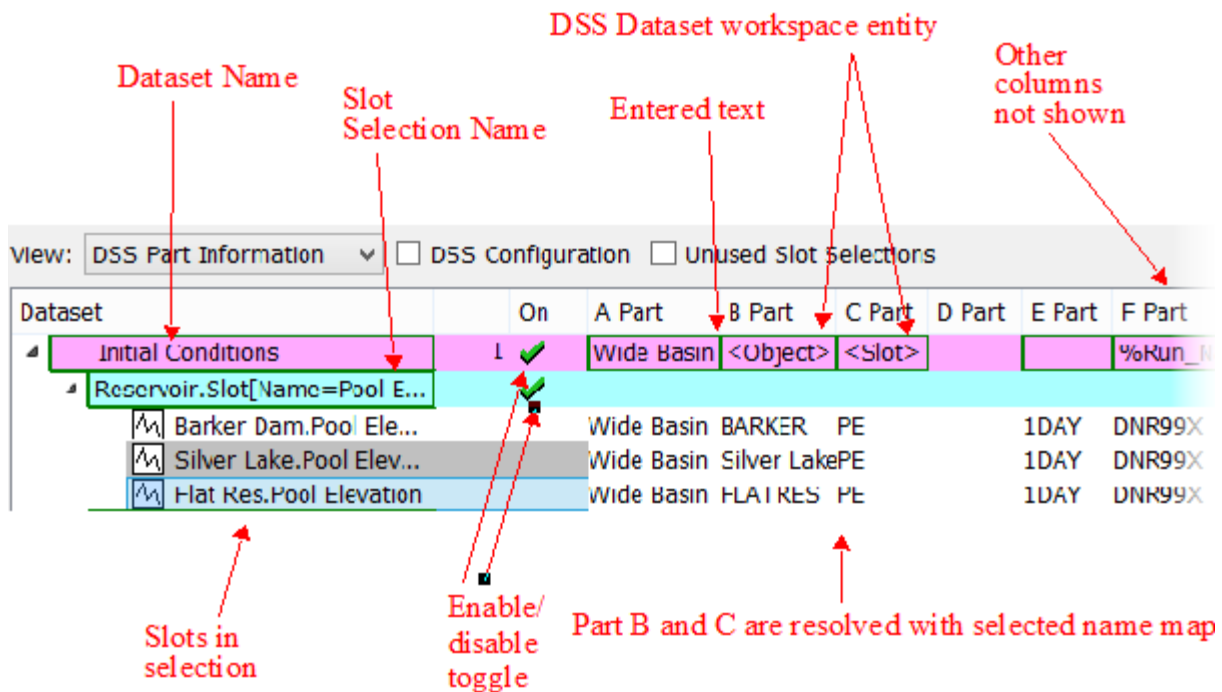


Figure 4.8 shows the right portion of the dialog.

Figure 4.8 Screenshot of Database DMI with DSS dataset (right portion)

E Part	F Part	Begin	End	Units	Data Type
	%Run_Name%	Start Tim...	Start Time...		
1DAY	DNR99X			ft	
1DAY	DNR99X			ft	
1DAY	DNR99X	20 Nov 19...	20 Nov 199...	ft	

Enabling and Disabling for Debugging

Datasets and slot selections can be enabled and/or disabled for debugging purposes. In the **On** column, there is a green check if it is enabled/on. A red X indicates it is disabled/off. The purpose of this feature is to allow you to debug a DMI by disabling datasets or slot selections and then running the DMI to test other aspects of the DMI. This behaves as follows:

- Turning off a dataset causes its slot selections to be turned off.
- Turning off all slot selections causes the dataset to be shown as off.
- The dataset is shown as a tristate (on/green, off/red, partially on/orange) based on the state of its slot selections.
- The on/off state is preserved in the model file when saved.

Tip: Use the right-click context menu on a slot selection to **Turn On this Slot Selection** but leave the rest in their current state. This is particularly useful to debug one slot selection. Disable all rows at the top-most dataset level and then use the right-click **Turn On this Slot Selection** menu to turn on only the desired slot selection row. Similarly, there is a right-click option to **Turn Off this Slot Selection**.

Use of Name Maps

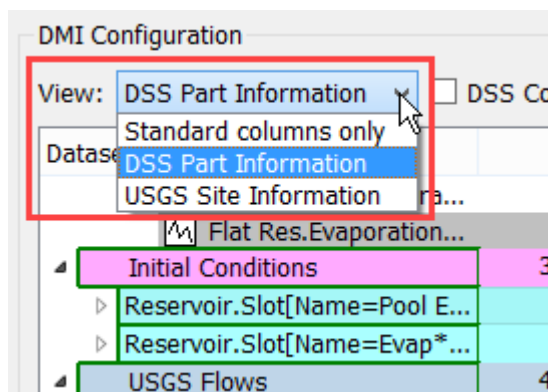
If you specify a Name Map when configuring the Dataset, then the following will occur:

- If the RiverWare Object is in the Name Map then its mapped name will be used for <Object> or <Parent> entities; otherwise, its RiverWare name will be used.
- If the Slot is in the Name Map then its mapped name will be used for <Slot> entities otherwise, its RiverWare name will be used.

- If a Table Slot column is in the Name Map then its mapped name will be used for Part C; otherwise, the Table Slot's RiverWare name will be used.

View Columns

Use the View menu in the DMI configuration to select the columns to show.

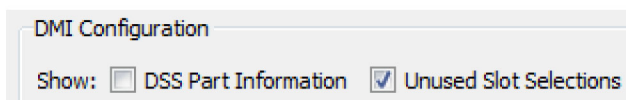


The following options are available depending on the types of datasets in use:

- Standard columns only - The On, Begin, End, and Units columns are shown. These are the only columns available for HDB and Excel Datasets.
- DSS Part Information - In addition to the standard columns, parts A-F columns are shown. For more information, see [“DSS Dataset Configuration”](#) on page 80.
- CDSS Site Information - In addition to the standard columns, Site Name and Site Number columns are shown. For more information, see [“CDSS Surface Water Time Series - Day”](#) on page 142.
- USGS Site Information - In addition to the standard columns, Site Name and Site Number columns are shown. For more information, see [“USGS Daily Values”](#) on page 150.
- CWMS Data API Name - In addition to the standard columns, a Name column is shown. For more information, see [“CWMS Data API Web Service”](#) on page 145.
- DIT Information - In addition to the standard columns, the SQLite File name and Scenario are shown. For more information, see [“DIT Tab”](#) on page 117.

Unused Slot Selections

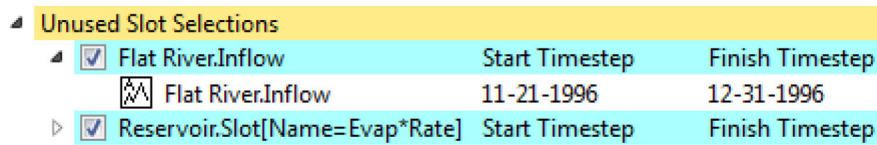
It is possible to have slot selections that are not used by any of the datasets. To show these selections, select **Unused Slot Selections**.



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The selections are then shown as a golden colored row titled Unused Slot Selections at the Dataset level. You can view and change the selections (and the slots selected). From right-click context menus, you can delete the selection. You can also drag and drop the slot selection onto a dataset to associate that slot selection with a dataset.



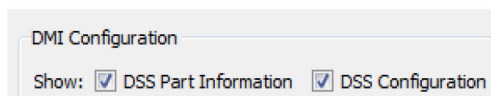
Unused Slot Selections		
☒ Flat River.Inflow	Start Timestep	Finish Timestep
 Flat River.Inflow	11-21-1996	12-31-1996
☒ Reservoir.Slot[Name=Evap*Rate]	Start Timestep	Finish Timestep

DSS Dataset Configuration

For DSS datasets, an additional panel allows you to fully edit the dataset.

Note: This is not available for Excel, DIT, HDB, or web service datasets. For those datasets, you can double-click the dataset and it will open the dataset editor.

When there is a DSS dataset in the DMI, the DSS checkboxes are displayed. Select the **DSS Configuration** checkbox to show general configuration settings.



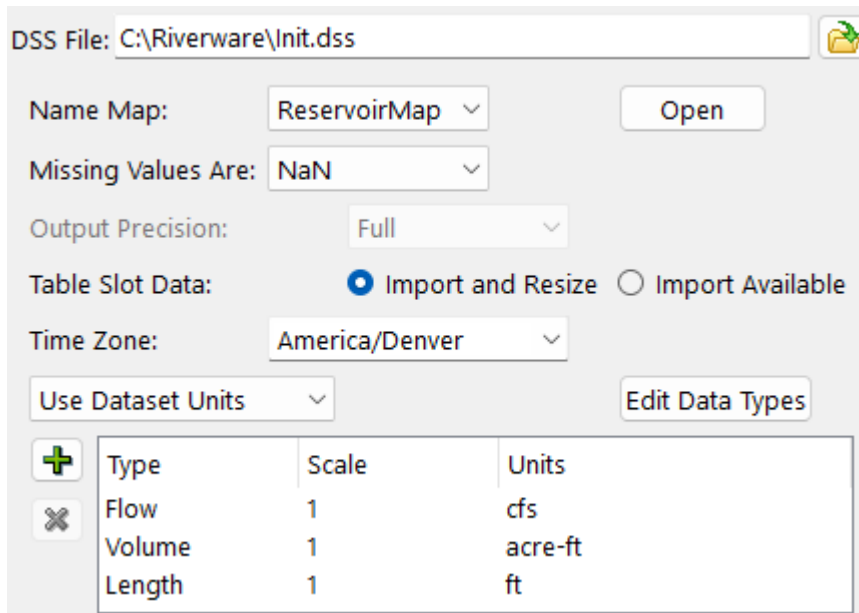
This shows a view similar to the dataset. When you have a DSS dataset selected, the general configuration becomes enabled and allows you to:

- Specify the DSS file for a DSS Dataset.
- Specify the name map using the pull down menu and/or **Open** the name map manager.
- Specify the desired behavior when missing values are encountered.
- Specify the level of precision for output DMIs.
- Specify the time zone of the data being imported, as necessary. See below.
- Specify the units for the selected DSS Dataset.
- Configure Data Types by selecting the **Edit Data Types** button.

See “[Datasets Overview](#)” on page 94 for descriptions of these fields.

Tip: DSS DMIs are not compatible with MRM as there is no way to change the part information for a particular run in an MRM.

Note: Each of these specifications only works on the *selected dataset*. In other words, you must highlight a dataset row from the middle portion of the dialog before it allows you to change units, scale, name map to use, or how missing values are treated.



DSS File: C:\Riverware\Init.dss

Name Map: ReservoirMap Open

Missing Values Are: NaN

Output Precision: Full

Table Slot Data: ☒ Import and Resize ☐ Import Available

Time Zone: America/Denver

Use Dataset Units Edit Data Types

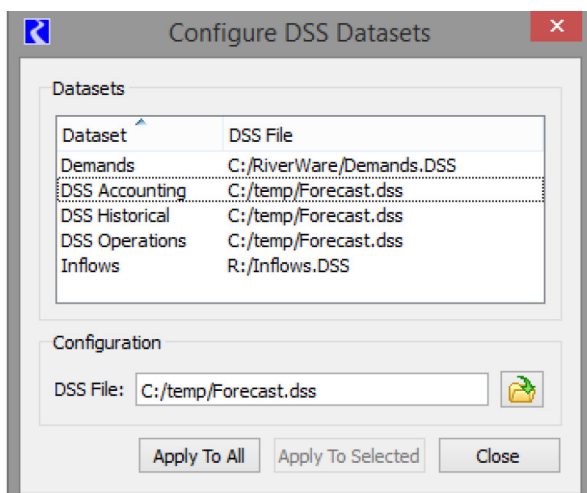
Type	Scale	Units
Flow	1	cfs
Volume	1	acre-ft
Length	1	ft

Setting Multiple DSS Dataset Files

You can configure the DSS file path for multiple DSS datasets through the Configure DSS Datasets dialog. The dialog is opened by

- Right selecting a DSS dataset in the Database DMI editor dialog and selecting the **Set Multiple DSS Dataset Files** menu item, or
- Selecting a DSS dataset in the database DMI and then using the **Datasets**, then **Set Multiple DSS Dataset Files** menu.

The Configure DSS Datasets dialog appears.



Configure DSS Datasets

Dataset	DSS File
Demands	C:/RiverWare/Demands.DSS
DSS Accounting	C:/temp/Forecast.dss
DSS Historical	C:/temp/Forecast.dss
DSS Operations	C:/temp/Forecast.dss
Inflows	R:/Inflows.DSS

Configuration

DSS File: C:/temp/Forecast.dss

Apply To All Apply To Selected Close

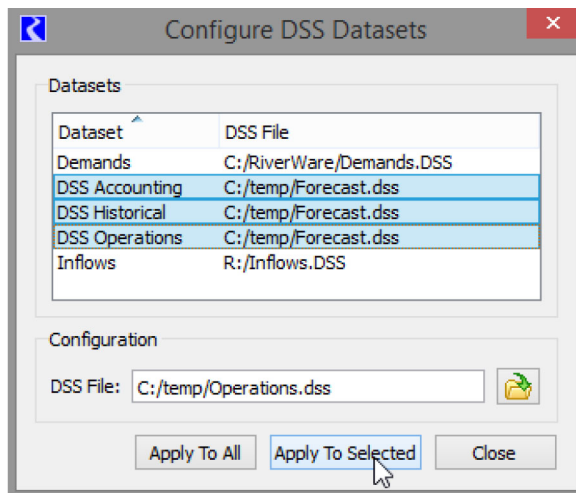
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The top portion of the dialog shows all of the DSS Datasets in the model and their DSS File in sortable columns. The bottom portion of the dialog shows the file path for the dataset that was selected when the dialog opened.

Specify an alternative DSS file if desired. Then:

- Select one or more datasets to modify and choose **Apply to Selected** as shown below.
- Select **Apply to All** to change every dataset to the new file path.



If a DSS dataset has been modified in another dialog, the DSS file cannot be set. You are notified with the following message: “The following datasets cannot be modified because they are being modified in another dialog”. Apply the changes to that dialog and you can now use the Configure DSS Datasets dialog again.

DSS DMI User Parameters

For DSS, you can specify values to be used for the user parameters. The parameters must be previously configured using the Parameter Dialog; see “[DMI Parameters](#)” on page 17 for details.

Edit a parameter value by double-clicking the Value cell to activate a control appropriate for the parameter’s type. To use the parameters in the DSS part information, specify the value with the percent sign; use the following syntax:

%Parameter%

Thus if you have a parameter named Run_Name with a text value and you wanted to put the value in part F, you would double-click part F on the desired Dataset row and type %Run_Name%. In the lower portion, double-click the value column and specify the integer to use. The resolved information is shown for each slot that is a part of the dataset. [Figure 4.9](#) shows this example.

Tip: Parameters values apply for the entire DMI; there is currently no way to change them for a particular run in an MRM run. DSS DMIs are not compatible with MRM.

Figure 4.9

t	E Part	F Part	Begin
		%Run_Name%	
			Start Timestep - 1 T
			Start Timestep
1DAY	DNR99X		21 Nov 1996 24:00
1DAY	DNR99X		21 Nov 1996 24:00

User Parameters		
Name	Type	Value
Run_Name	Text	DNR99X
Year	Integer Range	1965
HydYearType	List	Low

Examples

Now that we have described the overall approach for creating a Database DMI, we will present a few example use cases of creating various types of Database DMIs.

Example 1: DSS DMI Using the Database DMI Editor

This example shows how to create a Database DMI to/from DSS using the Database DMI Editor. We assume there are no other Datasets or Name Maps already defined. You would take the following steps:

1. From the workspace, open the DMI Manager using the **DMI**, then **DMI Manager**.
2. Create a new Database DMI using the **DMI**, then **New Database DMI** menu.
3. Open the Database DMI Editor using the **Edit**, then **Edit** menu.
4. In the upper portion of the dialog, enter a name, whether it is an Input or Output DMI, whether to confirm warnings, and whether to record invocations; see [“Name and Type”](#) on page 71. Let’s assume we are creating an Output DMI. Leave everything else unchecked.
5. First, add a dataset using the **Datasets**, then **New DSS Dataset** menu.
6. Expand the Dataset column and double-click the Dataset0 name and edit it.
7. Next, we will select the slots that we wish to output, say all Reservoir.Storages and Reservoir.Outflows. In the middle portion of the window, right-click the dataset you added, then use the **New Slot Selection**. A row is added to the tree view below the dataset.
8. Double-click the word NONE to open the slot selector. Use the selector to choose Reservoir.Storage. Add another slot selection and select Reservoir.Outflow. See [“Creating a Name Map of All Reservoir.Inflow Slots”](#) in *User Interface* for an example of a similar selection.

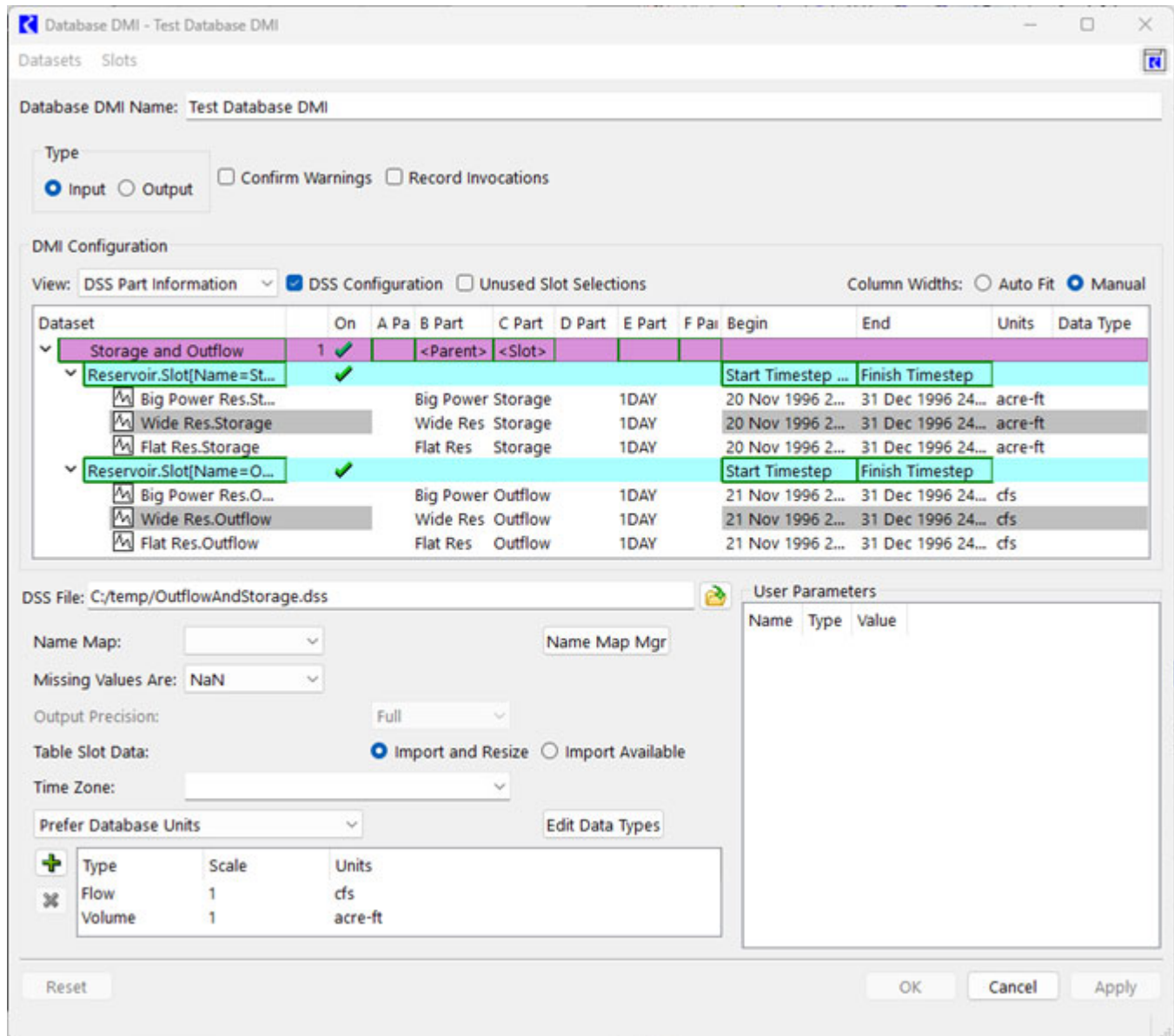
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9. For the Reservoir.Outflow row, specify the timestep by right-clicking once in the **Begin** field. Choose “Start timestep”. In the End column right-click once and choose “Finish timestep”. For the storage row, repeat and choose “Start Timestep - N” to “Finish Timestep”. Double-click the “N” in the Begin column and type “1” to specify 1 timestep.
10. On the dataset row, right-click in the B Part and select Workspace Entity. Choose <Parent>.
11. On the dataset row, right-click in the C Part and select Workspace Entity. Choose <Slot>.
12. Select the treeview symbol next to the Reservoir.Outflow row to see the exact slots you are going to output and all the part information, timestep, and units. We could change any of the part information by selecting the appropriate column on the Dataset row. For example, we could enter OBS for observed in part F by selecting that field on the dataset row and typing it.
13. Now configure the dataset. Select the dataset row. In the middle of the dialog, select the **DSS Configuration** toggle.
14. Specify the name of a **DSS File** in which to send the output either by typing it or using the selector. On an Input DMI, this file must already exist. On the Output DMI, it will be created if it does not exist.
15. If we have already configured a **Name Map**, we could select that we wanted to use it in the middle portion of the dialog; see “[DSS Dataset Configuration](#)” on page 80. A name map is created outside of this dialog; see “[Name Mapping](#)” on page 154. Because this is an Output DMI, we do not necessarily have to have a Name Map.
16. Now let’s specify the units to write. In the middle section, select the “+” button. Repeat to add two rows. See “[DSS Dataset Configuration](#)” on page 80 for details.
17. Double-click the word NONE under the type column to activate the pull-down menu. Choose **Volume**. On the Units column repeat and choose the desired unit, say **Acre-feet**. On the second row, Choose **Flow** and **cfs** as the units.
18. The Database DMI is now fully configured. Select **OK** to close the dialog and apply the changes. Invoke the DMI from the DMI manager using the **DMI**, then **Invoke** menu.

[Figure 4.10](#) shows the configuration for this example.

Figure 4.10 Screenshot of example DSS DMI



Example 2: Excel DMI to Export Data Using Headers

This example shows how to create a Database DMI to export data to Excel using the Header approach. We assume there are no other Datasets or Name Maps already defined. You would take the following steps:

1. From the workspace, open the DMI Manager using the **DMI**, then **DMI Manager**.
2. Choose **Utilities**, then **Datasets**
3. Choose **Dataset**, then **New Excel Dataset**
4. Double-click the newly created dataset to open it.

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5. You will create an output DMI so specify a name in the **Name:** field. Say “Excel Header PE Sheet”.
6. Select the **Excel** tab and type in or navigate to a worksheet. Because it is an output DMI and you don’t have an existing sheet, you will have to type in a file name in the selector.
7. Now you will need to choose which approach you wish to use: Ranges or Headers. If you choose Ranges, you will need to create Name Maps. See [“Slot Mapping—Spreadsheet Ranges Approach”](#) on page 125 for similar samples, based on the configuration. Let’s use the header approach (see [“Slot Mapping—Header Text and Sheet Names Approach”](#) on page 129). Select the **Map with Header Text and Sheet Names** option.
8. You are exporting series slot data, so select the Series Slots option.
9. Specify the Header orientation and specify a sheet name, for example PE. In this sample, use the Timestep, Slots, Runs option.
10. Optionally, specify to **Use Unit Name with the Slot Name**. [Figure 4.11](#) shows the setup for the Excel tab. With this setup, you do not have to use Name Maps, but you could. Thus, we have a fully configured dataset.

Figure 4.11 Screenshot of the Excel Dataset

Excel Dataset - Excel Header PE Sheet

File

Type: Excel

Name: Excel Header PE Sheet

General Excel

Workbook: C:/Temp/PEOutput.xlsx

Slot Mapping

☐ Map by Spreadsheet Ranges

☒ Map with Header Text and Sheet Names

Header Specification

☒ Series Slots ☐ Table / Periodic / Scalar Slots

☐ Specify Header Offset from Upper Left Spreadsheet Corner

Rows: 0 Columns: 0

Header Orientation

Rows	Columns	Sheets
☒ Timesteps	☐ Slots	☐ Run(s)
☐ Timesteps	☐ Run(s)	☐ Slots
☐ Slots	☐ Timesteps	☐ Run(s)
☐ Run(s)	☐ Timesteps	☐ Slots

Run Name Type

☒ Single Run Name: PE

☐ Append Suffix to Run Name 0

☐ Specify Single Run Name when DMI is invoked

☐ "Run" with MRM run number (i.e. Run0, Run1, etc.)

☐ "Trace" with MRM trace number (i.e. Trace5, Trace6, etc.)

☐ Use Begin Timestep Time Instead of End of Timestep

☒ Clear Dates and Values for Output

☒ Use Unit Name with Slot Name

☐ Open and Save Workbook More Information...

OK Apply Reset Cancel

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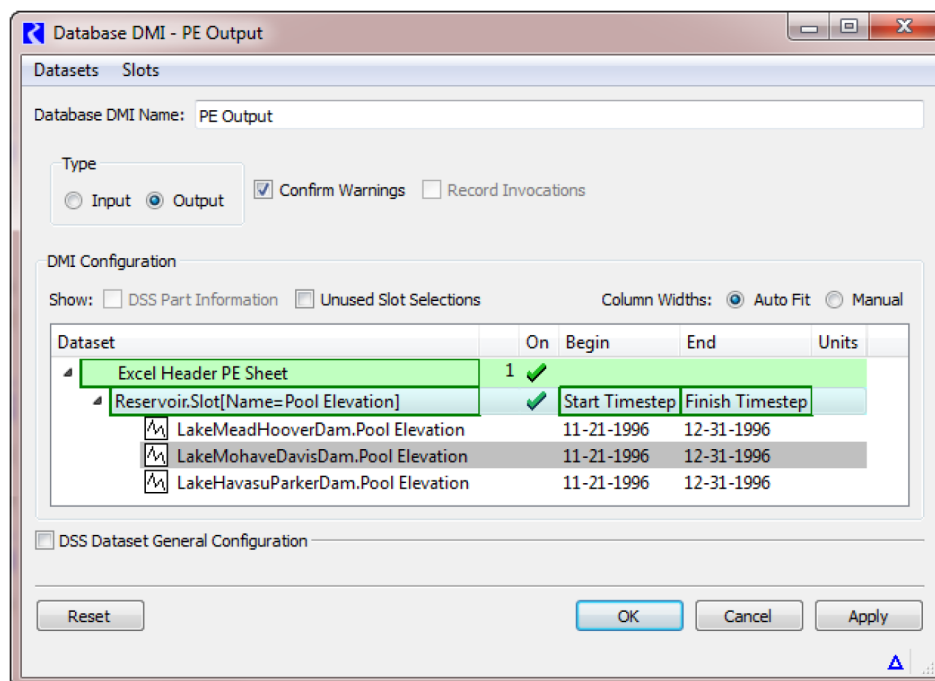
Database DMI

11. Select **OK** to accept the changes and close the window.
12. Back in the DMI Manager, create a new Database DMI using the **DMI**, then **New Database DMI** menu.
13. Open the Database DMI Editor using the **Edit**, then **Edit** menu.
14. In the upper portion of the dialog, enter a name, that it will be an Output DMI, and whether to confirm warnings. See [“Name and Type”](#) on page 71 for details.
15. Now add the dataset you configured earlier. Choose **Datasets**, then **Add Existing** to open the Dataset Selector dialog.
16. Choose the “Excel Header PE Sheet” dataset and select **OK**.
17. Next, we will select the slots that we wish to output, say all Reservoir Pool Elevations. With the Excel Header PE Sheet row highlighted, use the **Datasets**, then **New Slot Selection** menu.
18. Double-click the word NONE to open the slot selector. Use the selector to choose all Reservoir.Pool Elevations. See [“Creating a Name Map of All Reservoir.Inflow Slots”](#) in *User Interface* for an example of a similar selection.
19. For the Reservoir.Pool Elevation row, specify the begin timestep by right-clicking once in the **Begin** field. Choose “Start timestep”. In the **End** column right-click once and choose “Finish timestep”.

[Figure 4.12](#) shows this setup.

This database DMI is now fully configured. You can execute it from the DMI manager.

Figure 4.12



[Figure 4.13](#) shows the resulting spreadsheet for our three-reservoir model.

Figure 4.13

	A	B	C	D
1		Big Power Res.Pool Elevation (ft)	Wide Res.Pool Elevation (ft)	Flat Res.Pool Elevation (ft)
2	11/21/1996	1192.208238	635.1760648	NaN
3	11/22/1996	1192.241859	635.1318182	NaN
4	11/23/1996	1192.302769	634.934238	NaN
5	11/24/1996	1192.335546	634.9504454	NaN
6	11/25/1996	1192.365509	635.0791799	NaN
7	11/26/1996	1192.395472	635.2073427	NaN
8	11/27/1996	1192.425435	635.3355054	NaN
9	11/28/1996	1192.487753	635.2882072	NaN
10	11/29/1996	1192.517716	635.41637	NaN
11	11/30/1996	1192.570186	635.422473	NaN
12	12/1/1996	1192.629831	635.3347425	NaN
13	12/2/1996	1192.631801	635.4430706	NaN
14	12/3/1996	1192.674705	635.5513986	NaN
15	12/4/1996	1192.71761	635.6597266	NaN
16	12/5/1996	1192.760515	635.7680547	NaN
17	12/6/1996	1192.80342	635.8763827	NaN
18	12/7/1996	1192.846325	635.9847107	NaN
19	12/8/1996	1192.88923	636.0923286	NaN
20	12/9/1996	1192.932135	636.1998297	NaN

Example 3: Excel DMI to Import Data Using Ranges

This example shows how to create a Database DMI to import data from Excel using the Ranges approach. We assume there are no other Datasets or Name Maps already defined. You would take the following steps:

1. [Figure 4.14](#) is screenshot of the desired data in Excel. It is not in the format necessary for the Headers approach. So we will use the Ranges approach. We would like to import the Mountain Storage IN column to the slot Mountain Storage.Inflow for just the run range (Jan 28-Feb 6th). That is, we want to import the range G14:G23.

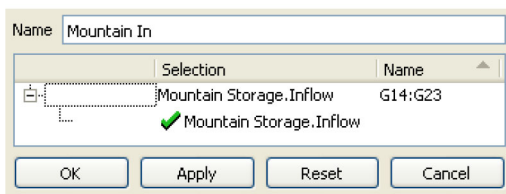
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Figure 4.14

	C	D	E	F	G	H	I
5							
6		MOUNTAIN STORAGE				BIG CAN	
7		PE	STOR	EVAP	IN	PE	STOR
8	Date	ft	1000 AF	in/d	1000 cfs	ft	1000 AF
9	23-Jan	784.5	85000	0.02	126.5	642.1	1693.6
10	24-Jan	784.6	87513	0.02	185.5	642.1	1693.6
11	25-Jan	784.8	88152	0.02	199.5	642.1	1693.6
12	26-Jan	784.9	88810	0.02	265.5	642.1	1693.6
13	27-Jan	785	89032	0.02	352.9	642.1	1693.6
14	28-Jan			0.02	378.1		
15	29-Jan			0.02	504.2		
16	30-Jan			0.02	2520.8		
17	31-Jan			0.02	2016.7		
18	1-Feb			0.05	1890.6		
19	2-Feb			0.05	1638.5		
20	3-Feb			0.05	1260.4		
21	4-Feb			0.05	1008.3		
22	5-Feb			0.05	756.2		
23	6-Feb			0.05	630.2		
24	7-Feb			0.05	513.1		
25	8-Feb			0.05	399.4		
26	9-Feb			0.05	316.5		
27	10-Feb			0.05	260.2		
28	11-Feb			0.05	200.4		
29							

- From the workspace, open the DMI Manager using the **DMI**, then **DMI Manager**.
- From the DMI Manager choose **Utilities**, then **Name Maps**.
- Choose **Name Map**, then **New** or select the + button.
- Double-click the newly created Name Map.
- Change the name to “Mountain In”
- Select **Name Map**, then **New** to add a row to the list.
- Click-pause-click on the word **NONE** and choose **Select Slot**.
- Using the selector, choose the reservoir slot **Mountain Storage.Inflow**. Select **OK**.
- In the Name Map, in the **Name** column type “G14:G23” (without quotes).



- Select **OK** in the Name Map and **OK/Cancel** in the Name Map Manager to close.
- In the DMI Manager, choose **Utilities**, then **Dataset**.
- Choose **Dataset**, then **New Excel Dataset**.
- Double-click the newly created dataset to open it.

15. You will create an input DMI so specify a name in the **Name:** field: “Input Mountain Inflow”.
16. In the Name Map section, choose the Name Map you created “Mountain In” from the pull down menu.
17. Select **Units**, then **New**
18. Under the **Type** column, click-pause-click and choose **Flow** from the pull down menu.
19. Change the **Scale** to 1000 and the **Units** to cfs.
20. Select **Apply**.
21. Select the **Excel** tab and type in or navigate to the worksheet where the data resides. In the sample, it is in C:/Temp/MountainStorageData.xls.
22. Select the **Map by Spreadsheet Ranges** button.
23. Choose the **Map to Excel Range Specification with Name Map**.
24. The data is all on the same worksheet and we didn’t enter it in the Name Map, so you will need to enter it here. In the **Worksheet** field, enter the name of the sheet, “ReservoirData”.
25. The dataset is fully configured, as shown in [Figure 4.15](#). Select **OK** to apply and close the dialog.

Figure 4.15 Excel Dataset Screenshot

Excel Dataset - Input Moutain Inflow

File

Type: Excel

Name: Input Moutain Inflow

General Excel

Workbook: C:/Temp/MountainStorageData.xlsx

Slot Mapping

- ☒ Map by Spreadsheet Ranges
- ☐ Map with Header Text and Sheet Names

Range Specification

- ☐ Map to Excel Named Ranges with Name Map
- ☒ Map to Excel Range Specifications with Name Map
 - ☒ Ranges on same worksheet
 - Worksheet: ReservoirData
 - (Name Map specification of form B2:B50)
 - ☐ Ranges on different worksheets
 - (Name Map specification of form Sheet1!B2:B50)

☐ Specify Range Offset for each Run of Multiple Run

Rows: 0 Columns: 0

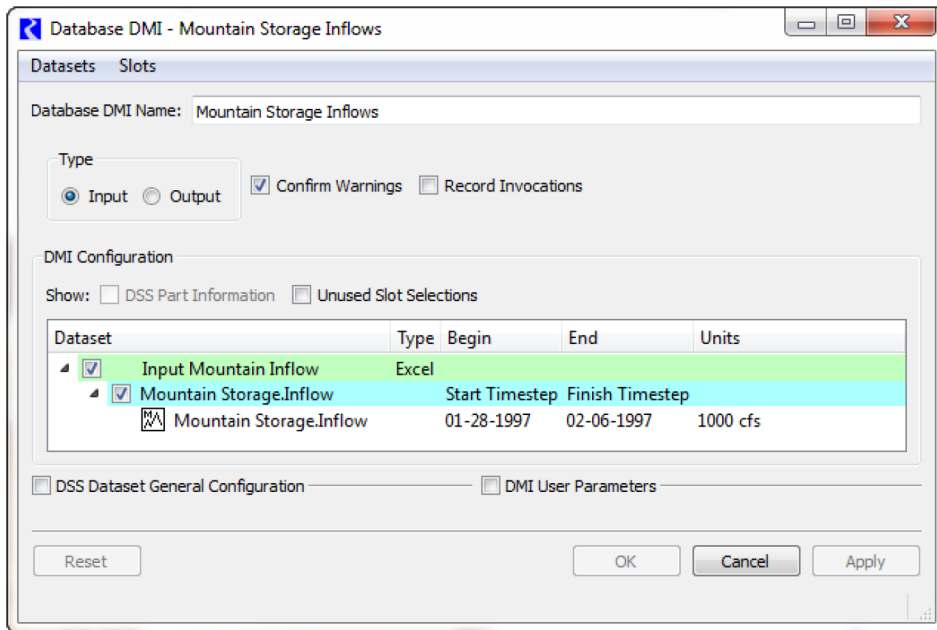
☐ Open and Save Workbook [More Information...](#)

OK Apply Reset Cancel

26. Back in the DMI Manager, create a new Database DMI using the **DMI**, then **New Database DMI** menu.

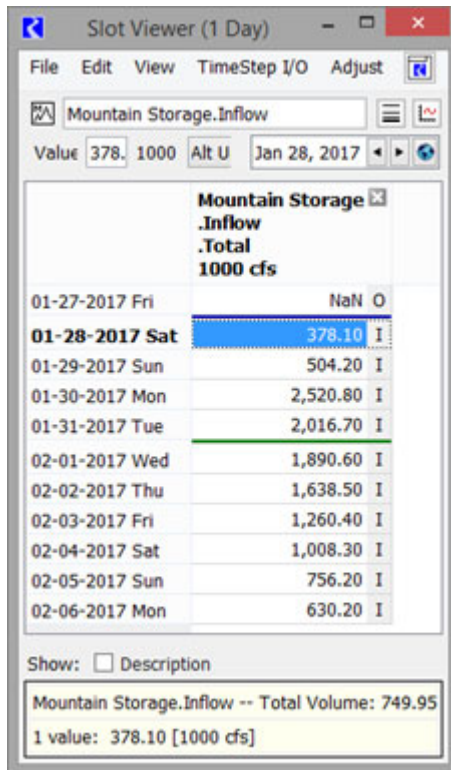
27. Open the Database DMI Editor using the **Edit**, then **Edit** menu.

28. In the upper portion of the dialog, enter a name and that it will be an **Input DMI**. See “[Name and Type](#)” on page 71 for details.
29. Now add the dataset you configured earlier. Choose **Datasets**, then **Add Existing** to open the Dataset Selector dialog.
30. Choose the **Input Mountain Inflow** dataset and select **OK**.
31. We will select the slots that we wish to input. With the Input Mountain Inflow row highlighted, select **Datasets**, then **New Slot Selection** menu.
32. Double-click on the word **NONE** to open the slot selector. Use the selector to choose Mountain Storage.Inflow.
33. For the Mountain Storage.Inflow row, specify the timestep by right-clicking once in the **Begin** field. Choose “Start timestep”. In the **End** column right-click once and choose “Finish timestep”.



34. This database DMI is now fully configured. You can execute it from the DMI manager. [Figure 4.16](#) shows the resulting slot.

Figure 4.16

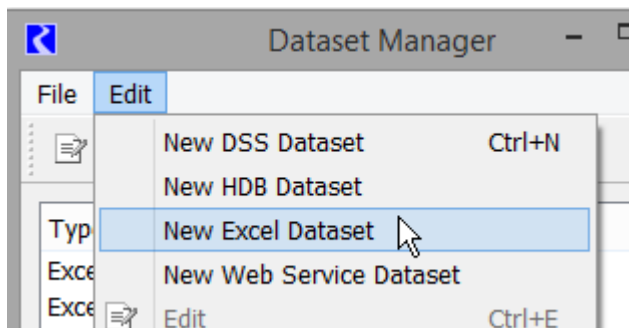


Datasets Overview

A dataset encapsulates all RiverWare knowledge about the database. In general, there are two types of knowledge contained in a dataset, configuration information and the database specification.

In RiverWare, datasets are defined and configured in the Dataset Manager as shown in [Figure 4.17](#). The Dataset Manager is accessed from the **Utilities**, then **Dataset** menu on the DMI Manager. Selecting **Edit**, then **New Excel Dataset**, for example, adds a new dataset of that type.

Figure 4.17 Screenshot of the Dataset Manager Edit menu



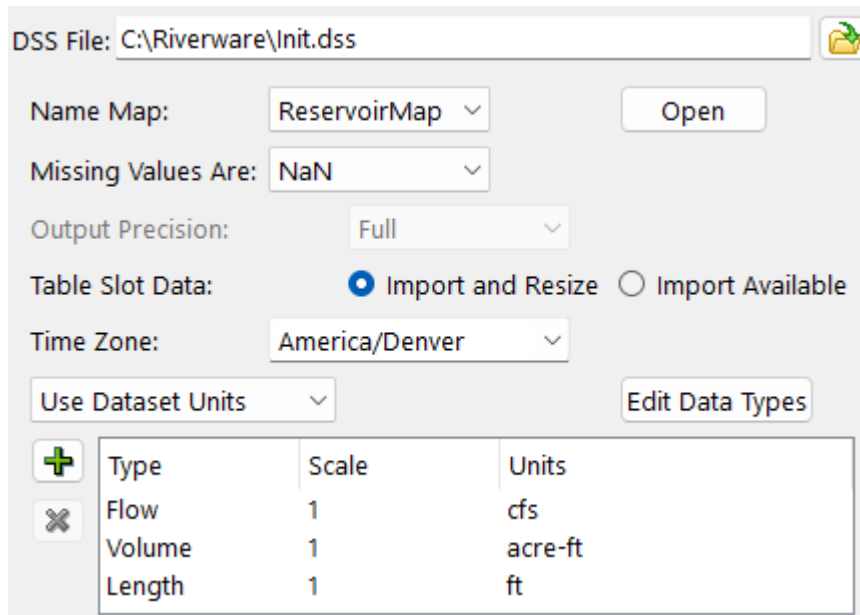
DSS Datasets

In general, all of the DSS dataset configuration can (and should) be performed from the Database DMI Editor. The DSS Dataset editor shows the results of configuration but the Database DMI has much more functionality. This section describes the functionality of the Database DMI Editor with respect to DSS datasets.

Note: A DSS dataset must be selected for this area to become active.

DSS File

Specify the DSS file either by typing in the file path or selecting it using the open file icon . If not already created, an export DMI will create the specified DSS file.



Type	Scale	Units
Flow	1	cfs
Volume	1	acre-ft
Length	1	ft

The DSS File name may contain user parameters as %ParamName% (the same as for the DSS part information). The DSS file name may contain multiple parameters but the parameters must be typed in; they are not available from a context menu. See [“DSS DMI User Parameters”](#) on page 82 for details.

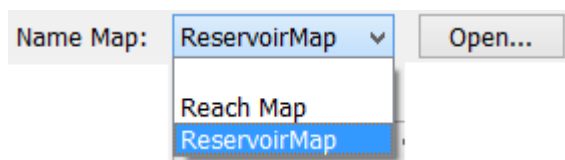
The DSS file name text shows the user parameters while a tooltip shows the substituted values. The user parameters can be all or part of the DSS file name and can be used in conjunction with environment variables, for example:

- A single parameter: %DssFile%
- A path and a parameter: C:\model\data\%DssFile%
- Multiple parameters: %ModelDir%\data\%DssFile% or %DataDir%\%DssFile%.dss
- Environment variable and a parameter: \$(DATA_DIR)\%DssFile%.dss

Note: You can set the file path for multiple DSS Datasets in one action using the Configure DSS Datasets dialog; see [“Setting Multiple DSS Dataset Files”](#) on page 81 for details.

Name Map

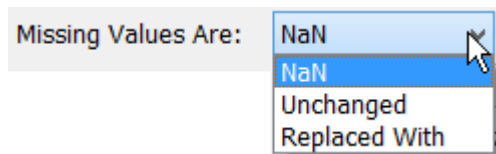
Optionally, select the desired Name Map from the **Name Map** menu. The selections are based on the Name Maps configured in the Name Map Manager. See [“Name Mapping”](#) on page 154 for details on Name Maps. If you have no name map selected, select the **Name Map Mgr** button to open the Name Map Manager. If you do have a Name Map selected, the button says **Open** to open that Name Map.



Missing Values

Specify how missing values are represented in the database. The choices are

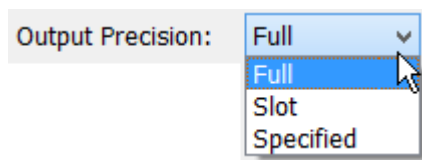
- **NaN.** RiverWare default, missing values are NaN in the database
- **Unchanged.** The DMI will not change missing values on import, uses NaNs on export
- **Replace With.** Provides a user input value that is substituted on import and export.



Output Precision

For Output DMIs, specify the precision of values that are sent to the database, that is, the number of digits to the right of the decimal point. The options are:

- **Full.** Output all available digits of precision. This is the default.
- **Slot.** Use the slot's display precision from the active unit scheme. Note, changing the active unit scheme could affect values sent out via the DMI.
- **Specified.** Use the specified precision.



Values are converted to the specified units, as described in [“Units”](#) on page 98, before rounding.

The rounding algorithm is: $\text{round}(\text{value} * 10^N) / 10^N$ where value is after unit conversion, N is the number of digits of precision, and round() is a standard function which rounds its argument to the nearest integer value, rounding halfway cases away from zero.

Caution: With this approach you could potentially see misleading results. For example, if a slot shows 1.23cfs, you export it using specified dataset units of cms, it becomes 0.0348cms. If you then reduce the precision to 2 digits, it will send 0.03cms as the output. If you then show the value in your database viewer using cfs and 2 digits of precision, it shows it as 1.06cfs. The value in the database does not match the value in RiverWare due to conversions and rounding. Be careful!

Also, to be clear, all values exported from a Database DMI are floating point numbers regardless of precision. For example, an expression slot might compute a value of 7538.21. Exported with 0 digits of precision, this would appear to be 7538, but technically, it was exported as a floating point number, 7537.9999999974 to 15 significant digits. This fact is really only important if you database displays 15-17 digits.

Table Slot Data

The user specifies how paired data should be imported to table slots in terms of overwriting existing data or leaving extra data unchanged.

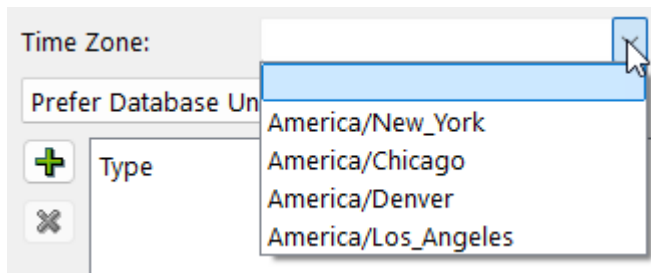
This option is only enabled for Input DMIs. The options are:

- **Import and Resize** - a DSS DMI will import paired data by resizing the table to match the imported table. This is the default option. For example if the table slot currently has 20 rows, but the DSS data only contains 14 rows, then after import, the table slot will now have 14 rows. The 6 rows at the bottom of the table slot will be removed.
- **Import Available** - a DSS DMI will expand a table to accommodate the imported table but it won't contract a table. This leaves any extra data in the table without change. For example if the table slot currently has 20 rows, but the DSS data only contains 14 rows, then 6 rows at the bottom of the table slot will remain unchanged. Only the first 14 rows will have the newly imported data.

Table Slot Data: ☒ Import and Resize ☐ Import Available

Time Zone

Specify the time zone for the data that will be imported.



Select one of the common American time zones or type in the IANA canonical or linked time zone for any time zone. See <https://www.iana.org/time-zones>.

Note: The time zone is currently only used for pseudo-regular timeseries. Pseudo-regular data is typically stored in the database in UTC but represents a measurement taken at the same local time each day, regardless of whether daylight saving time is in effect. For example a value might be stored

Chapter 4
Database DMI

on Oct 02, 2023 at 05:00 UTC but represents Oct 01, 2023, 24:00 local time. If you are not importing this type of data, the Time Zone field can remain blank.

Pseudo-regular time series are identified by a "~1Day" in the DSS E-Part. Only the duration of 1Day is supported; other timestep sizes will result in an error.

Also note that the DSS DMI only supports reading pseudo-regular time series. Writing pseudo-regular data results in an error.

Units

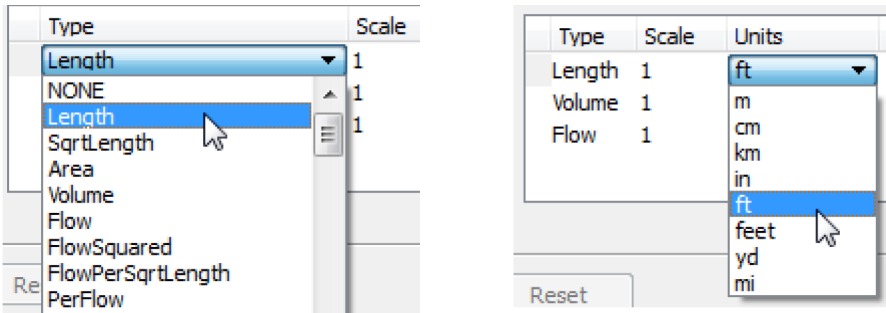
The units that are used in the DSS database are specified in the Dataset dialog. A pull-down menu allows the user to specify that the Database DMI should **Use Database Units**, **Prefer Database Units**, or **Use Dataset Units**. [Figure 4.18](#) shows an example.

- **Use Database Units.** The database contains all units (This option is not applicable to DSS because units are not required to be defined in DSS files, so can't be fully depended on)
- **Prefer Database Units.** Use the units in the database, if missing, use the specified Units
- **Use Dataset Units.** No units are specified in the database, use the specified units.

If you chooses either of the second two options, you must specify the units in the window below the pull-down menu. Select the **+** icon to add a new unit type:

This adds an entry to the list. Double-click the word **NONE** under the **Type** to change the type. [Figure 4.18](#) illustrates. Change the **Scale** and **Units** in a similar manner. These units are used to specify the units that will be written to the database if an Output DMI is used and the units that the database contains if an Input DMI is used.

Figure 4.18



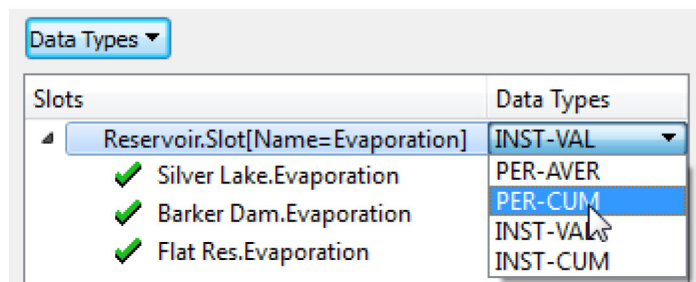
[Figure 4.19](#) shows an example of specified units.

Figure 4.19

Type	Scale	Units
Length	1	ft
Volume	1	acre-ft
Flow	1	cfs

Data Types

Select the **Edit Data Types** button to edit the data types. In the resulting dialog, you can specify that a selection of slots should have a certain data type. The DSS data types are: PER-VAL, PER-AVER, INST-VAL, INST-CUM. The selection is similar to other slot selections as it is dynamic and can make full use of wildcarding.



HDB Datasets

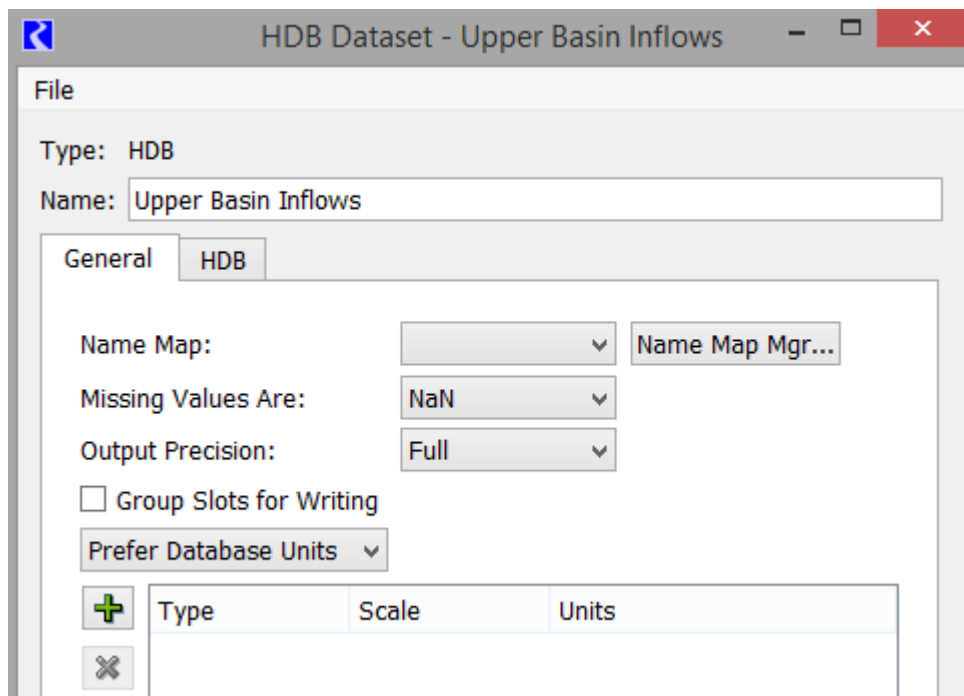
The HDB Dataset described in this section allows direct connections to and from an HDB database.

Note: To import from the HDB web server, use a Web Server dataset and then select the HDB web server. See [“HDB Web Service”](#) on page 146 for more information.

Double-clicking an HDB dataset in the Dataset Manager opens the HDB Dataset editor. In this dialog, the dataset can be renamed and configured. It has two tabs, one for general configuration and one specific to the HDB implementation.

General Tab

This section describes the general tab for the HDB dataset dialog box.



Name Map

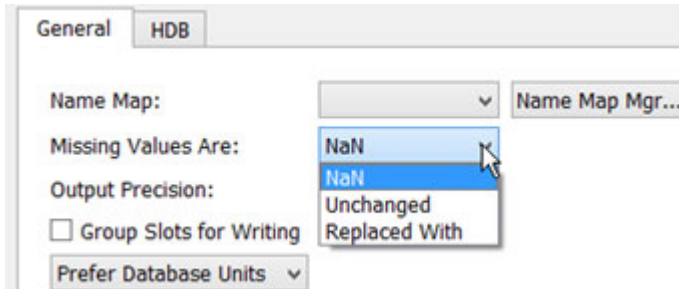
Mappings of RiverWare names to HDB IDs are contained in database tables, so Name Maps are not needed for configuring a dataset for HDB. Specifying which database mapping to use is described later in the HDB tab of the configuration dialog.

Missing Values

You specify how missing values are handled when exchanging data with HDB. In general, a NaN in RiverWare is not represented as a data record in HDB. The current choices are as follows:

- **NaN.** On export, no data record is created in HDB for NaNs in RiverWare, and any corresponding existing data record in the real tables is deleted (there would be no existing data record in the model tables). On import, missing values in HDB are made NaNs in RiverWare.
- **Unchanged.** On export, any existing data in the HDB real tables is left unchanged for a corresponding NaN value (there would be no existing data record in the model tables). On import, RiverWare data is left unchanged for a missing value in HDB.

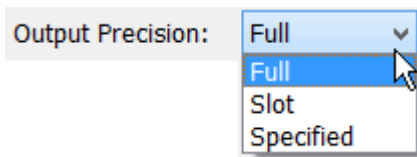
- **Replaced With.** User provides an input value that is substituted for NaNs on export or missing values on import.



Output Precision

For Output DMIs, specify the precision of values that are sent to the database, that is, the number of digits to the right of the decimal point. The options are:

- **Full.** Output all available digits of precision. This is the default.
- **Slot.** Use the slot's display precision from the active unit scheme. Note, changing the active unit scheme could affect values sent out via the DMI.
- **Specified.** Use the specified precision.



Values are converted to the specified units, as described in “Units” on page 102, before rounding.

The rounding algorithm is: $\text{round}(\text{value} * 10^N) / 10^N$ where value is after unit conversion, N is the number of digits of precision, and round() is a standard function which rounds its argument to the nearest integer value, rounding halfway cases away from zero.

Caution: With this approach you could potentially see misleading results. For example, if a slot shows 1.23cfs, you export it using specified dataset units of cms, it becomes 0.0348cms. If you then reduce the precision to 2 digits, it will send 0.03cms as the output. If you then show the value in your database viewer using cfs and 2 digits of precision, it shows it as 1.06cfs. The value in the database does not match the value in RiverWare due to conversions and rounding. Be careful!

Also, to be clear, all values exported from a Database DMI are floating point numbers regardless of precision. For example, an expression slot might compute a value of 7538.21. Exported with 0 digits of precision, this would appear to be 7538, but technically, it was exported as a floating point number, 7537.9999999974 to 15 significant digits. This fact is really only important if your database displays 15-17 digits.

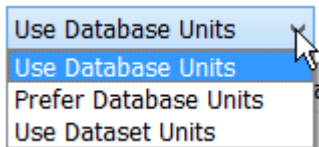
Group Slots for Writing

The **Group Slots for Writing** check box configures the DMI to group slots together when writing to the database. This option is particularly useful when you have many slots but few values in those slots. Grouping the slots minimizes the network traffic. This can significantly improve performance over a network.

Units

A menu in the Dataset dialog allows you to specify how units should be handled for a dataset. The choices are as follows:

- **Use Database Units**
- **Prefer Database Units**
- **Use Dataset Units**



Units are defined in HDB for each piece of data, so **Use Database Units** is always applicable. If the unit type of the data in HDB does not match the unit type of the slot in RiverWare and the units cannot be converted, the slot is skipped and a warning message is given. The exception is if the unit type of the slot in RiverWare is NOUNITS. Here the unit type matching between HDB and RiverWare is skipped and the data is allowed to be exchanged with no conversions being applied.

In addition, to improve performance you can configure slot units in the Dataset and Use Dataset Units. This prevents additional calls across the network to get the units from the database.

Configuration specific to the HDB database is specified in the **HDB** tab of the HDB Dataset editor.

HDB Tab

The screenshot shows a Windows-style dialog box titled "HDB Dataset - Dataset 2". It has two tabs: "General" and "HDB", with "HDB" currently selected. The "Type" is set to "HDB" and the "Name" is "Dataset 2". In the "HDB Database" field, "LCTEST" is entered. Below this is a "Mapping ID" field with an empty text box and a "Select" button. A checkbox labeled "Use water year tables in HDB for yearly timestep data" is unchecked. The "HDB Table Type" section contains three radio buttons: "Model" (which is selected), "Real", and "Ensemble". The "Model Run ID" section has two radio buttons: "Select ID when DMI is invoked" and "Select ID:" (which is selected). Below "Select ID:" is another empty text box with a "Select" button. At the bottom of the dialog are four buttons: "OK", "Apply", "Reset", and "Cancel".

HDB Database

The **HDB Database** field specifies which database the dataset will interact with. The entry in this field should be a TNS name or connect string that will allow connection through Oracle Networking Services (SQL*Net) to the desired database. You need to have Oracle networking available on your computer through Oracle Client or a full Oracle database installation.

Note: The HDB Database DMI connectivity requires either Oracle Client 19 or Oracle Instant Client 19. CADSWES tested with Oracle Client 19.3 and Oracle Instant Client 19.15.

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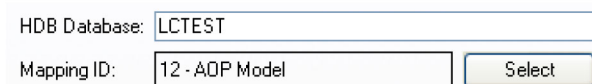
The first time in a RiverWare session that a specified database is connected to, you are prompted to provide a user name and password to log in. For a user to retrieve metadata or read and write data to HDB, they must be granted the roles `model_priv_role` and `app_role` in the database to have the necessary permissions.



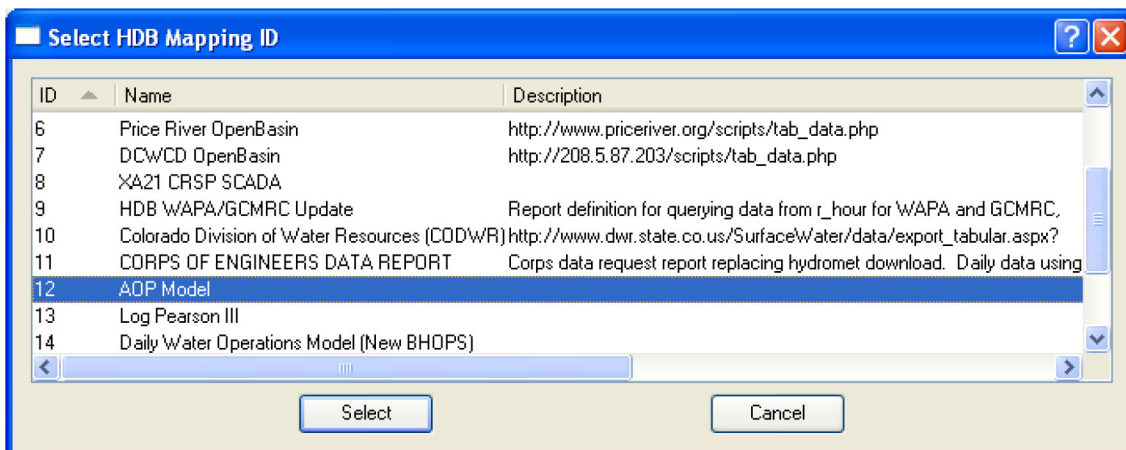
To run in batch mode, the login information must be provided in a file. The code looks for a file named `dblogins` in your home directory, which is specified by the `HOME` environment variable on Solaris, and the combination of the automatically defined `HOMEDRIVE` and `HOME` environment variables on Windows. The format of the file should be a line for every database, with that line containing the database name, login, and password separated by whitespace (blanks or tabs). Blank lines and comment lines starting with `#` are permitted in the file. If you want to set up a different file, the `DBLOGINS` environment variable can be defined as a full path (directory and file) where the code will look for the login information.

Mapping ID

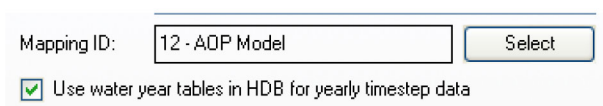
The **Mapping ID** field specifies what mapping will be used in the database to map RiverWare objects and slots to HDB sites and data types. You must select the adjacent **Select** button to pick a mapping from the database.



The **Select** button brings up the Select HDB Mapping ID dialog populated with all the available mapping IDs, their names, and descriptions from the `HDB_EXT_DATA_SOURCE` table in the database. You can then select a mapping, which populates the **Mapping ID** field on the HDB tab of the dataset dialog with the ID and the name of the mapping.



HDB maintains different tables for data of differing intervals (timesteps). Slots of varying timesteps can be freely intermixed in a list that is associated with a dataset, and the correct tables for each timestep will be automatically accessed in HDB. One case that does need specification is for the water year tables in HDB. There is no corresponding water year timestep in RiverWare, although some users have used slots with a 1 Year timestep to hold water year data in some of their RiverWare models. To handle this case, an option box on the dataset dialog to **Use water year tables in HDB for 1 Year timestep data** has been provided. If a user checks this box, all 1 Year timestep data read and written with the dataset will be moved to and from the water year tables in HDB instead of the year tables.



Mapping ID:

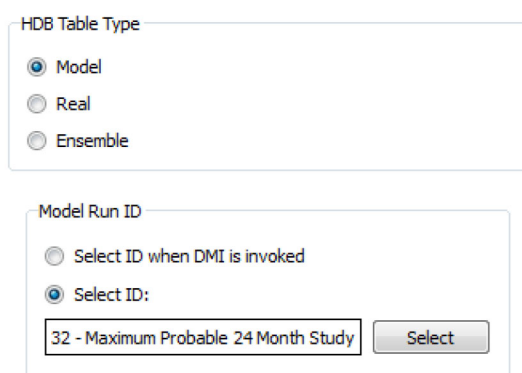
☒ Use water year tables in HDB for yearly timestep data

HDB Table Type

The HDB Table Type frame on the dataset dialog specifies which of the tables in HDB the dataset will interact with. The selection of **Model**, **Real**, or **Ensemble** determines the configuration options that become available in the rest of the dialog.

HDB Table Type—Model

A selection of **Model** in the HDB Table Type frame of the dataset dialog means that any data moved using the dataset will be read from or written to the model (m_ xxxx) set of tables in HDB. Model data is typically forecasted data, although there are no restrictions on its dates being future or historic. There can be multiple forecasts made for a piece of data at the same date and time, so data in the model tables must be associated with a model run ID that indicates from what run of a model the data originated. If data is written to a model run ID with a DMI, any previous data that was associated with that ID will be deleted from the model tables. Therefore, only the data from one particular run of a model is represented in a single model run ID. Reading or writing of model data requires metadata selection in the Model Run ID frame that appears with the selection of **Model** in the dataset dialog.



HDB Table Type

☒ Model
☐ Real
☐ Ensemble

Model Run ID

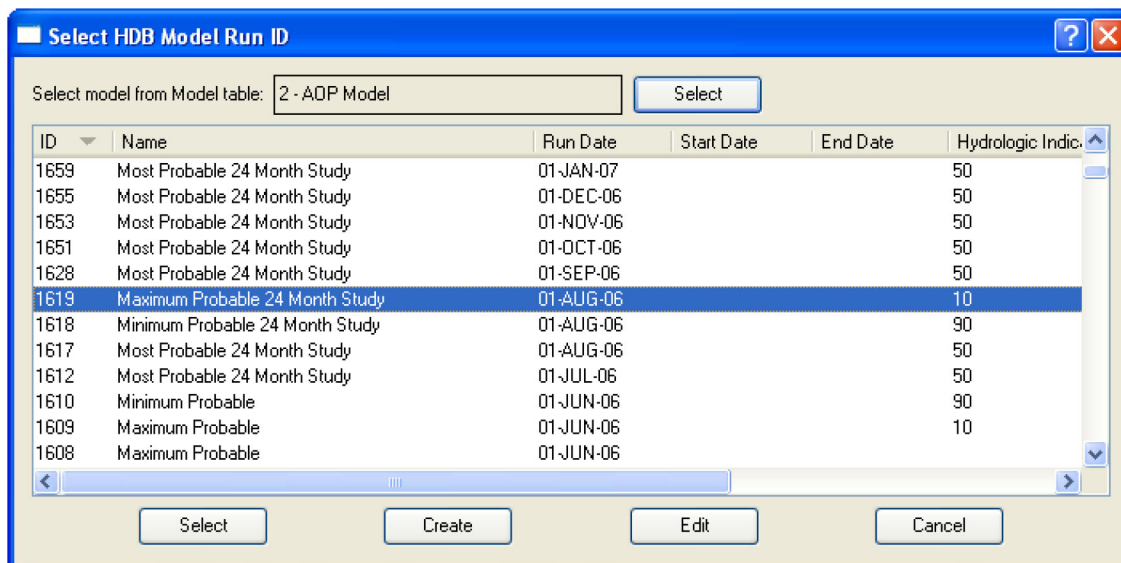
☐ Select ID when DMI is invoked
☒ Select ID:

The Model Run ID frame allows you to select either **Select ID** or **Select ID when DMI is invoked**. The **Select ID** option allows you to preselect a model run ID to associate with the dataset for reading and writing data with the

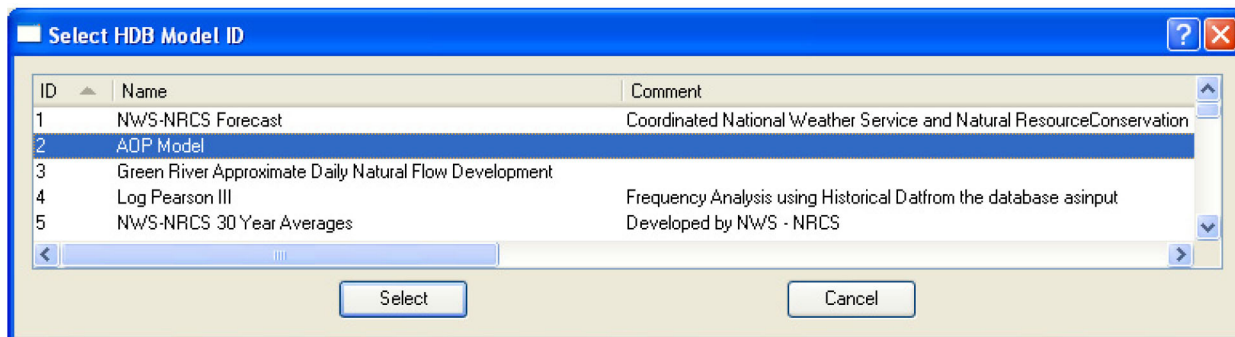
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model tables. To select a model_run_id, you must select the **Select** button adjacent to the model run ID field. This brings up the Select HDB Model Run ID dialog.



All model run IDs are associated with a particular model as defined in the database. The model run ID dialog is designed to display all the IDs in the database for a specified model. The model is selected via the **Select model from MODEL table** field at the top of the Model Run ID dialog. Selecting the **Select** button adjacent to the field brings up the Select HDB Model ID dialog populated with all the models defined in the HDB_MODEL table in the database.



You can then select a model, which populates the **Select model from MODEL table** field on the model run ID dialog with the model ID and name, as well as populating the model_run_id list in the dialog with information for all the model_run_ids defined for the model. You can then select an existing model run ID from the list, which closes the dialog and populates the model run ID field in the Model Run ID frame of the dataset dialog with the model run ID and name.

There are two other options for Model Run IDs on the Select HDB Model Run ID dialog. Instead of just selecting an existing ID, you can also edit an existing ID or create a new one. To edit or create a model run ID you still must have selected a model at the top of the dialog. An existing ID that is displayed can then be highlighted and edited by selecting the **Edit** button at the bottom. This brings up the Edit HDB Model Run ID dialog, which displays all

the information fields for the model_run_id and allows you to edit these fields. Selecting the **Save** button will then save the edited information for this model run ID to the database.

The **Create** button at the bottom of the Select HDB Model Run ID dialog allows you to create a model run ID that is associated with the model shown in the **Select model from MODEL table** field at the top of the dialog. Selecting **Create** brings up the Create HDB Model Run ID dialog, which contains fields where information can be entered for the new model run ID. The **Run Date** field is populated with the start time of the run, but can be changed by selecting any portion of the date and typing or using the spinners. At least the required fields marked with an asterisk must be filled out. Selecting the **Create** button will then save entered data to the database as a new model run ID where it will be assigned the next available model run ID number. The new ID will then be added to the list and highlighted back on the Select HDB Model Run ID dialog.

The Model Run ID frame of the dataset dialog has a second option for specifying model run IDs by selecting **Select ID when DMI is invoked**. When this option is selected, you do not preselect a model run ID that is always

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used with the dataset, but rather you are prompted for one when a DMI containing the dataset is executed. This option could be useful in a case where the DMI is run to record forecast data on a regular basis with the data always being preserved in separate model run IDs in the model tables. To limit the potential for you to accidentally overwrite the previously saved data by forgetting to change the ID; you must select the ID each time the DMI is run.

The image shows a software interface with two sections. The top section, titled 'HDB Table Type', contains three radio buttons: 'Model' (which is selected), 'Real', and 'Ensemble'. The bottom section, titled 'Model Run ID', contains two radio buttons: 'Select ID when DMI is invoked' (which is selected) and 'Select ID:'. Below the 'Select ID:' radio button is a text input field and a 'Select' button.

Selecting an ID under the **Select ID when DMI is invoked** option utilizes the same dialogs as does the preselect ID option. The Select HDB Model Run ID dialog appears where you can select a model and see its existing IDs. An existing ID can then be selected, an existing ID can be edited and then selected, or a new model run ID for the model can be created then selected.

Note: If the same dataset is used multiple times in a DMI, you are only prompted once to associate a model run ID with the dataset; the same dataset cannot be used with multiple model run IDs in a single DMI or DMI group execution.

HDB Table Type—Real

A selection of **Real** in the Table Type frame means that any data moved using the dataset will be read from or written to the real (r_ xxxx) set of tables in HDB. Real data are typically current or historical observed values;

future data cannot be loaded into the real tables. A write of real data to HDB requires that some other metadata be selected by you in the Real frame that appears with the selection of a **Real** table type in the Dataset dialog.

The screenshot shows the 'HDB Dataset - Dataset 2' dialog box. The 'HDB' tab is selected. The 'Name' field is 'Dataset 2'. The 'HDB Database' field is empty. The 'Mapping ID' field is empty with a 'Select' button. There is a checkbox 'Use water year tables in HDB for yearly timestep data'. The 'HDB Table Type' section has three radio buttons: 'Model', 'Real' (which is selected), and 'Ensemble'. Below this is a checkbox 'Read and display validation character'. Under this checkbox is a section titled 'Required for Writing to Real Tables' which contains 'Agency ID' (with a 'Select' button), 'Collection Sys ID' (with a 'Select' button'), and a checkbox 'Use Overwrite Flag'.

Validation Characters

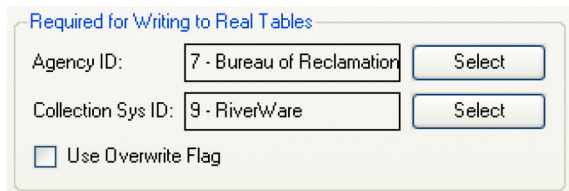
HDB has a per-value Validation field in the R_DAY, R_HOUR and R_INSTANT tables which typically indicates whether the data is provisional or approved. The field is a 1 byte character which is usually a “P” for provisional and an “A” for approved.

The **Read and display validation character** option allows you to import validation characters from HDB into Notes on Series Slot, which are then shown on slot and SCT dialogs. The Notes Manager dialog enables you to easily see all slots and timesteps with a particular note, for example to easily see all provisional slots and timesteps. See [“Notes on Series Slots” in User Interface](#) for details on Notes.

When the **Read and display validation character** is checked, the Validation field values are also imported through the DMI and set on a Series Slot Note, associated with the particular slot and timestep. The Series Slot Note Group which holds the validation characters is named “HDB Validation Group”. The group's name is fixed; you can't edit it. In the Edit Note Group, the label above the name is Fixed Note Group Name and the name is read-only. You can edit the group's color.

Required For Writing to Real Tables

The remainder of the controls for the Real table type are described in this topic.



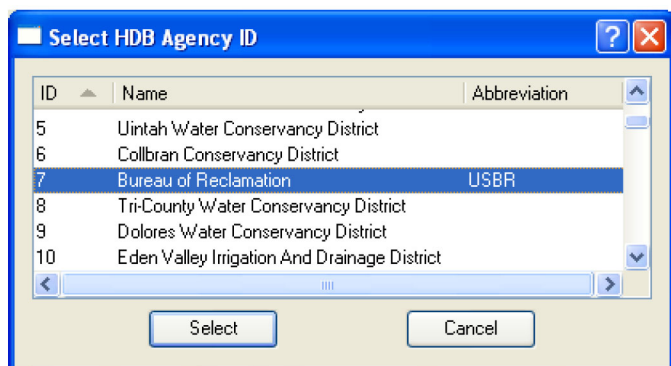
Required for Writing to Real Tables

Agency ID: 7 - Bureau of Reclamation

Collection Sys ID: 9 - RiverWare

☐ Use Overwrite Flag

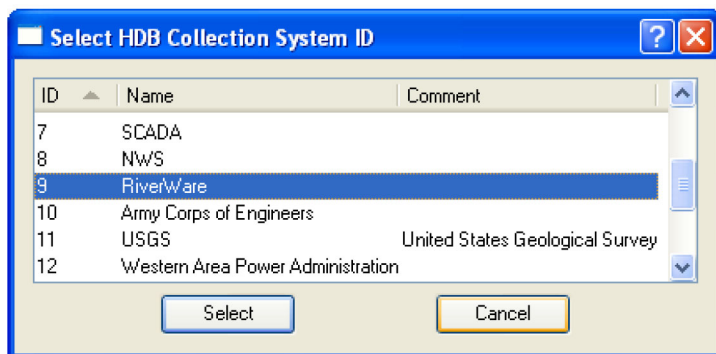
The **Agency ID** field specifies what agency ID will be associated with the data when it is written to the real tables. You must select an agency ID by selecting the **Select** button adjacent to the field. This will bring up the Select HDB Agency ID dialog populated with all the available agency IDs, their names, and abbreviations from the HDB_AGEN table in the database. You can then select an agency, which populates the **Agency ID** field on the dataset dialog with the ID and name of the agency.



Select HDB Agency ID

ID	Name	Abbreviation
5	Uintah Water Conservancy District	
6	Collbran Conservancy District	
7	Bureau of Reclamation	USBR
8	Tri-County Water Conservancy District	
9	Dolores Water Conservancy District	
10	Eden Valley Irrigation And Drainage District	

The **Collection Sys ID** field specifies what collection system ID will be associated with the data when it is written to the real tables. You must select a collection system ID by selecting the **Select** button adjacent to the field. This will bring up the Select HDB Collection System ID dialog populated with all the available collection system IDs, their names, and comments from the HDB_COLLECTION_SYSTEM table in HDB. You can then select a collection system, which populates the **Collection Sys ID** field on the dataset dialog with the ID and name of the collection system.



Select HDB Collection System ID

ID	Name	Comment
7	SCADA	
8	NWS	
9	RiverWare	
10	Army Corps of Engineers	
11	USGS	United States Geological Survey
12	Western Area Power Administration	

The **Use Overwrite Flag** check box specifies whether or not data is written to the real tables with an overwrite flag. This flag has some special meaning for how the data is handled in the database, namely that when the value is

written to its real table, it cannot be overwritten by aggregations of data from a shorter interval. Normally this box would not be checked.

HDB Table Type—Ensembles

A selection of **Ensemble** in the Table Type frame means that any data moved using the dataset will be determined through the REF_ENSEMBLE table in the database. An ensemble has a number of traces associated with it, and each trace has an associated model run ID. An ensemble would be used with MRM so that each run of the multiple run would correspond to a trace in the ensemble, and the data for that trace would reside under the trace's model run ID in the model (m_ xxxx) set of tables in HDB. See [“HDB Ensemble Configuration” in Solution Approaches](#) for details.

The image shows two overlapping dialog boxes. The top dialog, titled 'HDB Table Type', has three radio buttons: 'Model', 'Real', and 'Ensemble', with 'Ensemble' selected. The bottom dialog, titled 'Ensemble Configuration', has two radio buttons: 'Select ID when MRM is started' and 'Select ID', with 'Select ID' selected. Below these are three input fields with 'Select' buttons: 'Ensemble ID' (containing '1 - Historic Hydrology'), 'Trace ID to use outside MRM:' (containing '3'), and 'Agency ID to use when writing:' (empty).

The Ensemble ID frame has options for you to **Select ID** or **Select ID when MRM is started**. The **Select ID** option allows you to preselect an ensemble ID to associate with the dataset for reading or writing data with the model tables. You must select an ensemble ID by selecting the **Select** button adjacent to the ensemble field. This will bring up the Select HDB Ensemble ID dialog populated with all the available ensemble IDs, their names, the agency, the domain for the traces, and comments from the REF_ENSEMBLE table in the database. You can then select an ensemble, which populates the **Ensemble ID** field on the dataset dialog with the ID and name of the ensemble.

The image shows the 'Select HDB Ensemble ID' dialog box. It has a title bar with a blue icon and a close button. The main area is titled 'Select Ensemble ID for dataset HDB Hydrology Ensemble'. It contains a table with the following data:

ID	Name	Agency	Trace Domain	Comment
1	Historic Hydrology	Bureau of Reclamation	Historical hydrology	1960 - 1970
2	Projectd Hydrology	Bureau of Reclamation	Base Year	1960 - 1970, Reduced by 10%
3	High Demand	Bureau of Reclamation	Demand Level	High Demand Projection
4	High Demand;Historic Hydrology	Bureau of Reclamation	Demand Level;Historic...	High Demand Projection;1960

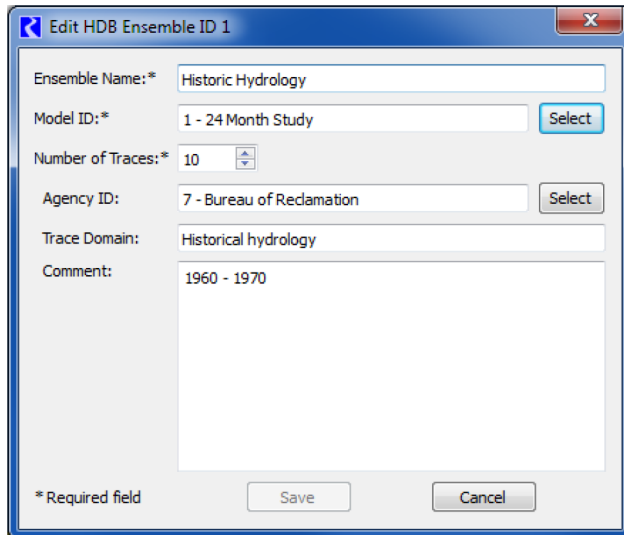
At the bottom of the dialog are four buttons: 'Select', 'Create', 'Edit', and 'Cancel'.

There are two other options for ensemble IDs on the Select HDB Ensemble ID dialog. Instead of just selecting an existing ID, you can also edit an existing ID or create a new one. An existing ID that is displayed can be highlighted and edited by selecting the **Edit** button at the bottom. This brings up the Edit HDB Ensemble ID dialog,

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Database DMI

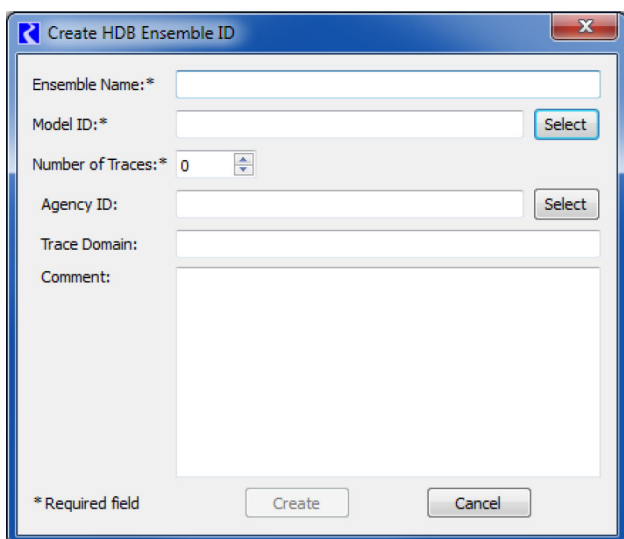
which displays all the information fields for the ensemble id and allows you to edit these fields. Selecting the **Save** button will then save the edited information for this ensemble ID to the database.



The 'Edit HDB Ensemble ID 1' dialog box contains the following fields and controls:

- Ensemble Name:***: Text input field containing 'Historic Hydrology'.
- Model ID:***: Text input field containing '1 - 24 Month Study' with a **Select** button to its right.
- Number of Traces:***: Spin box set to '10'.
- Agency ID:**: Text input field containing '7 - Bureau of Reclamation' with a **Select** button to its right.
- Trace Domain:**: Text input field containing 'Historical hydrology'.
- Comment:**: Large text area containing '1960 - 1970'.
- * Required field**: Legend at the bottom left.
- Save** and **Cancel** buttons at the bottom right.

The **Create** button at the bottom of the Select HDB Ensemble ID dialog allows you to create an ensemble ID. Selecting this brings up the Create HDB Ensemble ID dialog, which contains fields where information can be entered for the new ensemble ID. At least the required fields marked with an asterisk must be filled out. Selecting the **Create** button at the bottom will then save entered data to the database as a new ensemble ID. The new ID will then be added to the list and highlighted back on the Select HDB Ensemble ID dialog.

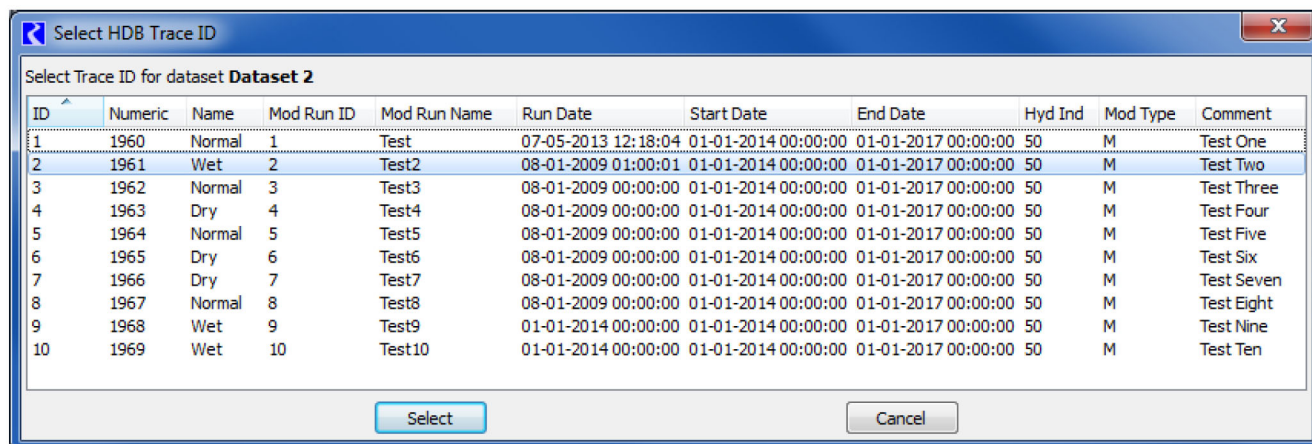


The 'Create HDB Ensemble ID' dialog box contains the following fields and controls:

- Ensemble Name:***: Empty text input field.
- Model ID:***: Empty text input field with a **Select** button to its right.
- Number of Traces:***: Spin box set to '0'.
- Agency ID:**: Empty text input field with a **Select** button to its right.
- Trace Domain:**: Empty text input field.
- Comment:**: Empty large text area.
- * Required field**: Legend at the bottom left.
- Create** and **Cancel** buttons at the bottom right.

It may be convenient to use an ensemble dataset outside of a multiple run to pull a particular trace from the ensemble for use in a single run or to manually test a DMI. Selecting the **Select** button adjacent to the **Trace ID to Use Outside MRM** field will open the Select HDB Trace ID dialog that is populated with all the traces for your selected ensemble ID specified for the dataset. The dialog shows descriptive information for each trace from the

REF_ENSEMBLE_TRACE table and also information for the trace's model run ID from the REF_MODEL_RUN table. You can then select a trace, which populates the trace ID into the dataset dialog.



The Ensemble ID frame of the dataset dialog has a second option for specifying ensemble IDs by selecting **Select ID when MRM is started**. When this option is selected, you do not preselect an ensemble ID that is always used with the dataset, but rather you are prompted for one when an MRM run containing this dataset in an input or output ensemble is started. This option could be useful, for example, where you want to select a different output ensemble for each execution of the multiple run so that data from earlier runs is not overwritten.

HDB Table Type

☐ Model
☐ Real
☒ Ensemble

Ensemble Configuration

Ensemble ID

☒ Select ID when MRM is started
☐ Select ID

1 - Historic Hydrology

Trace ID to use outside MRM: 3

Agency ID to use when writing:

Selecting an ID under the **Select ID when MRM is started** option utilizes the same dialogs as does the preselect ID option. The Select HDB Ensemble dialog appears when you start the multiple run where you can see existing ensemble IDs. An existing ID can then be selected, an existing ID can be edited and then selected, or a new ensemble ID for can be created then selected.

Note: If an ensemble id is being chosen for an output ensemble, the edit and create ensemble dialogs in this case have an additional trace control to **Use Number of MRM Input Traces (xx)**, where xx indicates that number of traces. This provides an easy way for the edited or newly created output

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Database DMI

ensemble to contain the correct number of traces to hold the run data as defined by the input ensembles.

Ensemble Name: * Historic Hydrology

Model ID: * 1 - 24 Month Study Select

Number of Traces: * 10 ☒ Use Number of MRM Input Traces (10)

Agency ID: 7 - Bureau of Reclamation Select

Trace Domain: Historical hydrology

Comment: 1960 - 1970

* Required field Save Cancel

In addition to the ensemble id options discussed above, there is also an option under the Ensemble Configuration frame to select an **Agency ID to use when writing**. Selecting **Select** will open the Select HDB Agency ID dialog where one of the agency ids defined in HDB can be chosen. If chosen, that agency id will be written to the dataset ensemble when that dataset is executed as part of an output ensemble. If one is not chosen, the behavior will be to preserve any agency id that is already defined in the database for the output ensemble.

Select Agency ID for dataset HDB Hydrology Ensemble

ID	Name	Abbreviation
1	Bostwick Park Water Conservancy District	
2	Carlsbad Irrigation District	
3	BIA (Ute-Indain Tribe)	
4	Central Utah Water Conservancy District	
5	Uintah Water Conservancy District	
6	Collbran Conservancy District	
7	Bureau of Reclamation	USBR
8	Tri-County Water Conservancy District	
9	Dolores Water Conservancy District	

Select Cancel

An HDB ensemble can have metadata that describe the ensemble as well as metadata that describe each trace. Metadata is a keyword associated with a value, such as “comment”/”Historic Hydrology”. In the HDB database, values for the “domain”, and “comment” keywords are held in the trace_domain, and cmmt fields of the REF_ENSEMBLE table. Other keyword/value pairs to describe ensembles are held in the REF_ENSEMBLE_KEYVAL table.

For trace metadata, the values for the “name” and “numeric” keywords are held in the trace_name and trace_numeric fields of the REF_ENSEMBLE_TRACE table. Other keywords to describe traces are held in the REF_MODEL_RUN_KEYVAL table.

See “[HDB Ensemble Configuration](#)” in *Solution Approaches* for details on using Ensembles in MRM.

DIT Datasets

The DIT Dataset allows direct connections to import data from the Demand Input Tool (DIT) SQLite database. For more information on this tool, see the DIT web site.

Double-clicking a DIT dataset in the Dataset Manager opens the DIT Dataset editor. In this dialog, the dataset can be renamed and configured. It has two tabs, one for general configuration and one specific to the DIT implementation.

General Tab

This section describes the general tab for the DIT dataset dialog box.

DIT Dataset - Demand Data

File

Type: DIT

Name: Demand Data

General | DIT

Name Map: Name Map Mgr...

Missing Values Are: NaN

Prefer Database Units

Type	Scale	Units
Flow	1	acre-ft/day
Concentration	1	mg/l
Mass	1	tons

OK Apply Reset Cancel

Name Map

Mappings of RiverWare names to DIT is typically not necessary for particular slots. See [“Slots Typically Imported”](#) on page 118 for more information on the default mapping. If you want to map other slots to values in the DIT, the following are the names of values in the database and would be the text to use in the name map:

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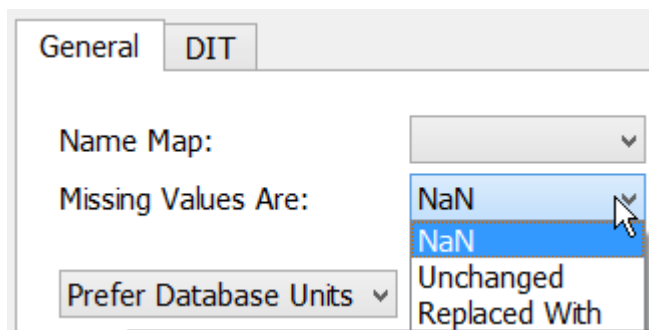
Database DMI

- **DiversionValue** - Diversion requests or schedules in the DIT.
- **DepletionValue** - Depletion requests or schedules in the DIT.
- **AggDivSalinityValue** - Salinity mass or concentration in the DIT.

Missing Values

Specify how missing values are handled when importing data from the DIT. In general, a NaN/NULL in the DIT is represented as a NaN in RiverWare. The current choices are as follows:

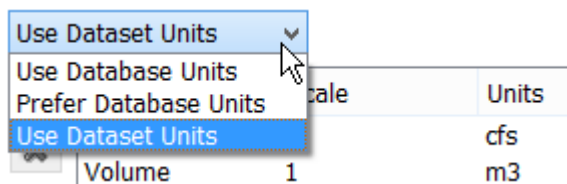
- **NaN.** On import, missing values in the DIT are made NaNs in RiverWare.
- **Unchanged.** On import, RiverWare data is left unchanged for a missing value in DIT.
- **Replaced With.** Provide a value that is substituted for NaNs or a missing values on import.



Units

A menu in the Dataset dialog allows you to specify how units should be handled for a dataset. The options are as follows:

- **Use Dataset Units**
- **Prefer Database Units**
- **Use Dataset Units**

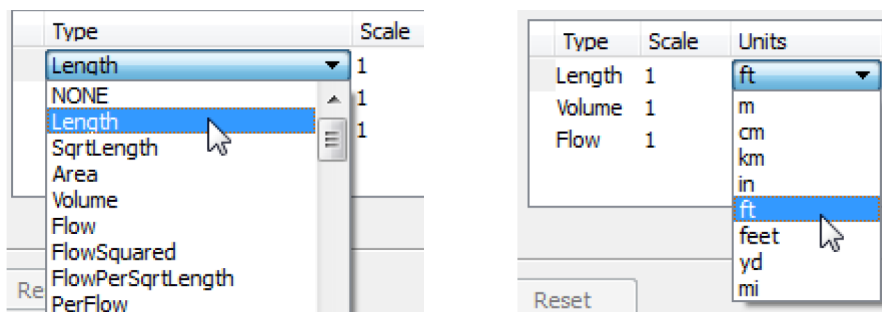


Units are currently not represented or read from the DIT. **Use Dataset Units** or the default **Prefer Database Units** are the most applicable selections.

If you chooses either of the second two options, you must specify the units in the window below the pull-down menu. Select the **+** icon to add a new unit type:

This adds an entry to the list. Double-click the word **NONE** under the **Type** to change the type. [Figure 4.20](#) illustrates. Change the **Scale** and **Units** in a similar manner. These units are the units that the database contains

Figure 4.20



[Figure 4.21](#) shows an example of specified units.

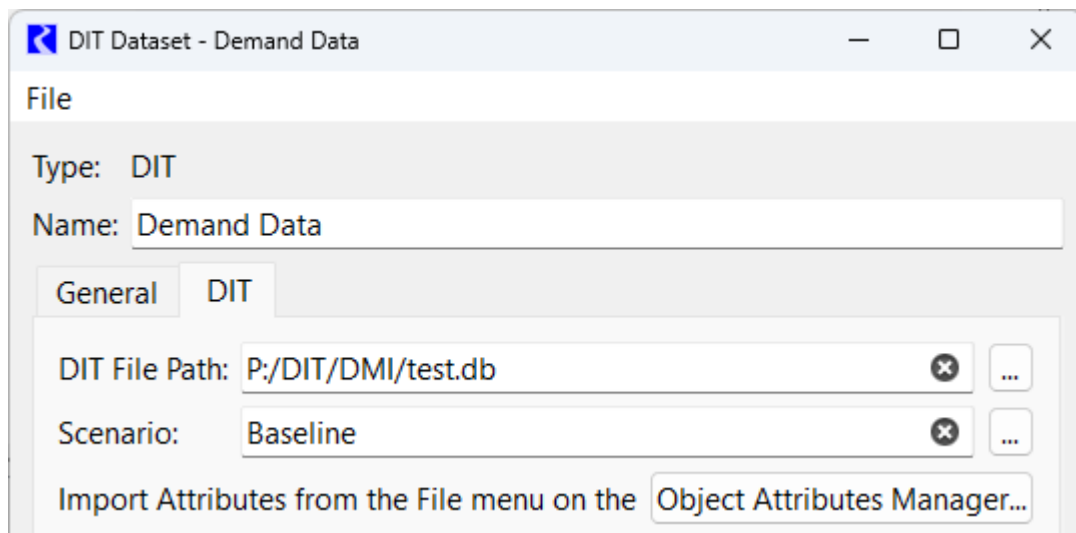
Figure 4.21

Type	Scale	Units
Length	1	ft
Volume	1	acre-ft
Flow	1	cfs

Configuration specific to the DIT database is specified in the DIT tab of the DIT Dataset editor.

DIT Tab

The following is a screenshot of the DIT tab of a DIT dataset:

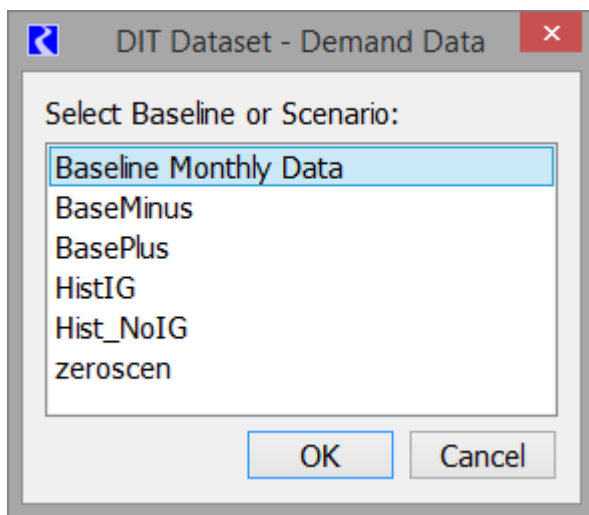


DIT File Path

First, specify the path to the database. Select the “...” button next to the **DIT File Path** field to open a standard file selector dialog and then locate the database file. Common SQLite file extensions are db, sqlite, sqlite3. Environment variable are also supported in the file path. Use the `$VARIABLE`, `$(VARIABLE)` or `${VARIABLE}` syntax.

Scenario

Once a valid DIT file is specified, select the “...” button next to the **Scenario** field to open the dialog below. Then, select which scenario's data should be imported. This dialog only opens if a valid DIT database file has been selected.



Import Attributes

The DIT may also contain object attributes and/or values that you would like to import as well. This import step is performed from the Object Attributes Manager. Use the button to access that utility.

Import Attributes from the File menu on the [Object Attributes Manager...](#)

See “[Import and Export Attributes and Values](#)” in *User Interface* for more information.

Slots Typically Imported

As described in “[Name Map](#)” on page 115, name maps can be used. The DIT dataset provides automatic mapping from certain slots to variables in the database. The table below shows the default slots in RiverWare and how they are mapped to data in the DIT. Importing to any other slot, without a name map, will result in an error.

Table 4.3 Default mapping of slot names to data in the DIT

Object.Slot in RiverWare	Data in the DIT	Unit type
WaterUser.Diversion Requested	DiversionValue	Flow
WaterUser.Diversion Schedule	DiversionValue	Flow
WaterUser.Depletion Requested	DepletionValue	Flow
WaterUser.Depletion Schedule	DepletionValue	Flow
AggDiversionSite.Return Flow Salinity Pickup Mass	AggDivSalinityValue	Mass
AggDiversionSite.Return Flow Salinity Pickup Conc	AggDivSalinityValue	Concentration
AggDiversionSite.Diversion Salt Concentration	AggDivSalinityValue	Concentration
AggDiversionSite.Salt Mass Removal	AggDivSalinityValue	Mass

For the last four items, the DMI and dataset checks that the slot and unit type matches the computation used in the DMI. This helps to ensure that mass/concentration data in the DIT gets imported to mass/concentration slots in RiverWare.

If you need to import DIT to other slots, use a name map to map the slot in RiverWare to the data in DIT. See [“Name Map”](#) on page 115 for more information.

Excel Datasets

Excel datasets provide user configuration options to specify how you wish the Database DMI to work with Microsoft Excel. Excel should be available on the system as RiverWare may start a non-visible copy of Excel in the background and access the specified workbook during DMI execution. There are two important things to remember:

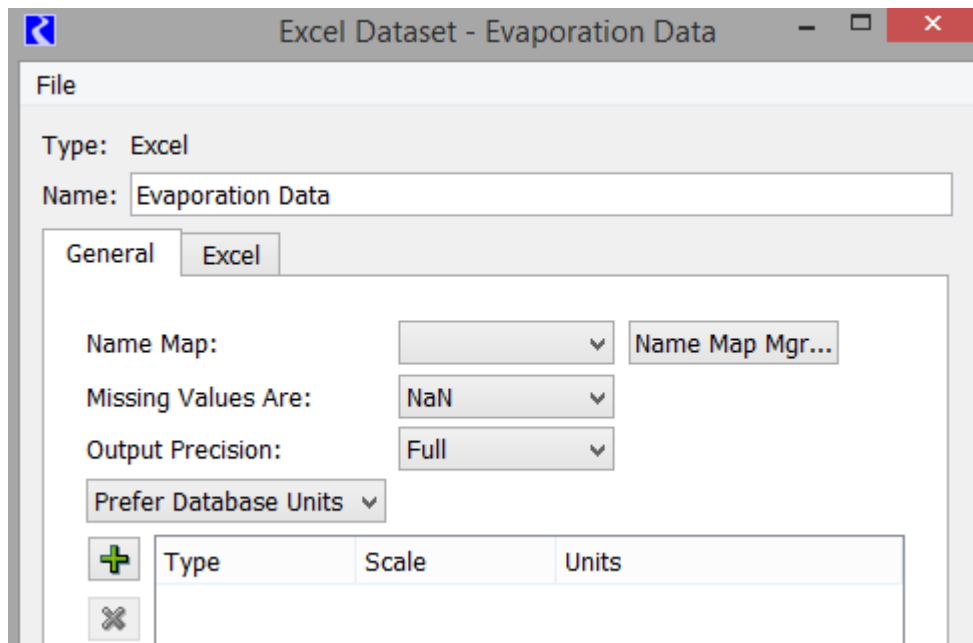
Note: You should not have the specified workbook open in a different session of Excel as this may lock out RiverWare from accessing the workbook.

Note: RiverWare is operating on the Excel file, so you will not see any unsaved changes to the file that may be made in a different Excel session.

See [“Example 2: Excel DMI to Export Data Using Headers”](#) on page 85 for a use case of creating a DMI to Excel.

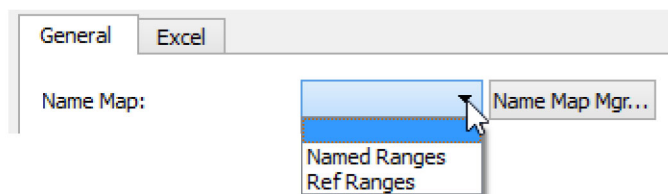
General Tab

Double-clicking an Excel dataset in the Dataset Manager opens the Excel Dataset editor. In this dialog, the dataset can be renamed and configured. It has two tabs, one for general configuration and one specific to the Excel implementation. Following is a description of the configuration options.



Name Map

Select the desired Name Map from the **Name Map** menu. The selections are populated based on the Name Maps configured in the Name Map Manager. See [“Name Mapping”](#) on page 154 for details on creating Name Maps. Name maps have special relevance for moving data using range specifications and header specification. See [“Slot Mapping—Spreadsheet Ranges Approach”](#) on page 125 and [“Using Name Maps”](#) on page 129 for details.

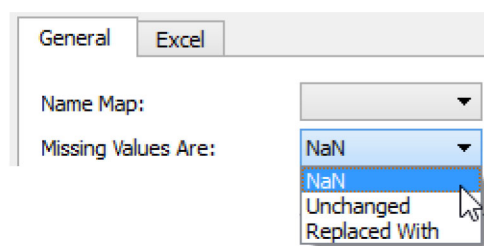


Missing Values

The user specifies how missing values are handled with Excel. The current choices are

- **NaN.** NaN value in RiverWare is written as NaN to Excel, NaN or blank value in Excel is read as NaN into RiverWare
- **Unchanged.** The DMI will not change missing values on import, uses NaNs on export

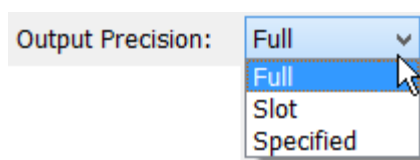
- **Replaced With.** Provides a user input value that is substituted on import and export



Output Precision

For Output DMIs, specify the precision of values that are sent to the database, that is, the number of digits to the right of the decimal point. The options are:

- **Full.** Output all available digits of precision. This is the default.
- **Slot.** Use the slot's display precision from the active unit scheme. Note, changing the active unit scheme could affect values sent out via the DMI.
- **Specified.** Use the specified precision.



Values are converted to the specified units, as described in “Units” on page 121, before rounding.

The rounding algorithm is: $\text{round}(\text{value} * 10^N) / 10^N$ where value is after unit conversion, N is the number of digits of precision, and round() is a standard function which rounds its argument to the nearest integer value, rounding halfway cases away from zero.

Caution: With this approach you could potentially see misleading results. For example, if a slot shows 1.23cfs, you export it using specified dataset units of cms, it becomes 0.0348cms. If you then reduce the precision to 2 digits, it will send 0.03cms as the output. If you then show the value in your database viewer using cfs and 2 digits of precision, it shows it as 1.06cfs. The value in the database does not match the value in RiverWare due to conversions and rounding. Be careful!

Also, to be clear, all values exported from a Database DMI are floating point numbers regardless of precision. For example, an expression slot might compute a value of 7538.21. Exported with 0 digits of precision, this would appear to be 7538, but technically, it was exported as a floating point number, 7537.9999999974 to 15 significant digits. This fact is really only important if your database displays 15-17 digits.

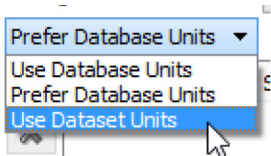
Units

The units that are used in interacting with Excel are specified in the Dataset dialog. A pull-down menu allows the user to specify that the Database DMI should **Use Database Units**, **Prefer Database Units**, or **Use Dataset Units**.

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- **Use Database Units.** Since Excel contains no units associated with a value in a cell, the slots' units (i.e. from the Unit Scheme) as displayed in RiverWare are substituted for Database Units and will be used for import and export under this option
- **Prefer Database Units.** For Excel, this option uses the slots' units (i.e. from the Unit Scheme) in RiverWare unless a unit is specified in the Dataset, then the specified unit is used
- **Use Dataset Units:** Uses the units specified in the dataset only



See “Units” on page 98 for instructions on specifying the units to use.

Note: For import and export of series slots that have the DateTime unit type, the DMIs translate to and from the Excel date representation. This allows dates to display correctly in both RiverWare and in Excel. But, Days/months/years in RiverWare are at the end of the timestep, which would be represented in Excel at the beginning of the next timestep. For example 24:00 June 1, 2015 in RiverWare would translate to 00:00 June 2, 2015 in Excel. Also, partial datetimes and dates before 1900 are negative values in Excel, so are just displayed with # signs as out of range. However the representation is still consistent and these date values will round trip correctly when exported to Excel and imported back into RiverWare.

FullDateTime		
12-15-2015 Tue	24:00 January 1, 2015	I
12-16-2015 Wed	24:00 June 1, 2015	I
12-17-2015 Thu	24:00 September 1, 2015	I

	A	B
1		Basin Data.Key System Dates
2	12/15/2015	1/2/15 0:00
3	12/16/2015	6/2/15 0:00
4	12/17/2015	9/2/15 0:00
5		

Note: Text Series Slot values can be imported or exported using Excel Datasets and Database DMIs. These behave just like series slots but import or export the text strings. Be aware that a NaN in Excel will be interpreted as a string, not as an invalid value. Use blank cells in Excel to prevent importing NaN text strings. See “Text Series Slots” in *User Interface* for details.

Excel Tab

Configuration specific to Excel is specified in the **Excel** tab of the Excel Dataset editor.

The screenshot shows the 'Excel Dataset - Input Moutain Inflow' dialog box with the 'Excel' tab selected. The 'File' section at the top shows 'Type: Excel' and 'Name: Input Moutain Inflow'. Below this are two tabs: 'General' and 'Excel'. The 'Excel' tab contains the following settings:

- Workbook:** C:/Temp/MountainStorageData.xlsx
- Slot Mapping:**
 - ☒ Map by Spreadsheet Ranges
 - ☐ Map with Header Text and Sheet Names
- Range Specification:**
 - ☒ Map to Excel Named Ranges with Name Map
 - ☒ Ranges on same worksheet
 - Worksheet:** ReservoirData
 - (Name Map specification of form B2:B50)
 - ☐ Ranges on different worksheets
 - (Name Map specification of form Sheet1!B2:B50)
 - ☐ Map to Excel Range Specifications with Name Map
- ☐ Specify Range Offset for each Run of Multiple Run
 - Rows:** 0
 - Columns:** 0

At the bottom of the dialog, there is a checkbox for 'Open and Save Workbook' and a 'More Information...' button. The bottom of the dialog features four buttons: 'OK', 'Apply', 'Reset', and 'Cancel'.

Workbook

In all cases, specify the Workbook that the dataset will read data from or write data to. The path and workbook name can be typed into the **Workbook** text box, or you can select the adjacent button to select the file. When executing an input DMI, this workbook must already exist. For an output DMI, the workbook will be created if it does not already exist.

Slot Mapping

Two approaches are provided for mapping data between RiverWare and Excel.

- Ranges: Specify the Excel range for each RiverWare slot. Choose the **Map by Spreadsheet Ranges** approach,
- Headers: Use a row header, a column header, and a worksheet name to map the Excel data with a RiverWare slot. The **Map by Header Text and Sheet Name** approach.

[Table 4.4](#) compares these approaches. Then more information is presented on the configuration options for both the Range and Header approach.

Table 4.4 Comparing Excel dataset approaches

Question	Ranges Approach	Headers Approach
What is the format of the Excel data?	Your data is in any format in Excel, but you must then specify how the data maps to RiverWare slots.	Your data in Excel has row header names, column header names, and sheet names that identify the data in a way that matches one of the orientations in the dataset.
How is the data mapped?	A DMI Name Map specifies how the data in RiverWare slots “maps” to data in an Excel sheet. Name Maps can specify an Excel absolute cell reference (A1:A23) or a named range.	The text in header cells in Excel and the name of the sheet “map” the data to RiverWare slots.
Can I have more timesteps of data in Excel than I want to import/export?	Yes, but the mapped Excel range must match the number of timesteps specified in the DMI. That is, the size of the range you specify must exactly match the number of values in the DMI.	Because this approach includes timestep information, the Excel sheet can have more data in it than you are processing via the DMI
Can I import and export table, periodic, or scalar slots?	Yes, your named ranges or cell references can map to non-series data.	Yes, a dataset using the Header approach can either deal with Series slots OR Table, Periodic or Scalar slots, but not both at the same time.
Can or do I use Name Maps?	You must use name maps to specify where the data will import from/export to in Excel.	Name Maps can optionally allow you to have different names of objects or slots in RiverWare than the header text in Excel.
How are MRM runs handled?	Data for each MRM run is offset from the previous run by a certain number of rows and columns, starting with the mapped range.	MRM runs is one of the variables you specify in the Excel orientation. MRM runs can be on sheets, columns or rows. Their names can optionally be Run0, Run1, Run2, etc. or trace number as set up in the MRM configuration (i.e. Trace5, Trace6, Trace 7, etc.)
How do I specify the Excel sheet?	Either specify a single sheet name or include the sheet name in the Name Map.	Sheet names depend on the selected orientation and can be run, slot, or timestep names.

Table 4.4 Comparing Excel dataset approaches (Continued)

Question	Ranges Approach	Headers Approach
What is the format of the timestep in Excel?	No timestep information is read or written.	You specify to either use the end of timestep (E.g. Jan, 1, 2011 23:59) or the beginning of timestep format (E.g. Jan 1, 2011 0:00).
How do I specify the units for Excel?	No unit information is read or written. Units in Excel are assumed to be the slot's units in RiverWare (i.e. from the active Unit Scheme). The General tab of the dataset can be used to specify a mapped unit by type if the units in Excel are different than the RiverWare slot units.	You can optionally write the unit name with the slot name to Excel or can read a slot name from Excel if it has a unit name with it. However the assumed units for values and the possibility for mapped units work the same as in the Range approach.
In general, why would I use this approach?	This approach is more work because a map entry must be created in a name map for every slot. Use this format if you are taking the data out of or inserting it into a spreadsheet that does not have the header information and if the amount of data is constant. Perhaps you have an existing spreadsheet that cannot be changed.	This approach is recommended for series data, particularly if you want to import or export data that will change length over time. For example, you have an Excel sheet with 100 years of daily data, but your model is for the current year and is updated every year. This approach is also very useful for Table, Periodic and Scalar slots that can be represented in the spreadsheet in the required format.
Where can I find more information	"Slot Mapping—Spreadsheet Ranges Approach" on page 125	"Slot Mapping—Header Text and Sheet Names Approach" on page 129

Slot Mapping—Spreadsheet Ranges Approach

The Map by Spreadsheet Ranges approach presents several options for specifying ranges associated with slots. In all cases, the Name Map functionality of Database DMIs is used to map individual slots to their range specifications. You can map Series, Table, Periodic, or Scalar slots using this approach.

Slot Mapping

- ☒ Map by Spreadsheet Ranges
☐ Map with Header Text and Sheet Names

Note: The size of the range specified must match the number of values to be read or written with the DMI. No header or other cells should be included in the range.

Note: Object name maps are not applicable to the range approach and are ignored. In this approach, the Excel range must be specified for each slot using a slot name map.

Discussions of the range specification options along with example Name Maps follow.

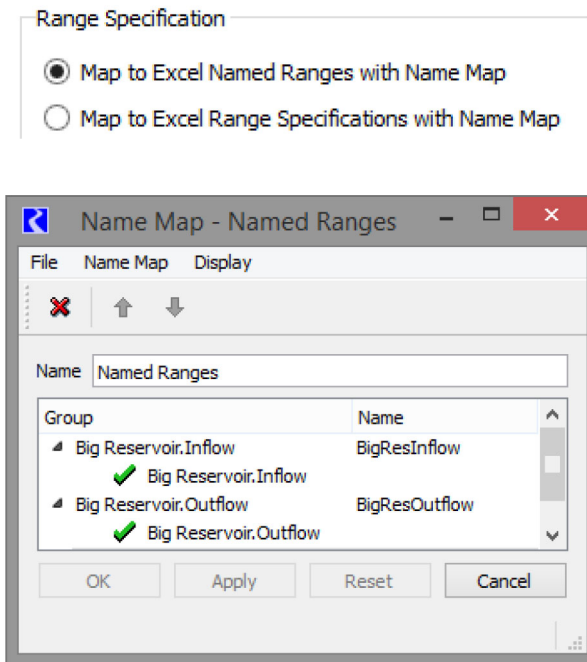
Map to Excel Named Ranges

Under the **Map to Excel Named Ranges with Name Map** option, the user would set up named ranges in their Excel workbook. Named ranges are a feature of Excel where a number of cells can be selected and assigned a name by the user to identify these cells. The Name Map created in RiverWare and selected in the **General** tab of the

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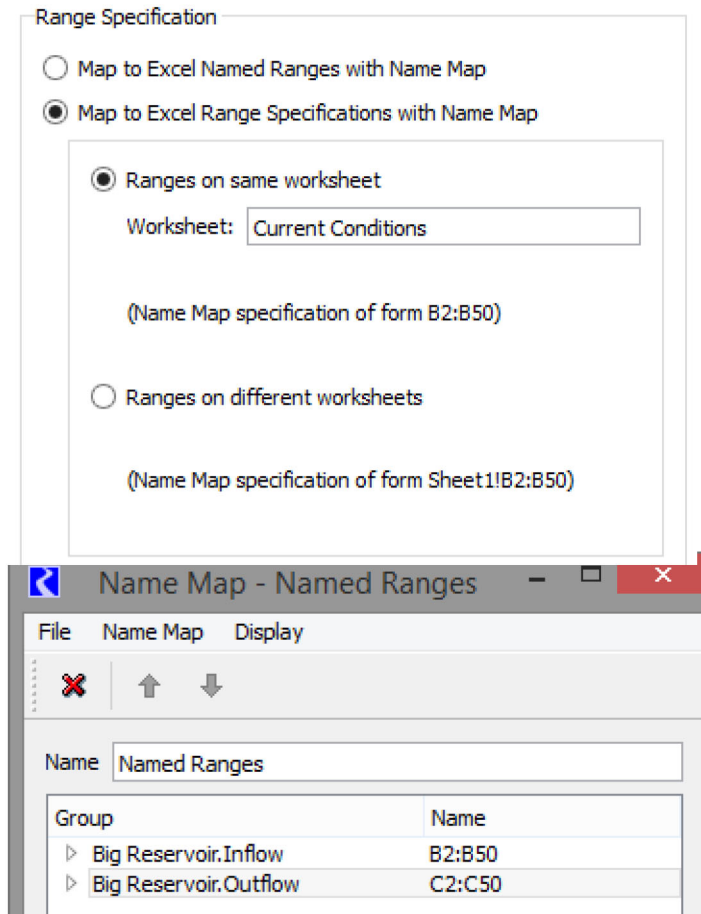
dataset dialog then will associate the RiverWare slot with the named range as seen in the following example name map. In this case, BigResInflow and BigResOutflow are named ranges in the Excel sheet.



Map to Excel Range Specifications

In the **Map to Excel Range Specifications with Name Map** option, there are two ways to specify ranges for slots. If all the ranges for the slots are on the same worksheet, then the **Ranges on same worksheet** option can be used. Here the worksheet name is specified in the **Worksheet** text box. Ranges specified in the name map can be of the form B2:B50, meaning the range will be from cell B2 to cell B50 of the specified worksheet. [Figure 4.22](#) shows an example of the type of Name Map created and selected on the **General** tab of the dataset dialog for this approach.

Figure 4.22



If ranges for the slots are on different worksheets then the **Ranges on different worksheets** option must be used. Under this option, ranges specified in the name map must include the worksheet name as part of the range specification in the form Sheet1!B2:B50. In this way, ranges referring to different worksheets can be included in the same name map. [Figure 4.23](#) shows an example of this type of Name Map.

Figure 4.23

Range Specification

☐ Map to Excel Named Ranges with Name Map
☒ Map to Excel Range Specifications with Name Map

☐ Ranges on same worksheet
 Worksheet:

 (Name Map specification of form B2:B50)

☒ Ranges on different worksheets

 (Name Map specification of form Sheet1!B2:B50)

Name Map - Named Ranges

File Name Map Display

☒ ☐ ☐

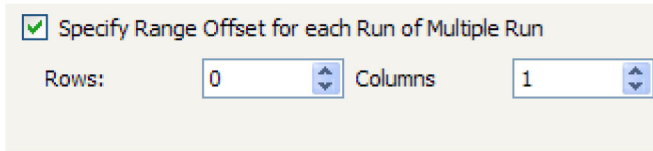
Name

Group	Name
▶ Big Reservoir.Inflow	Inflow!B2:B50
▶ Big Reservoir.Outflow	Outflow!B2:B50

Specify Range Offset

Under all variations of the Map by Spreadsheet Ranges approach, there is an option available to Specify Range Offset for each Run of Multiple Run. In a multiple run, the user may want to read different data into a slot for each run or write out the results for the slot under each run. If the option is checked, the mapped range for the slot is used for the first run, and the specified row and column offset is applied for each run thereafter. For example, the range for a slot specified in the dataset name map could be Sheet1!B2:B50 and the offset specified here could be 0 rows and 1 column. For an input DMI in a multiple run, the first run will read range B2:B50 on Sheet1 into the slot, the second run will offset one column and read range C2:C50 on Sheet1, the third range D2:D50, etc. For an output

DMI, the slot's results from the first run will be written to range B2:B50 on Sheet1, results from the second run will offset one column and be written to range C2:C50 on Sheet1, the third to range D2:D50, etc.



☒ Specify Range Offset for each Run of Multiple Run

Rows: Columns:

Slot Mapping—Header Text and Sheet Names Approach

The Map by Header Text and Sheet Name approach is the alternative to using ranges to move RiverWare slot data to or from Excel. The approach uses a row header, a column header, and a worksheet name to map Excel data with a RiverWare slot. [Figure 4.24](#) shows an example of the dataset dialog with this configuration selected.

Tip: When setting up an input DMI using this Excel dataset approach (especially the first time), it is often easiest to configure the DMI as an *Output* DMI first, run the DMI to create an Excel spreadsheet with the right formatting and desired structure. Edit that spreadsheet or use it as a guide for setting up your data in that format. Then, configure the DMI to be an *Input* DMI that will bring your data into RiverWare.

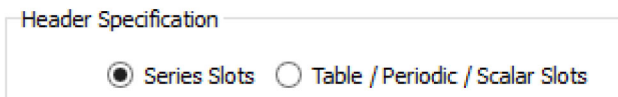
Using Name Maps

Name maps are not needed in the header approach. However they can be used to map to a different name in Excel. The following is performed for each slot:

- If there is no name map, the RiverWare object and slot name is used in Excel.
- If there is a slot mapping only, the specified mapped name is used in Excel.
- If there is an object mapping only, the mapped object is used for the object name followed by a period and the RiverWare slot name. This combined string is used in Excel.
- If there is a slot *and* object mapping, the mapped object name is used followed by a period and then the mapped slot name. This combined string is used in Excel.

Slot Types

An Excel dataset using the header approach can either import/export Series Slot data or Table / Periodic / Scalar Slot data, but not both. Choose the desired slots that will be accessed by this dataset.



Header Specification

☒ Series Slots ☐ Table / Periodic / Scalar Slots

Figure 4.24 Header Specification options

Header Specification

☒ Series Slots ☐ Table / Periodic / Scalar Slots

☐ Specify Header Offset from Upper Left Spreadsheet Corner

Rows: Columns:

Header Orientation

Rows	Columns	Sheets
☒ Timesteps	Slots	Run(s)
☐ Timesteps	Run(s)	Slots
☐ Slots	Timesteps	Run(s)
☐ Run(s)	Timesteps	Slots

Run Name Type

☒ Single Run Name:

☐ Append Suffix to Run Name

☐ Specify Single Run Name when DMI is invoked

☐ "Run" with MRM run number (i.e. Run0, Run1, etc.)

☐ "Trace" with MRM trace number (i.e. Trace5, Trace6, etc.)

☐ Use Begin Timestep Time Instead of End of Timestep

☐ Clear Dates and Values for Output

☐ Use Unit Name with Slot Name

Series Slots Configuration Options

When the Series Slots option is chosen, you can specify the format of the series slots in the spreadsheet.

The Header Orientation frame provides four orientation choices for how timesteps, slots, and runs will map to rows, columns, and sheets in Excel. For example, the first orientation as selected in the adjacent screenshot indicates that rows are timesteps, columns are slots, and worksheets are runs.

Note: Orientations for putting each timestep on a separate sheet are not supported.

Header Orientation

	Rows	Columns	Sheets
☒ Timesteps	Slots	Run(s)	
☐ Timesteps	Run(s)	Slots	
☐ Slots	Timesteps	Run(s)	
☐ Run(s)	Timesteps	Slots	

Figure 4.25 shows an Excel spreadsheet with the first orientation. In this orientation, timesteps are rows, so the timestep header is the first column with a time label for each row. Similarly, the slot header is the first row with each column labeled with the slot name. Sheets are runs, so the sheet is named with the single run name specified. With the orientation specified, the data in Excel associated with a slot over a time range can be found for reading via an input DMI, or can be found and overwritten, if existing, or be newly created by writing via an output DMI.

Figure 4.25

	A	B	C
1		Storage Reservoir.Inflow	Storage Reservoir.Outflow
2	1/2009	1000	800
3	2/2009	1100	800
4	3/2009	1200	800
5	4/2009	1300	1000
6	5/2009	1400	1300
7	6/2009	1500	1500
8	7/2009	1450	1500
9	8/2009	1300	1500
10	9/2009	1200	1400
11	10/2009	1000	1200
12	11/2009	900	1200
13	12/2009	800	1000
14	1/2010	600	1000
15	2/2010	500	800
16	3/2010	500	700
17	4/2010	500	600

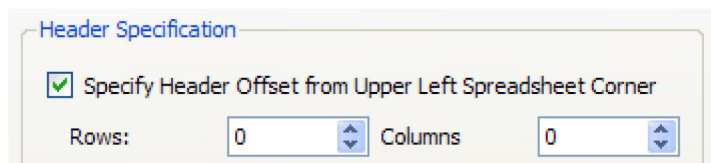
Note: On subsequent executions of an output DMI using the header approach, if there is already a matching header, data will be overwritten. If there is not already a matching header, the new header and data will be appended to any existing data. So in the example spreadsheet above, if you change the output slot from StorageReservoir.Outflow to Storage Reservoir.Storage and rerun the DMI (without deleting the.xlsx file), it will replace the inflow column with any new data and will append a StorageReservoir.Storage column after the Outflow column (i.e. it won't replace the outflow column with storage, but will instead append it).

The headers may not always be the first row and the first column. To accommodate these cases, an option is provided to specify how the headers are offset from the corner of the spreadsheet. By indicating how many rows

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and how many columns the header is offset from the upper left corner of the spreadsheet, the user can match how data resides in an existing spreadsheet or indicate where the headers and data should be written to a new sheet.



The image shows a dialog box titled "Header Specification". It contains a checked checkbox labeled "Specify Header Offset from Upper Left Spreadsheet Corner". Below this, there are two input fields: "Rows:" with a value of "0" and "Columns" with a value of "0". Both input fields have small up and down arrow buttons next to them.

Table / Periodic / Scalar Slots Configuration Options

When the **Table / Periodic / Scalar Slots** option is chosen, you are not allowed to specify the format of the slots in the spreadsheet. Instead, there is a very specific format that is used. [Figure 4.26](#) shows a sample workbook. Table slots have a header merged cell for the slot name, cells for column headers, and the columns that contain row headers and data columns. Periodic slots include a column of dates, one or more columns of data, and a header cell. Scalar slots are represented by a header cell and a value cell.

Note: The associated slots in RiverWare are also shown. The sheet name is Run0 because that option (described below) is used.

Figure 4.26

	A	B	C	D	E	F	G	H	I	J
1	Big Reservoir.Action Table				Big Reservoir.Periodic			Big Reservoir.Weekend Scalar (m3)		
2		Stage	Low Volume	High Volume		Min Flow	Avg Flow	Max Flow	550	
3	A	10	500	1000	Jan	56.6	126.6	176.6		
4	B	20	500	1000	Feb	84.8	154.8	204.8		
5	C	30	500	1000	Mar	113.1	183.1	233.1		
6	D	45	500	1000	Apr	141.3	211.3	261.3		
7	E	55	500	1000	May	169.6	239.6	289.6		
8	F	75	500	1000	Jun	197.8	267.8	317.8		
9	G	80	500	1000	Jul	162.5	232.5	282.5		
10	H	99	500	1000	Aug	127.2	197.2	247.2		
11					Sep	91.9	161.9	211.9		
12					Oct	56.6	126.6	176.6		

Big Reservoir.Action ...

Stage	Low Volume	High Volume
0: A	10.0	500
1: B	20.0	500
2: C	30.0	500
3: D	45.0	500
4: E	55.0	500
5: F	75.0	500
6: G	80.0	500
7: H	99.0	500

Big Reservoir.Peri...

Min Flow	Avg Flow	Max Flow
Jan	56.6	126.6
Feb	84.8	154.8
Mar	113.1	183.1
Apr	141.3	211.3
May	169.6	239.6
Jun	197.8	267.8
Jul	162.5	232.5
Aug	127.2	197.2
Sep	91.9	161.9
Oct	56.6	126.6

Big Reservoir.Week...
Value: 550 m3

Table / Periodic / Scalar Slot DMI Limitations

Functionality to import/export Table, Periodic, and Scalar slots from/to Excel has the following limitations:

- Multiple runs are supported by having sheets with either the RunN or TraceN format.
- Using an offset from the first row/column is not possible.
- The **Use Unit Name with Slot Name** is only supported for Scalar slots. Units in Table or Periodic columns are not supported.
- Column and row labels must be specified in Excel as strings, even when the labels are the string representation of numbers (which is common for row labels). Unfortunately, Excel tries to convert string numbers into Numbers. To prevent this, quote numeric column and row labels with a leading apostrophe, for example '10.
- Table or Periodic slots with the DATETIME unit type are not supported.

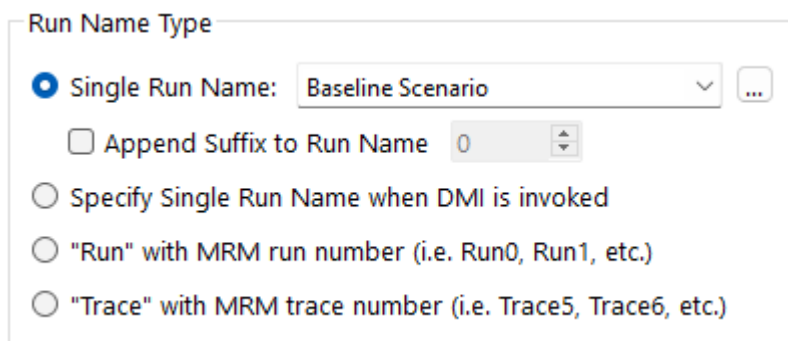
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- Importing a Table slot from Excel can only resize the number of rows in RiverWare. The number of columns cannot be changed.
- Importing a Periodic slot from Excel must match exactly the number of rows and columns. Changing the number of column or rows is not allowed.
- Periodic slots export includes row labels (i.e. dates) in the Excel sheet. But on import, the dates must match the row labels or an error will occur. It is not possible to change the periodic dates, interval or period based on data in Excel. These must all be configured in the slot before importing values from Excel.

Run Name Type

The **Run Name Type** options indicate how runs in the Excel workbook will be labeled.



Run Name Type

☒ Single Run Name: Baseline Scenario ...

☐ Append Suffix to Run Name 0

☐ Specify Single Run Name when DMI is invoked

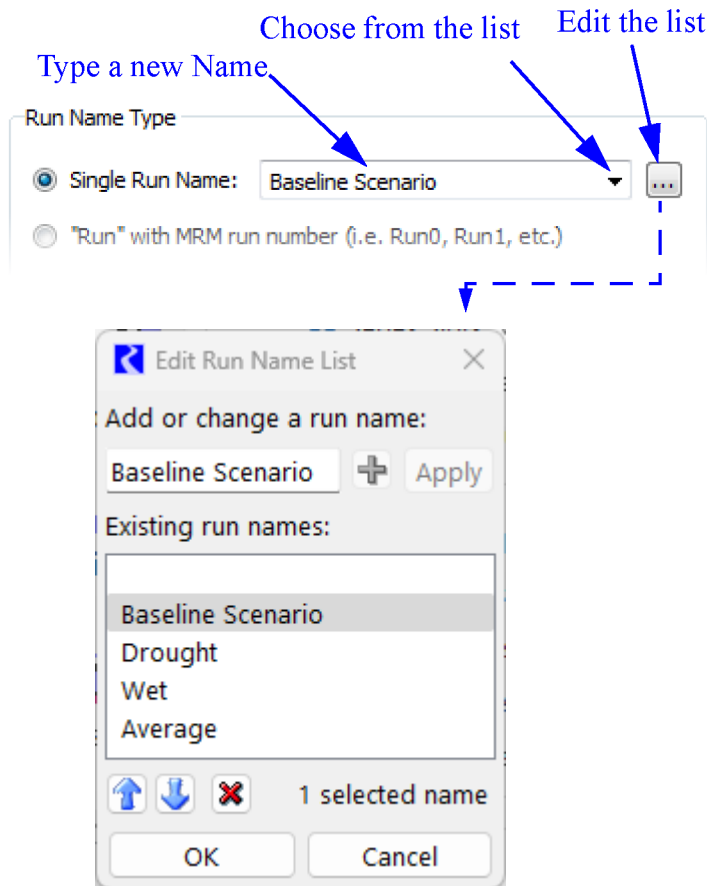
☐ "Run" with MRM run number (i.e. Run0, Run1, etc.)

☐ "Trace" with MRM trace number (i.e. Trace5, Trace6, etc.)

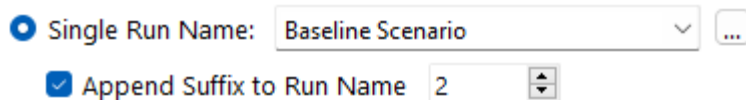
Use one of the following options:

- **Single Run Name.** For a single run, the **Single Run Name** must be entered, which will be used to map to the appropriate dimension in Excel depending on the orientation selected (either row, column, or sheet). Either type the text directly into the field or use the chooser to select from a list of possible values. Use the button to edit the list of possible values. In the Edit Run Name List dialog, add, change, or delete possible Run Names using the

buttons. A Run Name List can be copied from one Excel Dataset to another using copy/paste operations in the right-click context menu.



Also, below the Single Run Name, you can optionally specify a suffix to append for the run in the **Append Suffix to Run Name** field. Select the desired integer using the controls.



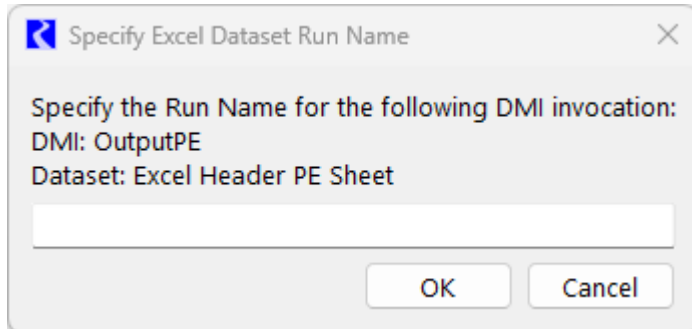
This value is appended to the single run name as a static value; it is **not** incremented within MRM. However, the Data Extractor Tool can increment this value during a data extraction. See [“Extract Data From Selected Model Files” in Solution Approaches](#). Note, this value can be set by a script using the Configure Excel Dataset action as described in [“Configure Excel Dataset” in Automation Tools](#).

- **Specify Single Run Name when DMI is Invoked.** For a single run, the **Single Run Name** must be entered when the DMI is invoked. The name is then used to map to the appropriate dimension in Excel depending on

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the orientation selected (either row, column, or sheet). When this option is selected, the name must be entered when the DMI is invoked, in the following window:



If a blank single run name is specified, or canceled, an error is raised and the DMI invocation stops.

Note: This option produces an error if the DMI is executed in MRM or Batch mode.

- **“Run” with MRM run number.** For multiple runs, a single run name is not adequate to label each run individually. Two options are provided for labeling multiple runs. If **“Run” with MRM run number (i.e. Run0, Run1, etc.)** is selected, then the run name used for mapping to Excel will be the word Run followed by the index number of the run within the multiple run.

Note: The index number used is zero-based to be compatible with how multiple run data has historically been written out of the ExcelWriter tool. The **Run** option cannot be used in the case of a distributed MRM run because run numbers start at zero in each distributed piece. A DMI error message is issued in this situation.

- **“Trace” with MRM trace number.** If **“Trace” with MRM trace number (i.e. Trace5, Trace6, etc.)** is selected, then the run name used for mapping to Excel will be the word Trace followed by the trace number of the run within the multiple run. Trace numbers are a one-based index into the runs, but the initial trace number for starting the multiple run can be specified by the user. Trace numbers can be used successfully with distributed MRM runs. See [“Input DMIs” in Solution Approaches](#) for details on setting up trace numbers in MRM.

Note: Excel worksheet names (i.e. the worksheet tab label) must be less than or equal to 31 characters; this limit is imposed by Excel. If you choose to have Runs on sheets and you type in text in the **Single Run Name** field, it is limited to 31 characters. If you choose to have Slots as sheets and your slot name is more than 31 characters, when the DMI is run the slot is not written/read and a message is posted. In this case, use Name Maps to map the slot names to shorter names that will be accepted by Excel.

Begin or End of Timestep Reference

For Series Slots, an option is provided to **Use Begin Timestep Time Instead of End of Timestep**.

☒ Use Begin Timestep Time Instead of End of Timestep

Timesteps in RiverWare are dated with the end of timestep time, that is, a 1 Day timestep for the last day of the year 2000 is dated 12/31/2000 24:00. If the begin timestep option is not checked, this date is written to (and is expected to be read from) Excel as 12/31/2000 23:59. A minute is subtracted compared to the RiverWare end of time so that the date appears in the correct month and year in Excel. (Excel has no concept of a 24:00 display, so 12/31/2000 24:00 is interpreted by Excel as 1/1/2001 00:00, which would appear to be the wrong month and year.)

If **Use Begin Timestep Time Instead of End of Timestep** is checked, then the timestep is written to (and is expected to be read from) Excel as the beginning of timestep in RiverWare. For the 12/31/2000 24:00 timestep in RiverWare at a 1 Day timestep size, the time in Excel would be 12/31/2000 00:00. This option may be particularly useful if the user is entering data into Excel and is typing in 1 Day timesteps as 12/30/2000, 12/31/2000, 1/1/2001, and so on, which end up by default with 00:00 hours and minutes in the Excel date format. With the begin timestep option checked, the data associated with these times will be moved into RiverWare for their correct timesteps. When the model timestep is 1 Hour, the 12/31/2000 24:00 timestep in RiverWare would have a time in Excel of 12/31/2000 23:00 if the Use Begin Timestep Time option is checked and an Excel time of 12/31/2000 23:59 if the option is not checked.

Clear Dates and Values for Output

Select the **Clear Dates and Values for Output** options to remove the dates and values before writing outputs.

☒ **Clear Dates and Values for Output**

This is valid and enabled only under the following conditions:

- The DMI is an Output DMI
- The dataset maps by headers, not ranges
- The dataset is for series data not table/periodic/scalar data
- The dataset must use the Timesteps Slots Run(s) Header option as shown below:

Header Orientation

Rows	Columns	Sheets
☒ Timesteps	☐ Slots	☐ Run(s)

- The **Read/Write the workbook directly** option in the **Connection Process** is selected

When this option is selected, both the dates and values will be deleted first and then the DMI will write new dates and values. For example, in the following figure, it will first clear out the dates and value for rows for 2 through 8, for 11/21 through 11/27, and then write values for 12/1 to 12/7 to get the screenshot on the right.

	A	B	C
		LakeMeadHooverDam.	LakeMohaveDavisDam.
		Pool Elevation (ft)	Pool Elevation (ft)
1			
2	11/21/1996	1192.2	635.2
3	11/22/1996	1192.2	635.1
4	11/23/1996	1192.3	634.9
5	11/24/1996	1192.3	635.0
6	11/25/1996	1192.4	635.1
7	11/26/1996	1192.4	635.2
8	11/27/1996	1192.4	635.3
9			
10	Average	1192.3	635.1



	A	B	C
		LakeMeadHooverDam.	LakeMohaveDavisDam.
		Pool Elevation (ft)	Pool Elevation (ft)
1			
2	12/1/1996	1192.6	635.3
3	12/2/1996	1192.6	635.4
4	12/3/1996	1192.7	635.6
5	12/4/1996	1192.7	635.7
6	12/5/1996	1192.8	635.8
7	12/6/1996	1192.8	635.9
8	12/7/1996	1192.8	636.0
9			
10	Average	1192.7	635.6

Warning: Blank lines should be used to separate dates and values from other content. Otherwise, values could be inadvertently cleared. For example, if there were values in row 9 above, they could be cleared, depending on the type of data present.

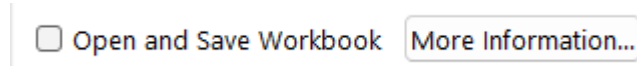
Include Unit Name

An option is also provided with the header approach to **Use Unit Name with Slot Name**. In an output DMI, this option will attach the unit name to the slot name when the slot name is written out to its appropriate row, column, or sheet, depending on the orientation selected. The format is the slot name, a space, and the unit name in parentheses, such as Storage Reservoir.Inflow (cfs). In an input DMI, this option means that the DMI will expect the unit name to be attached to the slot name as indicated above when it is looking for the slot in the Excel headers or sheets.

Open and Save Workbook

The final part of the Excel DMI tab is the choice of whether to have the DMI open and save the workbook as part of the process. The control is shown in [Figure 4.27](#).

Figure 4.27 Screenshot of the Open and Save Workbook selection



Whether checked or not, the DMI will read or write the workbook data directly. But, you can select whether to open and save the workbook. The choices are:

- Leave the **Open and Save Workbook** box *unchecked* to read or write the data directly, without opening Excel. The Excel file is read/written directly, and no formulas are evaluated. This is the default.
- *Check* the **Open and Save Workbook** option to open the workbook in Excel in the background. This may be necessary to evaluate formulas to recompute data or link to other workbooks. The behavior is as follows:
 - For an input DMI, the DMI will open and then save the workbook before importing data to RiverWare.
 - For an output DMI, the DMI will write the data to the Excel file and then open and save the workbook.

In both the input and output cases, formulas can be evaluated when the Excel file is opened, as configured in the Excel workbook.

Select the **More Information** button get a quick description of the two options.

Web Service Datasets

A web service dataset provides a connection to a website that accesses the data in a database. The dataset creates the query URL and sends it to the website, which then accesses the data and returns it in the appropriate format. RiverWare then reads the data and imports it into the slots. Web service datasets support Input DMIs only.

A web service URL uses the form:

base_URL?query_string

The *query_string* is a series of keyword=value pairs separated by “&”.

Your organization may use multiple web services, each with a different base URL. The following web services are supported:

- [“CDSS Surface Water Time Series - Day”](#) on page 142

- [“CWMS Data API Web Service”](#) on page 145
- [“HDB Web Service”](#) on page 146
- [“Reclamation Hydromet”](#) on page 149
- [“USGS Daily Values”](#) on page 150

You can access the Web Service Dataset dialog from either of the following:

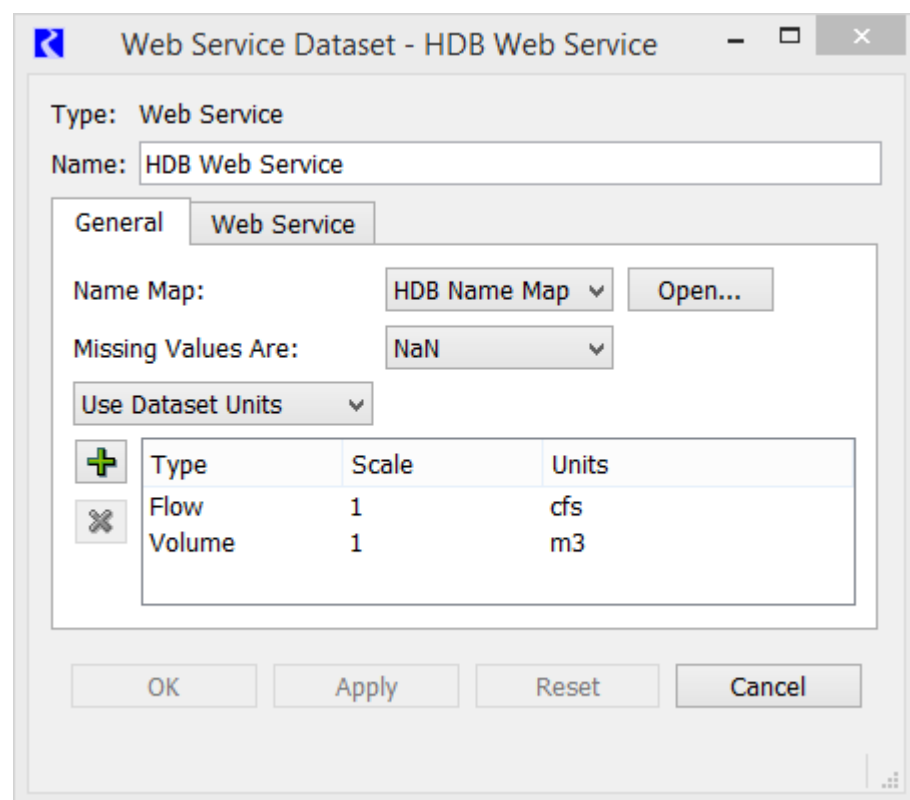
- Database DMI
- Dataset Manager

In the Web Service Dataset dialog, you can rename and configure the dataset. The dialog has the following tabs:

- [General Tab](#), page 139
- [Web Service Tab](#), page 141

General Tab

The General tab includes options general to nearly all datasets. This section describes how to use these options for a web service dataset.



Name Map

Select the desired name map from the Name Map menu. The selections are populated based on the name maps configured in the Name Map Manager. See [“Name Mapping”](#) on page 154 for details on creating name maps.

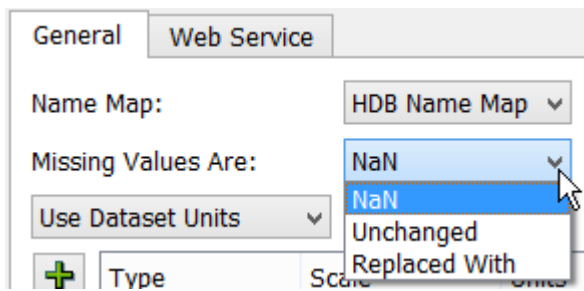
Name maps have special relevance for web service DMIs because they provide specific information on how a RiverWare slot maps to a location number or parameter code in the web database. See the following sections for details on specifying the names maps for each service:

- [“HDB Web Service Use of Name Maps”](#) on page 148
- [“CWMS Data API Use of Name Maps”](#) on page 146
- [“USGS Daily Values Use of Name Maps”](#) on page 153
- [“CDSS Daily Values Use of Name Maps”](#) on page 144

Missing Values

Specify how missing values are handled when data is imported. The options are as follows:

- **NaN**—on import, missing values in the database are set to NaNs in RiverWare.
- **Unchanged**—on import, RiverWare data is left unchanged for a missing value in the database.
- **Replaced With**—on import, missing values are replaced by a specified input value.



Time Zone

For CWMS Data API web service datasets (and DSS datasets), a Time Zone selector is shown. This time zone is only used for the import of pseudo-regular time series data. For more information:

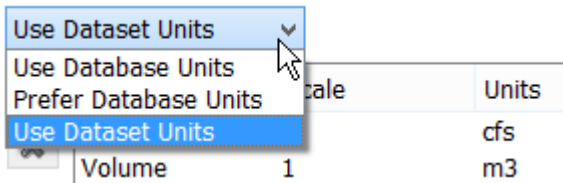
- CWMS Data API, see [“Time Zone”](#) on page 145
- DSS datasets, see [“Time Zone”](#) on page 97

Units

A menu in the Dataset dialog allows you to specify how units should be handled for a dataset. The options are as follows:

- **Use Database Units**
- **Prefer Database Units**

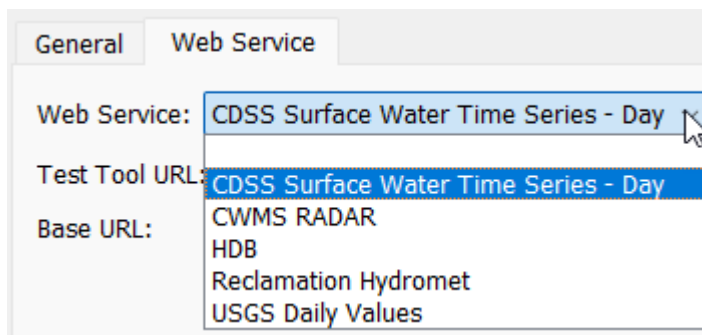
- **Use Dataset Units**



Units are currently not read from any web service; therefore, **Use Dataset Units** is the most applicable selection. If the unit type of the data does not match the unit type of the slot in RiverWare and the units cannot be converted, the slot is skipped and a warning message is given.

Web Service Tab

The Web Service tab includes the web service to use, a test tool URL, the base URL, and any other necessary information. Select the desired Web Service from the menu.



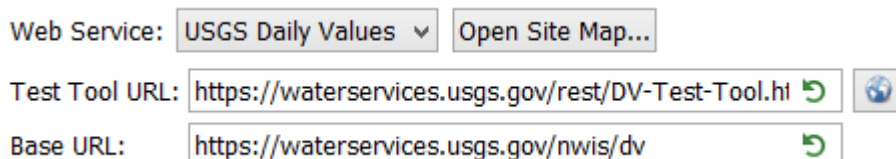
The dialog also includes the following common functionality.

- Test Tool URL (where supported)—URL that leads to the web service test query tool, which you can use to find locations and the desired parameters. Details for each web service are provided in the sections below. Use the

Web button to open a browser to the URL. 

- Base URL—URL on which all queries are formed.

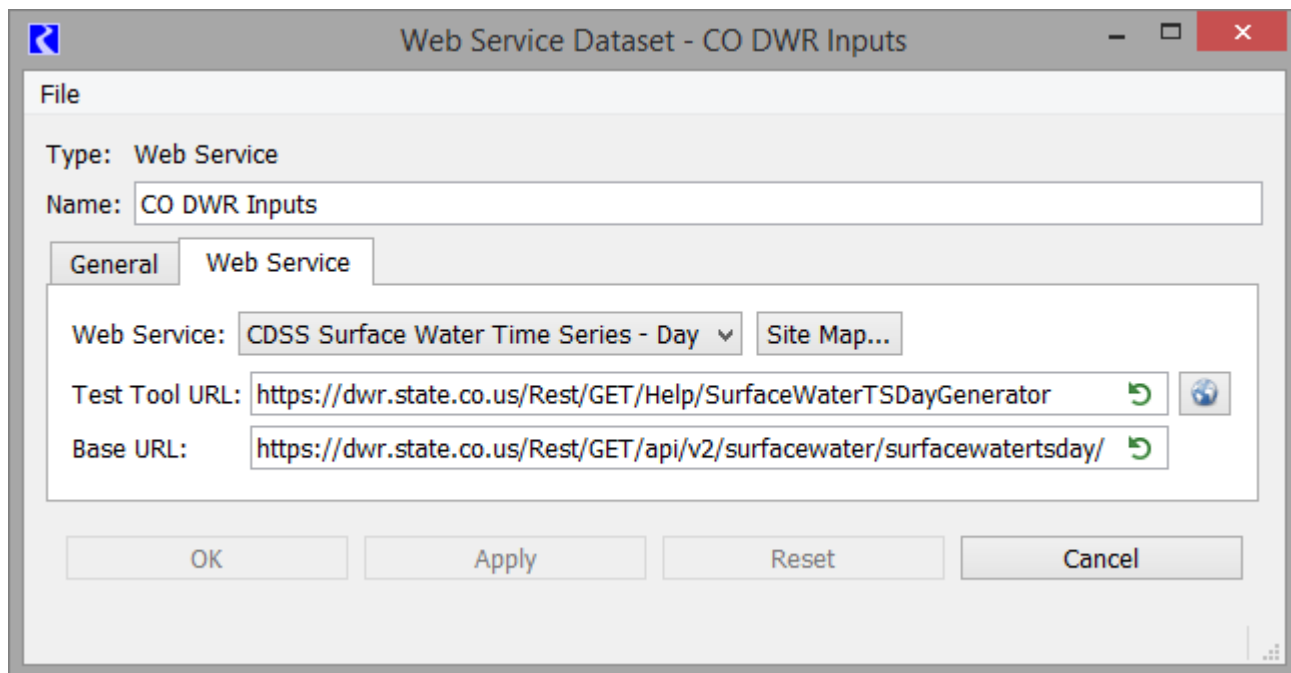
You can edit the URLs to change them, if necessary. Use the green arrow Reset button to set it to the default.



The following sections describe each web service.

CDSS Surface Water Time Series - Day

This web service connects to Colorado’s Decision Support Systems (CDSS). The web service is called CDSS Surface Water Time Series - Day.



The default URLs are as follows:

- Test query tool URL: <https://dwr.state.co.us/Rest/GET/Help/SurfaceWaterTSDayGenerator>
- Base URL: <https://dwr.state.co.us/Rest/GET/api/v2/surfacewater/surfacewatertsd/> This is editable if the Base URL changes.

Note: Since this is a daily values web service, data can only be imported to slots that have a 1 Day timestep size.

RiverWare identifies data by Object.Slot, while CDSS uses a site name and site ID. You must create a site map and/or a name map to specify how the Object.Slot are connected to the site ID.

See “[CDSS Site Maps](#)” on page 142 for details.

See “[CDSS Daily Values Use of Name Maps](#)” on page 144 for details.

Note: Date ranges are handled automatically by the specification of start and finish timesteps in the Database DMI.

CDSS Site Maps

The web service site map allows you to map object names to site IDs by querying the web service for potential site matches.

From the Web Service tab, select the **Site Map** button. The dialog lists all objects, by object type in the model. Navigate the tree view to find the desired object.

When you double click in the Site Name column, the site map queries the web service, in order, for the

1. **Alt Search Name** - For example, if your object is named, “South Platte at Denver”, it will not find it as the actual name is “South Platte River at Denver”, so you can enter an Alt Search Name, “South Platte” for example and it will look for that string.
2. **Object Name** - If your objects names closely match the Station Names, no Alt Search Name is necessary. It will search for the Object Name directly.

Also specify at the top of the dialog if you want to search in the Station Name or Abbreviated Name, as the example shows.

CDSS Surface Water Time Series - Day Site Map

Search Sites By: Example:
☒ Station Name SOUTH PLATTE RIVER AT DENVER
☐ Abbreviated Name PLADENCO

Test Tool URL:

Object Name	Alt Search Na	Site Name	Site ID
▷ DataObj			
▷ StorageReservoir			
▲ StreamGage			
Alamosa			
Otowi	Otowi Bridge	RIO GRANDE AT OTOWI BRIDGE NEAR SA...	2374

OK Cancel

Once it is finished, it shows a drop down list of the sites whose name contains the Alt Search Name or Object Name as a substring (in this example Alamosa):

Object Name	Alt Search Na	Site Name	Site ID
▷ DataObj			
▷ StorageReservoir			
▲ StreamGage			
Alamosa			
Otowi	Otowi Bridge	RIO GRANDE AT ALAMOSA ALAMOSA RIVER AB...E RESERVOIR, CO. ALAMOSA RIVER BE...E RESERVOIR, CO.	2374

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Select the desired name from the list and the Site ID is populated, for example 670:

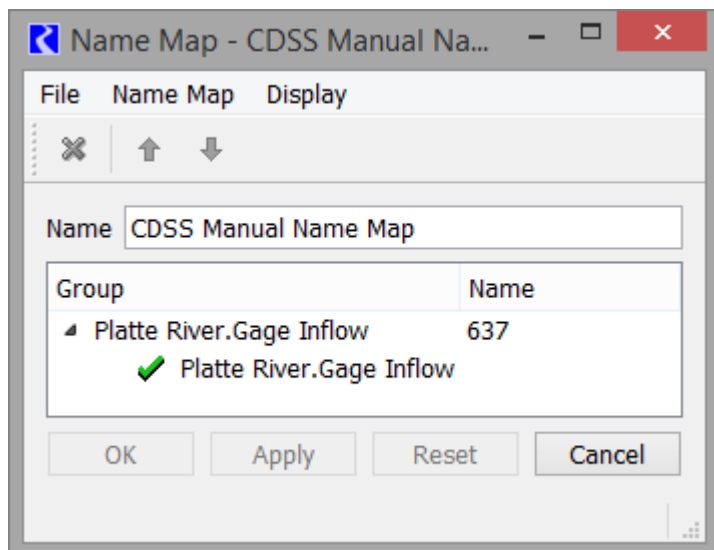
Object Name	Alt Search Na	Site Name	Site ID
▶ DataObj			
▶ StorageReservoir			
▲ StreamGage			
Alamosa		RIO GRANDE AT ALAMOSA	670
Otowi	Otowi Bridge	RIO GRANDE AT OTOWI BRIDGE NEAR SA...	2374

Tip: In the screenshot above, the gage Alamosa queried form the object name directly. For the other gage, Otowi, the user typed in Otowi Bridge as the Alt Search Name. This narrowed down the query to be more detailed.

Tip: The Site Name and Site ID field are editable, enabling you to enter the site name and ID if querying the web service didn't return the appropriate site. This would occur in the object name wasn't a substring of the site name, for example if the name in RiverWare is "Frying Pan" but the CDSS lists is as "Fryingpan". You could enter Fryingpan as the Site Name and then enter the correct Site ID or just enter Fryingpan as the Alt Search Name.

CDSS Daily Values Use of Name Maps

Site Maps, ("CDSS Site Maps" on page 142) map Object names to sites and are the recommended approach. If those don't work for some database reason, you can map Object names using normal Name Maps. Below is a screenshot of a name map that maps Platte River.Gage Inflow to Site ID 637.



Note: Unlike USGS gages that have multiple parameters, the CDSS sites are for a single parameter. For example Site ID 637 returns SOUTH PLATTE RIVER AT DENVER gage flows in cfs. For that reason, Name Maps are not required to map the slots to the parameters.

More information can be found in the service documentation at:

<https://dwr.state.co.us/Rest/GET/Help/SurfaceWaterTSDayGenerator>

CWMS Data API Web Service

This web service connects to the CWMS Data API (CDA) web service. This is a publicly available web service that pulls data from the public national CWMS database.

The URLs are as follows:

- Test Tool URL: <https://cwms-data.usace.army.mil/cwms-data/swagger-ui.html>
- Base URL: <https://cwms-data.usace.army.mil/cwms-data/timeseries>

USACE Office

For the CWMS Data API web service, configure the USACE Office that is responsible for the data. Five common offices are listed. If your office is not listed, double click in the office list and type your office code.

Time Zone

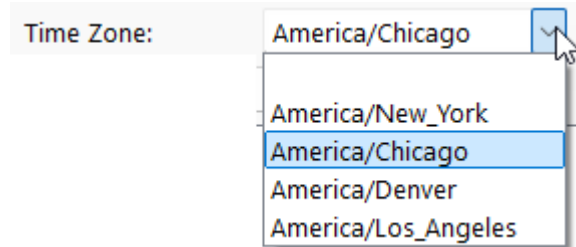
For the CWMS Data API web service, you may need to select the **Time Zone** on the **General** tab.

Note: The time zone is currently only used for pseudo-regular time series. Pseudo-regular data is typically stored in the database in UTC but represents a measurement taken at the same local time each day, regardless of whether daylight saving time is in effect. For example a value might be stored on Oct 02, 2023 at 05:00 UTC but represents Oct 01, 2023, 24:00 local time. If you are not importing this type of data, it can remain blank.

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Specify the time zone of the objects / slots that will be imported. Select one of the common North American time zones or type in the IANA canonical or linked time zone for any time zone. See <https://www.iana.org/time-zones>



CWMS Data API Use of Name Maps

RiverWare identifies data by Object.Slot, while CWMS uses a query string.

Note: In the CWMS Data API web service dataset, you must create a name map to map the Object.Slot to the correct location and parameter.

In the name map, create a slot selection for each slot. Then, in the Name column, specify the text that represents location, parameter, timestep, and other information. In [Figure 4.28](#), “Keystone.Pool Elevation” is mapped to KEYS.Elev.Inst.1hour.0.Ccp-Rev. Contact the developers of the CWMS Data API web service for more information on the query formats.

Figure 4.28 CWMS Data API name map example

Name CWMS Name Map	
Group	Name
Keystone.Pool Elevation	KEYS.Elev.Inst.1hour.0.Ccp-Rev
✓ Keystone.Pool Elevation	

Other Information

This CWMS Data API now explicitly uses version 2.

The Version 2 web service supports pagination, in which a large logical query could be split into multiple actual queries. For example, a logical query of 2,000 values could be split into four actual queries of 500 values. The default query size is 500. (If necessary you can adjust the page size by setting the environment variable WEBSVC_CDA_PAGESIZE to a non-negative integer.)

The presence of the "next-page" element in the JSON document informs the CDA web service that another actual query is required, and it performs the query by incorporating the next page's ID into the query URL. The CDA web service iterates until there isn't a "next-page" element.

HDB Web Service

This web service connects to the HDB web service.

Note: HDB datasets allow direct connections to and from an HDB database. See “HDB Datasets” on page 99 for details.

Web Service Dataset - HDB Web Service

Type: Web Service

Name: HDB Web Service

General Web Service

Web Service: HDB

Test Tool URL: https://www.usbr.gov/lc/region/g4000/riverops/_HdbWebQuery.html

Base URL: <https://www.usbr.gov/pn-bin/hdb/hdb.pl>

Desired HDB: Lower Colorado Regional Office

☒ Observed ☐ Model Run ID = 0

OK Apply Reset Cancel

The default URLs are as follows:

- Test tool URL: https://www.usbr.gov/lc/region/g4000/riverops/_HdbWebQuery.html
- Base URL: <https://www.usbr.gov/pn-bin/hdb/hdb.pl>

Configuration

When setting up an HDB web service dataset, specify the HDB to use and the type of data.

First, select the **Desired HDB**.

Desired HDB: Lower Colorado Regional Office

☒ Observed

- Lower Colorado Regional Office
- Upper Colorado Regional Office
- Eastern Colorado Area Office
- Yuma Area Office
- Lahontan Basin Area Office
- Pacific Northwest Regional Office
- Great Plains Regional Office

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Then, select either **Observed** or **Model Run ID**. For the latter, also specify the Model Run ID.

☒ **Observed** ☐ **Model Run ID** =

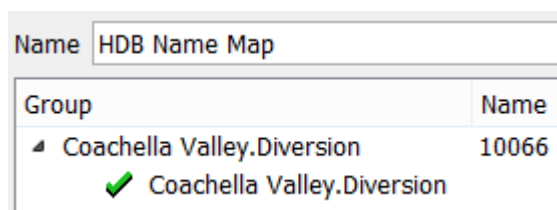
HDB Web Service Use of Name Maps

RiverWare identifies data by Object.Slot, while HDB uses Site / Data type IDs (SDIs) or other identifiers.

Note: In the HDB web service dataset, you must create a name map to map the Object.Slot to the SDI or other identifier.

In the name map, create a slot selection for each slot, then specify the SDI in the Name column. In [Figure 4.29](#), “Coachella Valley.Diversion” is mapped to SDI 10066.

Figure 4.29 HDB web service name map example



Group	Name
Coachella Valley.Diversion	10066

HDB Web Service Timesteps

The HDB Web Service returns values that include dates at the beginning of the timestep. The values are imported at the appropriate timestep in the slot. For example, for daily values using RiverWare dates 9/1/2019 24:00 through 10/1/2019 24:00, the query is:

```
https://www.usbr.gov/pn-bin/hdb/hdb.pl?svr=lchdb&sdi=10066&tstp=DY&t1=2019-09-01T00:00&t2=2019-10-01T00:00&table=R&mrid=0&format=json
```

Here the bold value indicates 9/1/2019 00:00 which is the beginning of the timestep 9/1 00:00 - 24:00. In RiverWare, this daily timestep is 9/1/2019 24:00. The HDB web service uses beginning of timestep notation, RiverWare uses end of timestep.

This can look misleading for longer timesteps. For example, the query for data for RiverWare monthly values December 2019 to January 2020 is:

```
https://www.usbr.gov/pn-bin/hdb/hdb.pl?svr=lchdb&sdi=1930&tstp=MN&t1=2019-12-1&t2=2020-1-1&table=R&mrid=0&format=json
```

It returns the two data point:

```
{"t":"12/1/2019 12:00:00 AM","v":"1090.49"},  
{"t":"1/1/2020 12:00:00 AM","v":"1094.68"}
```

The DMI then imports these beginning of timestep values into the rows for 12-2019 and 1-2020, which of course are represented internally in RiverWare as 12/31/2019 24:00 and 1/31/2020 24:00, respectively.

Figure 4.30 Sample HDB web service imported cells

12-2019	1,090.49	I
01-2020	1,094.68	I

Reclamation Hydromet

This web service connects to Reclamation’s Hydromet web service hosted by the Missouri Basin and Arkansas Rio Grande-Texas Gulf region. (Previously, the Great Plains region)

Web Service Dataset - Hydromet Web Service Dataset

File

Type: Web Service

Name: Hydromet Web Service Dataset

General Web Service

Web Service: Reclamation Hydromet

Test Tool URL: https://www.usbr.gov/gp/hydromet/archive_form.html

Base URL: <https://www.usbr.gov/gp-bin/arcread.pl>

The default URLs are as follows:

- Test query tool URL: https://www.usbr.gov/gp/hydromet/archive_form.html
- Base URL: <https://www.usbr.gov/gp-bin/arcread.pl> This is editable if the Base URL changes or other Reclamation offices have similar web services.

Note: Since this is a daily values web service, data can only be imported to slots that have a 1 Day timestep size.

Note: Date ranges are handled automatically by the specification of start and finish timesteps in the Database DMI.

Reclamation Hydromet use of Name Maps

RiverWare identifies data by Object.Slot, while Hydromet uses site and parameters.

Note: In the HDB web service dataset, you must create a name map to map the Object.Slot to the site/parameter.

In the name map, create a slot selection for each slot, then specify the Site and Parameter in the Name column. In [Figure 4.31](#), “Buffalo Bill.Outflow” is mapped to BBR QD.

Figure 4.31 Hydromet web service name map example

Name: Hydromet

Group: Buffalo Bill.Outflow

Name: BBR QD

Buffalo Bill.Outflow

More information can be found in the web service documentation at

https://www.usbr.gov/gp/hydromet/automated_retrieval.pdf

USGS Daily Values

This web service connects to the USGS Daily Values web service.

Web Service Dataset - USGS Input

File

Type: Web Service

Name: USGS Input

General Web Service

Web Service: USGS Daily Values Site Map...

Test Tool URL: https://waterservices.usgs.gov/test-tools/

Base URL: https://api.waterdata.usgs.gov/ogcapi/v0/collecti

☐ Specify API Key Info...

API Key:

The following fields and controls are available:

- Test Tool URL: <https://waterservices.usgs.gov/test-tools/>
- Base URL: <https://api.waterdata.usgs.gov/ogcapi/v0/collections/daily/items> This is editable if the Base URL changes.
- API Key: Use the controls to specify an API Key, if desired. Without an API Key, only 50 queries (i.e. slots) can be made per hour. With an API Key, 1000 queries per hour can be made. See “API Keys” on page 151 for more information.

Note: Since this is a daily values web service, data can only be imported to slots that have a 1 Day timestep size.

Tip: It is recommended to only have one USGS web service dataset per model. This can be shared by multiple DMIs. As a result, it is only necessary to have one Site Map and one Name Map that contains all of the mappings for the model.

RiverWare identifies data by Object.Slot, while USGS uses a site and parameter ID. You must create a site map and/or a name map to specify how the Object.Slot are connected to the gage number and parameter.

See “[USGS Site Maps](#)” on page 151 for details.

See “[USGS Daily Values Use of Name Maps](#)” on page 153 for details.

Note: Date ranges are handled automatically by the specification of start and finish timesteps in the Database DMI.

Note: The USGS web service will break up long queries into multiple pages. The default page size is 500, so 500 timesteps of a series slot can be imported with one query. If necessary, you can set the environment variable WEBSVC_USGS_LIMIT to a different non-negative integer. A smaller value will cause more paging and could lead to timeouts. A larger value (up to 10,000), will lead to fewer pages but runs the risk of transmitting a large amount of data for each query, which can lead to other issues.

API Keys

The USGS provides machine readable and automated access to water data through their Application Programming Interface (API). For more information, see <https://waterdata.usgs.gov/blog/api-whats-new-wdfn-apis/#introduction>.

Queries data with or without an API Key. Without an API key, the web service is limited to 50 requests per IP address per hour. With an API key, the limit is 1,000 requests per IP per hour. RiverWare queries the web service one slot at a time, so a limit of 50 requests per IP address per hour is a limit of 50 slots per hour. API keys are further discussed at <https://waterdata.usgs.gov/blog/api-whats-new-wdfn-apis/#api-keys>.

API keys are free and can be obtained at <https://api.waterdata.usgs.gov/signup/>.

Figure 4.32 shows the API controls. Check the box to specify the key. The Info button provides information similar to this section.

Note: The API Key must be specified per user and is stored in the registry. It is shared amongst all USGS web service datasets although it is recommended to only have one USGS web service dataset per model

Figure 4.32 USGS Web Service API key controls



USGS Site Maps

The web service site map allows you to map object names to site IDs by querying the web service for potential site matches.

From the Web Service tab, select the **Site Map** button. The dialog lists all objects, by object type in the model. Navigate the tree view to find the desired object.

When you double click in the Site Name column, the site map queries the web service, in order, for the

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1. **Alt Search Name** - For example, if your objects are named something different than the station name, it will not find it, so you can enter an Alt Search Name, "Heron Reservoir" for example and it will look for that string.
2. **Object Name** - If your objects names match the Station Names, no Alt Search Name is necessary. It will search for the Object Name directly.

Again, double click the Site Name field to initiate the query.

Object Name	Alt Search Name	Site Name	Site ID
> DataObj			
▼ StreamGage			
Alamosa	Rio Grande at Alamosa	RIO GRANDE AT ALAMOSA, CO.	USGS-08223000
Otowi		RIO GRANDE AT OTOWI BRIDGE, NM	

OK Cancel

Once it is finished querying, it shows a list of the sites whose name contains the object name as a substring (in this example Otowi, a location on the Rio Grande):

▼ StreamGage			
Alamosa	Rio Grande at Alamosa	RIO GRANDE AT ALAMOSA, CO.	USGS-08223000
Otowi		RIO GRANDE AT OTOWI BRIDGE, NM	

RIO GRANDE AT OTOWI BRIDGE, NM
SIROTOWITZ WEL...HUR SPRINGS FL
19N.08E.18.313 OTOWI SPRING, NM
19N.07E.18.241 OTOWI SEEP

Select the desired name from the list and the Site ID is populated:

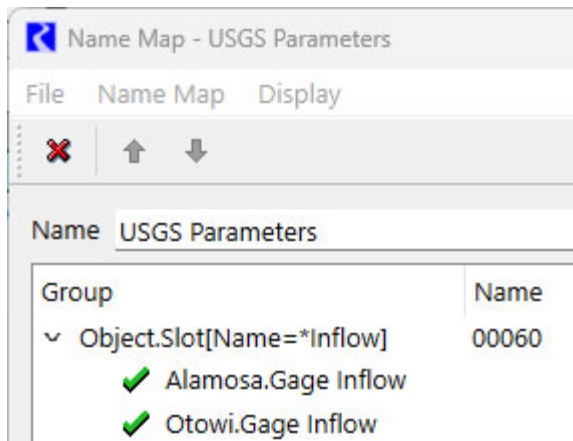
▼ StreamGage			
Alamosa	Rio Grande at Alamosa	RIO GRANDE AT ALAMOSA, CO.	USGS-08223000
Otowi		RIO GRANDE AT OTOWI BRIDGE, NM	USGS-08313000

Tip: The Site Name and Site ID field are editable, enabling you to enter the site name and ID if querying the web service didn't return the appropriate site. This would occur in the object name wasn't a substring of the site name, for example if the name in RiverWare is "Blue Water" but the USGS lists is as "Bluewater". You could enter Bluewater as the Site Name and then enter the correct Site ID, or just enter Bluewater as the Alt Search Name.

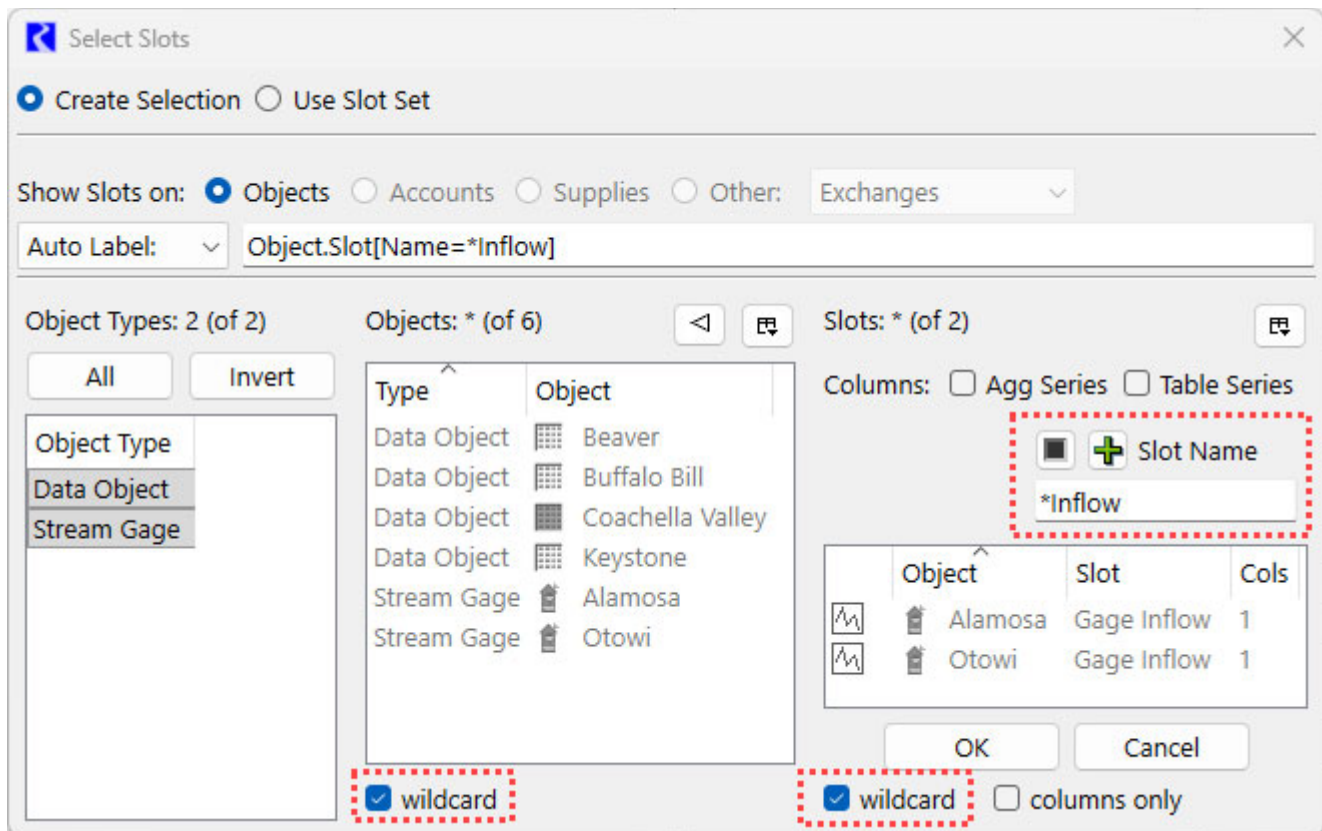
USGS Daily Values Use of Name Maps

Use Site Maps (“[USGS Site Maps](#)” on page 151) to map Object names to sites. Name Maps must be used to connect slot names to parameter IDs.

Most likely, similar slots (e.g., gage inflows, reservoir inflows) have the same parameter ID. As a result, use the name map's wildcarding capability to configure all similar slots with a single mapping.



This slot selection was created using the slot selector as follows; notice the wildcard and filters highlighted:



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If similar slots don't all have the same parameter ID the name map's prioritized selections make it easy to configure the general case with exceptions, see “[Name Map Ordering and Priorities](#)” on page 156.

Also, in the Name Map, specify the parameter ID in the Name column. Some common USGS parameter IDs include:

- Mean discharge in cubic feet per second = 00060
- Elevation of reservoir water surface above datum, feet = 00062
- Stage or gage height, feet = 00065

More information can be found in the service documentation at:

<https://waterservices.usgs.gov/test-tools/>

Although the Site Map utility and wild carding in Name Maps are the preferred approach, you can still use the original approach where the name map specifies the full query string. In the name map, create a slot selection for each slot, then specify the query strings in the Name column. These are typically in the following form:

&Argument=value&Argument2=value

Some common arguments are as follows:

- monitoring_location_id — USGS plus the eight-digit site location, E.g USGS-08313000
- parameter_code — data to return from the gage.

See [Figure 4.33](#) for an example, Otowi.Inflow is mapped to gage 08313000 and parameter 00060 (mean daily flow).

Figure 4.33 USGS Daily Values name map example using query syntax

Name USGS Sites and Parameters	
Group	Name
▼ Otowi.Gage Inflow	&monitoring_location_id=USGS-08313000¶meter_code=00060
✓ Otowi.Gage Inflow	

Note: Date ranges are handled automatically by the specification of start and finish timesteps in the Database DMI; they do not need to be specified in the name map. Similarly, format is also automatic.

Name Mapping

It is likely that RiverWare object or slot names do not match their associated database names. For example, in RiverWare, the reservoir inflow is called *Inflow* in, while in DSS Part C, it is called *RES-INFLOW*. Because different databases use different naming conventions, there is no way for RiverWare to know the database naming convention. The name mapping tool provides a way for you to specify the link between the two naming systems. The name mapping tool allows you to define multiple name mappings.

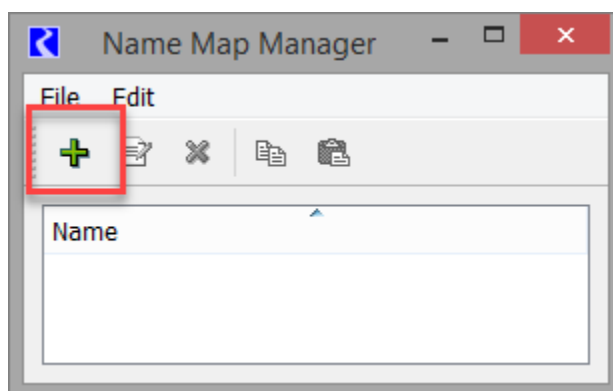
Note: If the names are mapped within the database itself, as they are in HDB, a Name Map is not needed.

Note: For DSS datasets, there is no need to create a name map if the map is strictly to specify a different letter case; this is taken care of automatically. For example, if a reservoir is called BigLake in RiverWare and BIGLAKE in the DSS database, a name map is not necessary.

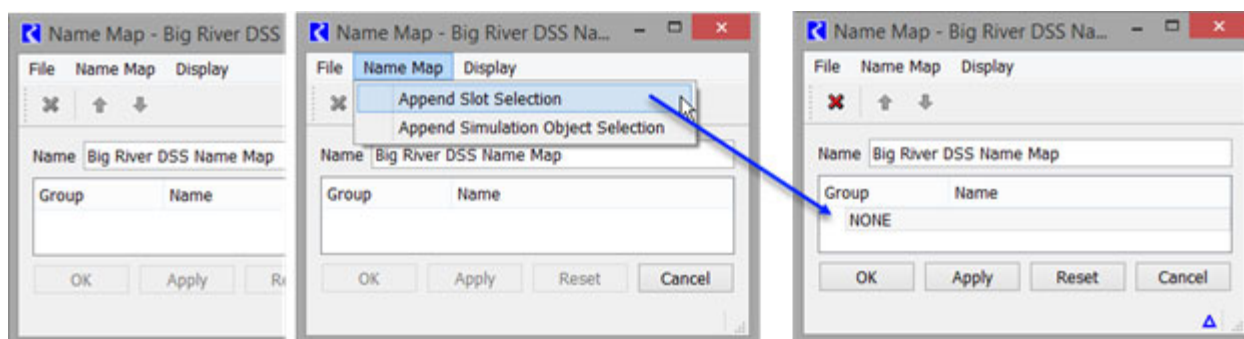
Configuring Name Mapping

To open the name mapping editor from the DMI Manager, select **Utilities**, then **Name Maps**.

This opens the Name Map Manager dialog. Select the **+** icon to add a new Name Map to the editor.



Double-click the newly added Name Map to open the Name Map dialog as shown in the following figure. In this dialog, the user is able to edit the name of the Name Map. Selecting **Name Map**, then **New** adds a new entry to the list.

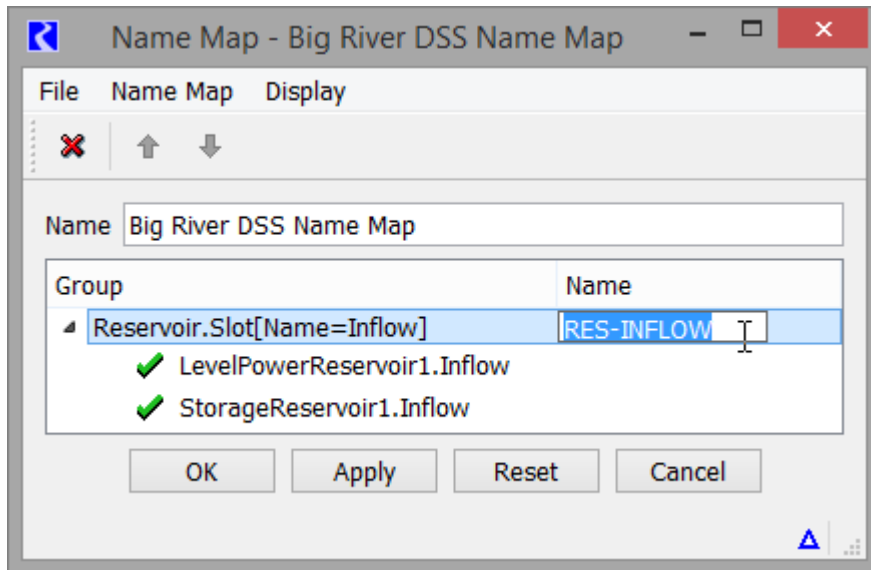


Selecting the newly selected entry opens the context menu options shown in [Figure 4.34](#). Either select **Append Slot Selection** or **Append Simulation Object Selection**. This action adds an item to the list. Double Click the “NONE” item to open the selector dialog for the appropriate selection.

Select the desired slots or objects that will be part of this Name Map. If accounting is enabled, supplies, accounts, exchanges, and paybacks are also available. Use the filters and wildcards to select generic slots. For example, the user could select Reservoir.Inflow as an entry. See [“Creating a Name Map of All Reservoir.Inflow Slots” in User Interface](#) for details on using the selector. The selections made here are considered dynamic root selections. That is, the selection is saved with the model file and is automatically updated if objects or slots are added to or removed from the model. This is especially useful when using wildcards.

To define the alternate name, that is, the name in the database, select the blank area to the right of the entry under the word Name. [Figure 4.34](#) shows an example.

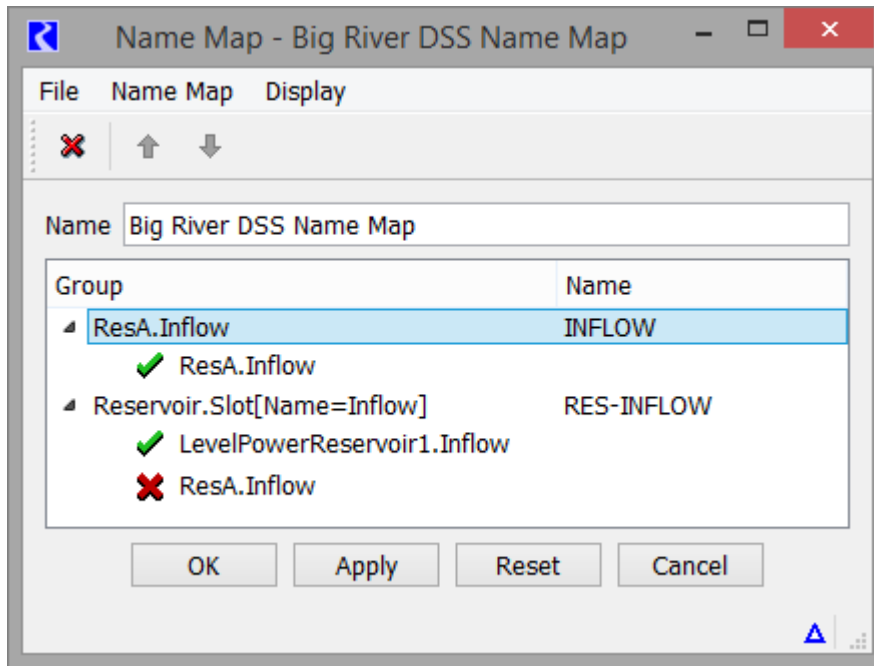
Figure 4.34 DMI Name Map with alternative name specified



Name Map Ordering and Priorities

The entries in the Name Map are prioritized. If there is a conflict between two entries, the entry with the highest priority takes precedence and the other entry becomes disabled with a red X. Thus, if there is a red X, there is a higher priority selection that controls the item. For example, in [Figure 4.35](#), the first entry for ResA.Inflow (INFLOW) takes priority over the second entry (RES-INFLOW). The green check marks show that the first item is controlling the selection. This functionality allows you to create a name map for a generic case but redefine it in special case.

Figure 4.35 Screenshot of DMI Name Map with entries impacted by the ordering



The priority order is based on the least exact match (the more general case, for example Reservoir.Inflow) to the more exact match (the more specific case, for example, ResA.Inflow)

Sometimes you might want to change the display order. Use the **Display** menu and selecting **Custom**. Once in

custom mode, select an entry and then use the up and down arrows   or **Display**, then **Move Up** or **Move Down** file choices to rearrange rows. To return to prioritized mode, the select **Display**, then **Priority**. Regardless of the ordering, the name maps are stored according to priority; custom order is just for display.

Creating a Name Map from a JSON file

Instead of creating the Name Map in the RiverWare interface, you might have tools/scripts to create a JSON text file that represents the name map. Or perhaps, you find it easier to work with text files. Once you have the JSON file created in the correct format, use the **File** and then **Import Slot Mapping from JSON File**.

When the JSON file is imported, the Name Map creates a slot selection mapped to its alternate name for each slot in the file. This is functionally equivalent to selecting each slot and entering its alternate name. The name map also provides detailed feedback on the import – which slot mappings were created, which slot mappings could not be created (for example, because a slot doesn't exist in the model) and any syntax errors in the JSON file.

The JSON file format supports simulation object, account and supply slots. The format is as follows:

```
{
  "slotList": [
    {
      "ObjectSlot": "<Object Name>.<Slot Name>",
      "AltName": "..."
    }
  ]
}
```

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```
    },
    {
      "ObjectAccountsSlot": "<Object Name>^<Account Name>.<Slot Name>",
      "AltName": "..."
    },
    {
      "SupplySlot": "<Supply Name>.Supply",
      "AltName": "..."
    }
  ]
}
```

Tip: Create a template JSON file (basically, the text above) that you can edit. Use the **File** and then **Create Empty Template Slot Mapping JSON File**.

The name map uses the term “AltName”, which is the name in the database, for example an SDID in HDB. You need to specify the <Object.Slot> and “...” for the AltName in the file. For example, the following file could be imported to map Coachella Valley. “Coachella Valley.Diversion” to location/parameter 10066 in the database.

```
{
  "SlotList": [
    {
      "ObjectSlot": "Coachella valley.Diversion",
      "AltName": "10066"
    }
  ]
}
```

Note: You cannot export a Name Map as a JSON file. To use a name map in another model, see [“Export”](#) on page 13. The export is available from the File and then Export Name Map.